2026 PRELIMS 99% MASTERY · CONSOLIDATED CHAPTER-WISE FACTS

ENVIRONMENT & ECOLOGY

पर्यावरण एवं पारिस्थितिकी — संपूर्ण उच्च-प्राथमिकता संकलन

Ecosystem Dynamics, Biodiversity Hotspots, National Parks, Biosphere Reserves, Ramsar Wetlands, Climate Conventions (UNFCCC, CBD), Pollution Acts.

UPPCS & UKPCS Prelims 2026 • High-Yield Master Revision Series • 99% Coverage Edition

Environment & Ecology — Master Chapter Index

Consolidated High-Yield Revision Matrix • All 46 Chapters

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Chapter 01: 00 — Executive Bird's-Eye Master Matrix

★ UPPCS & UKPCS Prelims Daily Fact Dump · Active Recall Cheat-Sheet

Exam Focus: UPPSC Prelims (Paper 1), UKPSC Prelims (Paper 1), RO/ARO, BEO
Frequency: Daily morning read (15–20 mins)
Question Types Covered: Environmental Acts vs Years, International Conventions & Protocols, 18 Biosphere Reserves & UNESCO MAB list, Wildlife Conservation Projects, Environmental Movements & Pioneers, Research Institutes, Ecological Pyramids & Food Chains, AQI pollutants, Ramsar Sites in UP & UK, Environmental Days, Ecological Concepts & Coining Pioneers, Toxic Pollutants & Occupational Diseases, 4 Biodiversity Hotspots of India.

1. Core Environmental Legislation in India (Exact Years & Articles)

LEGISLATION / POLICY NAME	YEAR OF ENACTMENT	CONSTITUTIONAL BACKING / KEY MANDATE	LANDMARK INSTITUTIONAL OUTCOME
Indian Forest Act	**1927** (earlier 1865, 1878)	Consolidated forest laws; categorized into **Reserved, Protected, and Village Forests**	State Forest Departments
Wildlife (Protection) Act (WPA)	**1972** (9 Sept 1972)	Framework for protecting wild animals, birds, and plants; **6 original schedules restructured into 4 schedules in 2022 Amendment**	National Board for Wildlife (NBWL, chaired by PM); NTCA; Wildlife Crime Control Bureau (WCCB)
Water (Prevention and Control of Pollution) Act	**1974**	Prevention and abatement of water pollution; maintains wholesomeness of water	Established **Central Pollution Control Board (CPCB)** & State PCBs
Forest (Conservation) Act (FCA)	**1980** (25 Oct 1980)	Prohibits diversion of forest land for non-forest purposes without prior central approval	Forest clearance mechanism
Air (Prevention and Control of Pollution) Act	**1981**	Prevention, control, and abatement of air pollution; **Noise pollution included as an air pollutant by 1987 Amendment**	Empowered CPCB and SPCBs to set ambient air standards
Environment (Protection) Act (EPA)	**1986** (23 May 1986)	**Umbrella legislation** enacted under **Article 253** in the wake of the **Bhopal Gas Tragedy (Dec 1984)**	Empowered Union Govt to coordinate state actions, set emission norms; CRZ notifications; EIA rules
Public Liability Insurance Act	**1991**	Mandatory public insurance for providing immediate relief to victims of accidents occurring while handling hazardous substances	Environment Relief Fund
National Environment Tribunal Act	**1995**	Strict liability for damages arising out of any accident while handling hazardous substances	Precursor to NGT
Biological Diversity Act (BDA)	**2002**	Realized commitments under CBD (1992); 3-tier structure for conservation and equitable benefit sharing	**National Biodiversity Authority (NBA, Chennai)**, State Biodiversity Boards (SBB), Biodiversity Management Committees (BMC)
Forest Rights Act (FRA)	**2006**	Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act; vests forest land rights	Gram Sabha is the statutory authority to initiate rights recognition process
National Green Tribunal (NGT) Act	**2010** (18 Oct 2010)	Created dedicated environmental tribunal under **Article 21 (Right to a healthy environment)**; mandated disposal within **6 months**	**Principal Bench: New Delhi**; 4 Regional Benches: **Bhopal, Pune, Kolkata, Chennai**
CAMPA Act	**2016**	Compensatory Afforestation Fund Act; manages funds collected for diversion of forest land (90% to State, 10% to Centre)	National & State Compensatory Afforestation Funds

2. Environmental Movements in India (Pioneers & Locations)

MOVEMENT	YEAR & PLACE	KEY LEADERS / ACTIVISTS	CORE ENVIRONMENTAL OBJECTIVE / STRATEGY
Bishnoi Movement	**1730**, Khejarli village (Jodhpur, Rajasthan)	**Amrita Devi Bishnoi**	King Abhay Singh ordered felling of **Khejri trees** (*Prosopis cineraria*); Amrita Devi and her 3 daughters hugged trees; **363 Bishnois sacrificed their lives**; Amrita Devi Bishnoi Wildlife Protection Award instituted in memory.
Chipko Movement	**1973**, Reni village, Chamoli (**Uttarakhand**)	**Sunderlal Bahuguna**, **Chandi Prasad Bhatt** (Ramon Magsaysay awardee), **Gaura Devi** (*Chipko Mother*)	Protest against commercial logging of ash trees in Alaknanda basin; villagers hugged trees to prevent felling; coined slogan: "*Ecology is permanent economy*"; led to 15-year ban on green felling in UP/UK hills.
Appiko Movement	**1983**, Salkani, Uttara Kannada (Karnataka)	**Pandurang Hegde**	Southern version of Chipko; hugged trees in Western Ghats to save tropical forests from commercial monoculture plantations.
Silent Valley Movement	**1978**, Palakkad (Kerala)	Kerala Sastra Sahitya Parishad (**KSSP**), Romulus Whitaker	Protest against proposed **hydroelectric project on Kunthipuzha River** that would submerge the pristine moist evergreen rainforest habitat of the endangered **Lion-Tailed Macaque**; project scrapped; declared Silent Valley National Park (1984).
Narmada Bachao Andolan (NBA)	**1985**, Narmada Valley (Gujarat, MP, Maha)	**Medha Patkar**, **Baba Amte**	Mass movement against large dams on Narmada (particularly **Sardar Sarovar Dam**); demanded comprehensive rehabilitation, environmental impact assessment, and rights of displaced tribal villagers.
Jungle Bachao Andolan	**1982**, Singhbhum (Bihar / Jharkhand)	Local tribal communities	Tribals agitated against government scheme to replace indigenous, multipurpose **Sal forests** with commercially profitable **Teak (Sagwan)** plantations; known as "Greed Game Political Populism".
Tehri Dam Protest	1980s–1990s, Tehri (**Uttarakhand**)	**Sunderlal Bahuguna**	Agitation against the construction of the mega Tehri Dam on the seismic fault lines of Bhagirathi River; multiple hunger strikes.

3. Environmental Research Institutes & Organizations (Match Goldmine)

INSTITUTE / ORGANIZATION	LOCATION / HEADQUARTERS	ESTABLISHED YEAR / KEY FUNCTION
Botanical Survey of India (BSI)	**Kolkata** (West Bengal)	**1890**; survey and conservation of plant resources of India
Zoological Survey of India (ZSI)	**Kolkata** (West Bengal)	**1916**; survey, exploration, and taxonomic research on animal diversity
Forest Research Institute (FRI)	**Dehradun (Uttarakhand)**	**1906** (Imperial Forest Research Institute); Greco-Roman architectural marvel
Wildlife Institute of India (WII)	**Dehradun (Uttarakhand)**	**1982**; premier autonomous institution for wildlife research and management
National Environmental Engineering Research Institute (NEERI)	**Nagpur** (Maharashtra)	**1958**; pioneer in environmental monitoring, sewage, industrial pollution solutions (CSIR)
Central Pollution Control Board (CPCB)	**New Delhi**	**1974**; statutory organization under Water Act 1974; sets national air/water norms
National Biodiversity Authority (NBA)	**Chennai** (Tamil Nadu)	**2003**; statutory body under Biological Diversity Act 2002
Sálim Ali Centre for Ornithology & Natural History (SACON)	**Coimbatore** (Tamil Nadu)	Dedicated to conservation of avifauna and wetland ecology
Indian Institute of Forest Management (IIFM)	**Bhopal** (Madhya Pradesh)	**1982**; premier forestry management and planning institute
G.B. Pant National Institute of Himalayan Environment	**Kosi-Katarmal, Almora (Uttarakhand)**	**1988**; apex research body on sustainable development and ecology of Indian Himalayas
Central Arid Zone Research Institute (CAZRI)	**Jodhpur** (Rajasthan)	**1959**; research on desertification and arid land agriculture
National Disaster Management Authority (NDMA)	**New Delhi**	**2005**; apex body under Disaster Management Act 2005; **Chaired by the Prime Minister**

4. Indian Wildlife Conservation Projects (Chronological Ladder)

PROJECT NAME	YEAR LAUNCHED	LAUNCH LOCATION / NODAL AGENCY	KEY OBJECTIVE / STATUS / PRELIMS FACT
Project Tiger	**1 April 1973**	**Jim Corbett National Park (Uttarakhand)**	Central scheme; initiated with 9 reserves under Kailash Sankhala ("Tiger Man of India"); statutory backing via NTCA (2006); **58 Tiger Reserves** (Madhav TR, MP, is the 58th). Corbett has among the highest tiger densities.
Project Crocodile	**1975**	Hyderabad / Odisha (with UNDP & FAO aid)	Conservation of 3 crocodilian species: Gharial (*Gavialis gangeticus*), Mugger crocodile, and Saltwater crocodile (Bhitarkanika, Odisha)
Project Hangul	**1970**	Dachigam National Park (J&K)	Conservation of Kashmir Stag (*Cervus elaphus hanglu* - critically endangered) with IUCN/WWF aid
Project Rhino (IRV 2020)	**1987 / 2005**	Assam (Kaziranga, Manas, Orang, Pobitora)	Indian Rhino Vision 2020 aimed to attain a wild population of at least 3,000 Greater One-Horned Rhinos in Assam
Project Elephant	**February 1992**	Jharkhand (Singhbhum)	Centrally sponsored; protects elephant habitats and corridors; addresses human-elephant conflict; **Mike sites** (Monitoring of Illegal Killing of Elephants); campaign: *Haathi Mere Saathi*
Project Sea Turtle	**1999**	Odisha coast (Gahirmatha / Rushikulya)	Conservation of **Olive Ridley turtles** (*Lepidochelys olivacea*) during synchronized mass nesting (**Arribada**) with MoEFCC & UNDP
Project Snow Leopard	**2009**	High-altitude Himalayan states (Hemis, Gangotri, Nanda Devi)	Conservation of high-altitude ecosystem and the elusive Ghost of the Mountains (*Panthera uncia*)
Project Dolphin	**2020**	Ministry of Environment, Forest & Climate Change	Gangetic Dolphin (*Platanista gangetica*) declared **National Aquatic Animal on 5 October 2009** (5 Oct celebrated as **National Dolphin Day**); blind; uses echolocation; Vikramshila Sanctuary (Bihar)
Project Cheetah	**September 2022**	**Kuno National Park (Madhya Pradesh)**	Intercontinental translocation of African cheetahs from Namibia (8) and South Africa (12); Asiatic Cheetah declared extinct in India in **1952**

5. Ramsar Wetlands in Uttar Pradesh & Uttarakhand

- **Uttarakhand**:
- **Asan Conservation Reserve** (Dehradun; confluence of Asan and Yamuna; **first Ramsar of Uttarakhand, Oct 2020**).
- **Jhilmil Jheel Conservation Reserve** (Haridwar, **2005**) is the state's swamp-deer / barasingha wetland — **not** the Ramsar first (that is Asan).
- **Uttar Pradesh (12 Ramsar sites)** — do not stop at the older list of 10: 1. **Bakhira Wildlife Sanctuary** (Sant Kabir Nagar). 2. **Haiderpur Wetland** (Muzaffarnagar–Bijnor; Solani / Hastinapur WLS). 3. **Nawabganj Bird Sanctuary** (*Chandra Shekhar Azad*, Unnao). 4. **Parvati Arga Bird Sanctuary** (Gonda). 5. **Saman Bird Sanctuary** (Mainpuri). 6. **Samaspur Bird Sanctuary** (Rae Bareli). 7. **Sandi Bird Sanctuary** (Hardoi). 8. **Sarsai Nawar Jheel** (Etawah; Sarus Crane). 9. **Sur Sarovar (Keetham Lake)** (Agra). 10. **Upper Ganga River** (Brijghat to Narora). 11. **Patna Bird Sanctuary** (Etah). 12. **Shekha Jheel** (Aligarh).

> **Traps:** Sultanpur NP = **Haryana**. Rudrasagar = **Tripura**. Samaspur and Sarsai Nawar **are** in UP.

5A. Protected areas of UP & UK — Declared / still in force

PA	STATE	FROM (YEAR)	HIGH-YIELD
Hailey → Ramganga → Jim Corbett NP / TR	**Uttarakhand**	NP **1936** (Hailey); Project Tiger launch site **1973**	India's first National Park. Nainital–Pauri belt.
Rajaji NP / TR	**Uttarakhand**	NP **1983**; tiger reserve **2015**	Elephant corridor; Haridwar–Dehradun–Pauri
Nanda Devi NP	**Uttarakhand**	**1982** (BR **1988**; UNESCO mixed serial with VoF)	Chamoli; Nanda Devi peak
Valley of Flowers NP	**Uttarakhand**	**1982** (UNESCO **1988**, expanded **2005**)	Smallest UK NP; Chamoli
Gangotri NP	**Uttarakhand**	**1989**	Largest UK NP (~2390 sq km); Uttarkashi
Govind NP / WLS	**Uttarakhand**	WLS **1955**; NP later	Snow leopard belt; Uttarkashi
Dudhwa NP / TR	**Uttar Pradesh**	NP **1977**; TR **1987**	Lakhimpur Kheri Terai; rhino from **1984**; barasingha
Pilibhit TR	**Uttar Pradesh**	**2014**	Terai; tiger + woodland swamp deer
Ranipur TR	**Uttar Pradesh**	**2022/2023**	Chitrakoot–Banda (Vindhyan); not Terai
Hastinapur WLS	**Uttar Pradesh**	**1986**	**Largest UP sanctuary**; Meerut–Bijnor belt; Haiderpur Ramsar inside
Katarniaghat WLS	**Uttar Pradesh**	**1975**	Terai Arc toward Nepal; gharial / dolphin

UP has **no** biosphere reserve and **no** marine PA. UK biosphere = **Nanda Devi (1988; UNESCO MAB 2004)**.

6. National Air Quality Index (AQI)

Launched in **October 2014** under Swachh Bharat Abhiyan by CPCB: - **Six Categories**: 1. *Good* (0–50) 2. *Satisfactory* (51–100) 3. *Moderately Polluted* (101–200) 4. *Poor* (201–300) 5. *Very Poor* (301–400) 6. *Severe* (401–500). - **Eight Monitored Pollutants (Frequently Tested in PCS!)**: 1. **Particulate Matter 10 (PM10)** 2. **Particulate Matter 2.5 (PM2.5)** 3. **Nitrogen Dioxide (NO2)** 4. **Sulphur Dioxide (SO2)** 5. **Carbon Monoxide (CO)** 6. **Ozone (O3)** 7. **Ammonia (NH3)** 8. **Lead (Pb)**. *(Notice: Carbon Dioxide (CO2) is NOT part of the AQI 8 pollutants!)*

7. The 18 Biosphere Reserves of India & 12 UNESCO MAB Reserves

NO.	BIOSPHERE RESERVE	YEAR SET UP	STATE / REGION	PART OF UNESCO MAB (WNBR)?
1	**Nilgiri**	**1986**	Tamil Nadu, Kerala, Karnataka	**Yes (2000 — First in India)**
2	**Nanda Devi**	**1988**	**Uttarakhand** (Chamoli, Pithoragarh, Bageshwar)	**Yes (2004)**
3	**Nokrek**	**1988**	Meghalaya (Garo Hills)	**Yes (2009)**
4	**Great Nicobar**	**1989**	Andaman and Nicobar Islands	**Yes (2013)**
5	**Gulf of Mannar**	**1989**	Tamil Nadu (marine)	**Yes (2001)**
6	**Manas**	**1989**	Assam	No
7	**Sundarbans**	**1989**	West Bengal	**Yes (2001)**
8	**Similipal**	**1994**	Odisha (Mayurbhanj)	**Yes (2009)**
9	**Dibru-Saikhowa**	**1997**	Assam (*Smallest Biosphere Reserve*)	No
10	**Dehang-Debang**	**1998**	Arunachal Pradesh	No
11	**Pachmarhi**	**1999**	Madhya Pradesh (Satpura Range)	**Yes (2009)**
12	**Khangchendzonga**	**2000**	Sikkim	**Yes (2018)**
13	**Agasthyamalai**	**2001**	Kerala and Tamil Nadu	**Yes (2016)**
14	**Achanakmar-Amarkantak**	**2005**	Madhya Pradesh and Chhattisgarh	**Yes (2012)**
15	**Great Rann of Kutch**	**2008**	Gujarat (*Largest Biosphere Reserve*)	No
16	**Cold Desert**	**2009**	Himachal Pradesh (Pin Valley, Kibber, Chandratal)	No
17	**Seshachalam Hills**	**2010**	Andhra Pradesh (Tirupati)	No
18	**Panna**	**2011**	Madhya Pradesh	**Yes (2020)**

8. International Environmental Conventions & Protocols

YEAR	CONVENTION / AGREEMENT	PLACE / CITY	PRIMARY OBJECTIVE / REGULATORY SCOPE
1971 (2 Feb)	**Ramsar Convention on Wetlands**	Ramsar (Iran)	Conservation and wise use of wetlands; entered into force 1975; **World Wetlands Day (2 Feb)**; India joined **1 Feb 1982**; Keoladeo & Chilika were first sites; **Montreux Record** (Keoladeo NP & Loktak Lake)
1972 (5–16 Jun)	**Stockholm Conference (UNCHE)**	Stockholm (Sweden)	UN Conference on the Human Environment; **Created UNEP (Nairobi, Kenya)**; celebrated as **World Environment Day (5 June)**
1973 (3 Mar)	**CITES (Washington Convention)**	Washington, D.C. (USA)	Convention on International Trade in Endangered Species of Wild Fauna and Flora; Appendix I, II, III; **World Wildlife Day (3 March)**
1979	**Bonn Convention (CMS)**	Bonn (Germany)	Convention on the Conservation of Migratory Species of Wild Animals
1982	**UNCLOS**	Montego Bay (Jamaica)	Law of the Sea: Territorial waters (12 nm), Contiguous zone (24 nm), Exclusive Economic Zone (200 nm)
1985	**Vienna Convention**	Vienna (Austria)	Framework convention for the protection of the stratospheric ozone layer
1987 (16 Sep)	**Montreal Protocol**	Montreal (Canada)	**Legally binding phase-out of Ozone Depleting Substances (ODS / CFCs)**; entered into force 1 Jan 1989; **World Ozone Day (16 Sept)**; **Kigali Amendment (2016)** phase out HFCs
1987	**Brundtland Report**	UN World Commission	Report titled ***Our Common Future***; formally defined **Sustainable Development**
1989	**Basel Convention**	Basel (Switzerland)	Control of transboundary movements of **Hazardous Wastes and their disposal**; entered into force 1992
1992 (3–14 Jun)	**Rio Earth Summit (UNCED)**	Rio de Janeiro (Brazil)	Landmark summit producing 3 Rio Conventions: 1. **UNFCCC** (Climate Change) 2. **CBD** (Biological Diversity) 3. **UNCCD** (Combating Desertification) Also produced: **Agenda 21**, Rio Declaration, and Forest Principles
1997 (11 Dec)	**Kyoto Protocol**	Kyoto (Japan)	UNFCCC COP-3; **Legally binding greenhouse gas emission reduction targets for Annex-I developed countries**; 6 greenhouse gases (CO2, CH4, N2O, HFCs, PFCs, SF6); Clean Development Mechanism (CDM); entered into force 2005; expired 2020
1998	**Rotterdam Convention**	Rotterdam (Netherlands)	Prior Informed Consent (PIC) procedure for certain hazardous chemicals and pesticides in international trade
2000	**Cartagena Protocol on Biosafety**	Montreal / Cartagena	Protocol to CBD governing the safe transfer, handling, and use of **Living Modified Organisms (LMOs)**
2001	**Stockholm Convention on POPs**	Stockholm (Sweden)	Elimination or restriction of **Persistent Organic Pollutants (POPs)** (original "Dirty Dozen" including DDT, Dioxins, PCBs)
2010	**Nagoya Protocol**	Nagoya (Japan)	Supplementary agreement to CBD on **Access to Genetic Resources and the Fair and Equitable Sharing of Benefits (ABS)**
2013	**Minamata Convention**	Kumamoto (Japan)	Global treaty to protect human health and the environment from adverse effects of **Mercury (Hg)**
2015 (12 Dec)	**Paris Agreement**	Paris (France)	UNFCCC COP-21; keep global temperature rise **well below 2°C** above pre-industrial levels, pursue efforts to limit rise to **1.5°C**; Nationally Determined Contributions (NDCs); entered into force **4 Nov 2016**
2015	**Sustainable Development Goals (SDGs)**	UN General Assembly	**17 Goals, 169 Targets (2015–2030)**; SDG 1: No Poverty, SDG 2: Zero Hunger, SDG 3: Good Health, SDG 4: Quality Education, SDG 5: Gender Equality, SDG 6: Clean Water & Sanitation, SDG 13: Climate Action, SDG 14: Life Below Water, SDG 15: Life on Land

YEAR	CONVENTION / AGREEMENT	PLACE / CITY	PRIMARY OBJECTIVE / REGULATORY SCOPE
2022	**Kunming-Montreal GBF**	Montreal (CBD COP-15)	Global Biodiversity Framework; includes landmark **"30x30 target"** (protect at least 30% of planet's land and ocean by 2030)

9. Fundamental Ecological Principles & Laws

A. Lindeman's 10% Energy Law (1942)

- Formulated by **Raymond Lindeman**.
- Only **approximately 10% of the energy** entering any trophic level is transferred and organized into organic matter at the next higher trophic level.
- 90% of energy is lost as metabolic heat, respiration, and catabolism.
- Explains why food chains rarely exceed 4 to 5 trophic levels.

B. Ecological Pyramids (Charles Elton, 1927)

1. **Pyramid of Energy**: - **ALWAYS UPRIGHT, WITHOUT EXCEPTION!** - Energy decreases progressively at each successive trophic level according to thermodynamics.
2. **Pyramid of Biomass**: - **Terrestrial Ecosystems (Grassland, Forest)**: **UPRIGHT** (Producer biomass > Herbivore > Carnivore). - **Aquatic Ecosystems (Pond, Lake, Ocean)**: **INVERTED** (Phytoplankton standing crop biomass is much smaller than zooplankton and predatory fish).
3. **Pyramid of Numbers**: - **Grassland & Cropland**: **UPRIGHT** (Millions of grasses support fewer herbivores, and even fewer carnivores). - **Tree / Parasitic Ecosystem**: **INVERTED** (A single large tree supports hundreds of birds, which host thousands of parasites and hyper-parasites). - **Forest Ecosystem**: **Spindle-shaped** (Few large trees support many herbivores, which are eaten by fewer carnivores).

C. Bioaccumulation vs Biomagnification

- **Bioaccumulation**: Increase in the concentration of a chemical pollutant in an **individual organism over time** (rate of intake > rate of excretion).
- **Biomagnification**: Progressive increase in the concentration of persistent, non-biodegradable, fat-soluble toxins (e.g., **DDT, Mercury, PCBs**) at **successively higher trophic levels** of a food chain.
- *Classic example:* Phytoplankton (0.00005 ppm) → Zooplankton → Small Fish → Fish-eating birds / Falcons (25 ppm causing egg-shell thinning and population collapse).

D. Greenhouse Gases: Global Warming Potential (GWP, 100-Year Horizon)

Ranked from highest warming impact per molecule to lowest (relative to CO2 = 1):

SF6 (23,500) > PFCs (7,000–9,000) > HFCs (1,400–3,000) > N2O (273) > CH4 (28) > CO2 (1)

- **Water Vapour**: Most abundant natural greenhouse gas contributing greatest share to natural greenhouse effect (~60%), but has short atmospheric residence time.
- **Carbon Dioxide (CO2)**: Primary anthropogenic driver of global climate change due to massive volume and century-long residence time.
- **Methane (CH4)**: Major sources: Paddy/rice fields, ruminant livestock enteric fermentation, landfills, wetlands, coal mining.

10. Important Environmental Days (UPPCS Direct Match)

DATE	ENVIRONMENTAL OBSERVANCE	CONNECTED MILESTONE / SIGNIFICANCE
2 February	**World Wetlands Day**	Signing of the Ramsar Convention on Wetlands (1971)
3 March	**World Wildlife Day**	Adoption of CITES (Washington Convention, 1973)
21 March	**International Day of Forests**	Promotes awareness of the importance of all types of forests
22 March	**World Water Day**	Focuses on the importance of freshwater
23 March	**World Meteorological Day**	Establishment of WMO (1950)
22 April	**Earth Day**	First held in 1970; global environmental movement
22 May	**International Day for Biological Diversity**	Adoption of the agreed text of the CBD (1992)
5 June	**World Environment Day**	Opening day of the historic Stockholm Conference (1972)
8 June	**World Oceans Day**	Concept originally proposed at Rio Earth Summit 1992
29 July	**International Tiger Day**	Instituted at the Saint Petersburg Tiger Summit (2010)
12 August	**World Elephant Day**	Dedicated to the preservation and protection of elephants
16 September	**World Ozone Day**	Signing of the Montreal Protocol (1987)
5 October	**National Dolphin Day (India)**	Declaration of Gangetic Dolphin as National Aquatic Animal (2009)
16 October	**World Food Day**	Founding of the FAO (1945)
5 December	**World Soil Day**	Promotes importance of healthy soil and sustainable soil management

11. Foundational Ecological Terms & Coining Scientists / Pioneers

TERM / ECOLOGICAL CONCEPT	COINED BY / PIONEER	YEAR	CORE DEFINITION / EXAM RELEVANCE
Ecology (*Oekologie*)	**Ernst Haeckel**	**1866**	Coined from Greek *oikos* (house/dwelling) + *logos* (study). Study of interactions between living organisms and their physical environment.
Ecosystem	**Arthur Tansley (A.G. Tansley)**	**1935**	The basic functional ecological unit consisting of biotic community together with its abiotic environment interacting as a system.
Biodiversity (Biological Diversity)	**Walter G. Rosen** (coined); popularized by **Edward O. Wilson (E.O. Wilson)**	**1985 / 1988**	Coined as contraction of "biological diversity". E.O. Wilson published landmark book *Biodiversity* (1988); known as "Father of Biodiversity".
Biodiversity Hotspots	**Norman Myers**	**1988**	Biogeographic regions characterized by exceptional levels of plant endemism and high levels of habitat loss (≥70% loss). Adopted by Conservation International.
Ecological Niche	**Joseph Grinnell** (habitat niche); refined by **Charles Elton** & **G.E. Hutchinson**	**1917 / 1927 / 1957**	The specific functional role and physical space occupied by an organism within its ecosystem (*address + profession of species*). Hutchinson introduced "fundamental vs realized niche".
Deep Ecology	**Arne Naess**	**1973**	Radical environmental philosophy rejecting anthropocentrism; advocates that all living beings have intrinsic moral value regardless of their utility to humans.
Ecological Footprint	**William Rees & Mathis Wackernagel**	**1992**	Measure of human demand on Earth's ecosystems; the area of biologically productive land and water required to produce goods consumed and absorb waste.
Gaia Hypothesis	**James Lovelock & Lynn Margulis**	**1972**	Proposes that living organisms interact with their inorganic surroundings on Earth to form a synergistic and self-regulating complex system that helps maintain conditions for life.
Acid Rain	**Robert Angus Smith**	**1872**	Discovered connection between atmospheric pollution (SO2 & NOx from coal burning) and acid precipitation in Manchester, UK. Normal rain pH ~5.6; acid rain pH < 5.6.
Smog (Smoke + Fog)	**H.A. Des Voeux**	**1905**	Coined at the Public Health Congress in London to describe the deadly mixture of coal smoke and dense winter fog (**Classical / London Smog — reducing in nature, contains SO2**).
Photochemical Smog (Los Angeles Smog)	**Arie Haagen-Smit**	**1950s**	Sunlight-induced photochemical reaction between Nitrogen Oxides (NOx) and Volatile Organic Compounds (VOCs). **Oxidizing in nature; major components: Ozone (O3), PAN (Peroxyacetyl nitrate), and Aldehydes**.
Discovery of Antarctic Ozone Hole	**Joe Farman, Brian Gardiner, Jonathan Shanklin**	**May 1985**	British Antarctic Survey scientists discovered drastic springtime thinning of the stratospheric ozone layer over Halley Bay, Antarctica, caused by Chlorofluorocarbons (CFCs).
Biocoenosis	**Karl Möbius**	**1877**	An interacting community of different living species occupying a common biotope/habitat.
Father of Indian Ecology	**Prof. Ramdeo Misra**	1950s–1970s	Established pioneering ecological research and education in India at Banaras Hindu University (BHU), Varanasi.
Ecotone & Edge Effect	**Frederic Clements**	1905	**Ecotone**: Transition zone between two distinct biomes/communities (e.g., mangrove between marine & terrestrial; marshland between aquatic & dry land). **Edge Effect**: Phenomenon where species richness and population density are significantly greater in the ecotone than in either adjoining community (e.g., birds in forest edge).

12. Major Environmental Pollutants, Toxic Heavy Metals & Occupational Diseases

TOXIC POLLUTANT / METAL	SOURCE / ENVIRONMENTAL MEDIUM	CHARACTERISTIC DISEASE / SYNDROME	CLINICAL PATHOLOGY / PRELIMS HIGHLIGHTS
Methylmercury (Hg)	Industrial effluent discharged into coastal waters; biomagnified in fish	**Minamata Disease**	First discovered in **Minamata Bay (Kumamoto, Japan, 1956)**; caused by Chisso Corp chemical plant; severe neurological syndrome, ataxia, loss of motor control, sensory impairment, convulsions, congenital deformities.
Cadmium (Cd)	Mining runoff, zinc smelting, contaminated irrigation water for paddy fields	**Itai-Itai Disease ("Ouch-Ouch Disease")**	First discovered in **Toyama Prefecture (Japan, 1912)**; severe osteomalacia (extreme bone softening and painful skeletal fractures) and kidney failure; *Itai-Itai* means "it hurts, it hurts" in Japanese.
Nitrate (NO_3^-)	Runoff from nitrogenous chemical fertilizers in groundwater drinking water	**Blue Baby Syndrome (Methemoglobinemia)**	Nitrate converts to nitrite in infant gut; binds with infant hemoglobin to form **methemoglobin**, which cannot release oxygen to tissues; skin turns blue due to asphyxiation; fatal if untreated.
Arsenic (As)	Natural geochemical leaching into shallow groundwater aquifers	**Blackfoot Disease, Hyperkeratosis & Arsenicosis**	Severe peripheral vascular gangrene ("black foot"), skin hyper-pigmentation, keratosis, and bladder/skin cancer. Rampant in **Ganga-Brahmaputra alluvial plains: West Bengal, UP, Bihar, and Bangladesh**. Permissible limit: **0.01 mg/L** (WHO).
Fluoride (F^-)	Over-extracted groundwater leaching from fluorapatite minerals	**Dental & Skeletal Fluorosis**	Permissible limit: **1.0 to 1.5 mg/L**. Excess causes mottling of teeth, calcification of ligaments, severe crippling skeletal deformity, and **Knock-Knee Syndrome (Genu Valgum)**. Widespread in Rajasthan, Telangana, UP.
Silica (SiO_2 dust)	Inhalation of crystalline silica dust in stone quarries, agate cutting, sandblasting	**Silicosis**	Incurable occupational fibrotic lung disease causing permanent pulmonary scarring and progressive respiratory failure.
Asbestos Fibers	Asbestos cement manufacturing, building demolition, brake lining manufacture	**Asbestosis & Mesothelioma**	Asbestos fibers penetrate deep into lungs causing pulmonary fibrosis; strongly carcinogenic, inducing **malignant mesothelioma** (cancer of pleural lining).
Coal Dust	Underground coal mining and extraction	**Pneumoconiosis (Black Lung Disease / Anthracosis)**	Chronic inflammation and irreversible progressive massive fibrosis of lungs due to accumulated inhaled coal mine dust.
Cotton Dust	Cotton mills, textile spinning, and ginning factories	**Byssinosis ("Brown Lung Disease" / Monday Fever)**	Occupational asthma and chest tightness occurring on the first workday after a weekend break (*Monday disease*).
Sugarcane Bagasse	Inhalation of dry bagasse fibers during sugarcane processing & paper milling	**Bagassosis**	Hypersensitivity pneumonitis triggered by thermoactinomycetes in moldy bagasse.
Lead (Pb)	Lead-acid batteries, leaded gasoline (historic), paints, smelters, e-waste	**Plumbism / Saturnism**	Neurotoxic; causes severe cognitive and developmental impairment in children, anemia (inhibits heme synthesis), kidney damage, and characteristic **Burtonian blue line along gums**.
E-Waste Hazardous Elements	Discarded computers, mobile phones, printed circuit boards (PCBs)	Multi-organ toxicity	Lead (solder & CRTs), Cadmium (rechargeable Ni-Cd batteries & chip resistors), Mercury (switches & flat screens), Brominated Flame Retardants (BFRs - endocrine disruption).

13. Biodiversity Hotspots of India & Global Criteria

A. Global Criteria for Biodiversity Hotspot (Norman Myers, 1988)

To qualify as an international biodiversity hotspot, a biogeographic region must fulfill **TWO STRICT QUANTITATIVE CRITERIA**: 1. **High Endemism**: It must contain at least **1,500 species of vascular plants (>0.5% of the world's total) as endemics** (species found nowhere else on Earth). 2. **High Degree of Threat**: It must have lost at least **70% of its original primary native vegetation** (i.e., has 30% or less of its original natural habitat remaining). - Globally, there are **36 recognized Biodiversity Hotspots** covering only 2.5% of Earth's land surface, but supporting over 50% of world's endemic plant species and 43% of endemic bird, mammal, reptile, and amphibian species.

B. The 4 Biodiversity Hotspots Covering India

HOTSPOT NAME	GEOGRAPHICAL EXTENT IN INDIA	ADJOINING INTERNATIONAL REGIONS	LANDMARK ENDEMIC SPECIES / KEY FEATURES
1. Himalaya Hotspot	Covers the **entire Indian Himalayan region**: Jammu & Kashmir, Ladakh, Himachal Pradesh, **Uttarakhand**, Sikkim, and Arunachal Pradesh.	Pakistan, Nepal, Bhutan, Tibet / Southern China	Snow Leopard, Himalayan Musk Deer, Himalayan Tahr, Blue Sheep (*Bharal*), Red Panda; vast alpine and temperate rhododendron-oak ecosystems.
2. Indo-Burma Hotspot	Covers all of **North-East India south of the Brahmaputra River** (Assam plains/hills, Meghalaya, Nagaland, Manipur, Mizoram, Tripura) and the **Andaman Islands**.	Myanmar, Thailand, Laos, Vietnam, Cambodia, and Southern China	Hoolock Gibbon (India's only ape), Golden Langur, Stump-tailed Macaque; extraordinary diversity of orchids, ginger species, and freshwater turtles.
3. Western Ghats & Sri Lanka Hotspot	Covers the continuous mountain chain along the western coast across **Gujarat, Maharashtra, Goa, Karnataka, Kerala, and Tamil Nadu**.	Island nation of Sri Lanka	**Lion-Tailed Macaque** (Silent Valley), **Nilgiri Tahr** (Eravikulam), Nilgiri Langur, Malabar Gliding Frog; exceptional concentration of endemic amphibians (~80% endemic), reptiles, and flowering plants.
4. Sundaland Hotspot	Covers the **Nicobar Islands** of India (Great Nicobar, Little Nicobar).	Indonesia (Sumatra, Java, Borneo), Malaysia, Singapore, Brunei, and Southern Thailand	Nicobar Megapode (mound-building bird), Nicobar Tree Shrew, Saltwater Crocodile; rich coastal coral reefs and dense tropical rainforests.

Crucial Exam Trap: Andaman Islands fall under **Indo-Burma Hotspot**, while Nicobar Islands fall under **Sundaland Hotspot**!

Chapter 02: Environment (पर्यावरण) Basics

🎯 Consolidated Must-Score Facts

1. **Ecology** (पारिस्थितिकी) is the **study** of organism–environment links. An **ecosystem** (पारिस्थितिकी तंत्र) is the **functional unit**. A **habitat** (वास स्थान) is the **place**. A **niche** is the **role**.
2. **Liebig's law** says growth is limited by the **scarcest** essential resource. **Shelford's law** says a species survives only inside a **min–max tolerance** range.
3. **IPAT** is Impact = Population (जनसंख्या) × Affluence × Technology. Hardin's **Tragedy of the Commons** dates to **1968**.
4. **Chipko** (चिपको) began in **1973** at **Reni, Uttarakhand (उत्तराखंड)**. **JFM** guidance is **1990**. **FRA** is **2006**.
5. **Article 48A** is the State DPSP (राज्य के नीति निदेशक तत्व) on environment. **Article 51A(g)** is the citizen's Fundamental Duty. **Article 48** is about agriculture and cows — not environment.
6. **Article 21** (अनुच्छेद 21) has been read to include a **healthy environment**. The **42nd (42वां) Amendment (1976)** added 48A and 51A(g).
7. **MoEF** began in **1985** and became **MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) in 2014**. Headquarters is New Delhi.
8. **Stockholm 1972** (स्टॉकहोम सम्मेलन) was the first UN human-environment conference and led to **UNEP** (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) at **Nairobi**.
9. **Ganga Action Plan** (गंगा एक्शन प्लान) began in **1985**. **Namami Gange** (नमामि गंगे) launched in **2014**.
10. **Carrying capacity** is the maximum load an environment can sustain indefinitely. Resource extraction must stay **below** regeneration, and waste must stay **within** absorption capacity — not the reverse.
11. **Ecological footprint** (पारिस्थितिक पदचिह्न) measures the land/water area needed to support consumption, in **global hectares (gha)**.
12. **Pedosphere** (soil) is usually grouped with the lithosphere or with edaphic abiotic factors. It is not counted as a fifth sphere.
13. **Micro-scale** environment covers immediate surroundings such as a room or a pond edge.
14. **Meso-scale** covers a village or city.
15. **Macro-scale** covers regional or global systems.
16. **Total environment** can be natural, social, economic, or cultural. Competitive papers usually default to **natural/physical** environment.
17. **UNEP** (United Nations Environment Programme) was created at Stockholm. Its headquarters is in **Nairobi, Kenya**.
18. **World Environment Day (WED)** is observed every year on **5 June**. The date marks the opening day of the Stockholm Conference in 1972.
19. **MoEF** was established in **1985**. It was renamed **MoEFCC in 2014** when climate change (जलवायु परिवर्तन) was added to its mandate.
20. **Article 48A** is a Directive Principle. The State shall protect the environment and safeguard forests and wildlife. The **42nd Amendment (1976)** added it.
21. **Article 51A(g)** is a Fundamental Duty. Every citizen shall protect and improve the natural environment. The same amendment added it.
22. **Article 21** guarantees the Right to Life. The Supreme Court expanded it to include the **right to a wholesome and healthy environment** in cases such as *M.C. Mehta* and *Subhash Kumar*.
23. **Liebig's Law of Minimum** states that growth is controlled by the **scarcest essential resource**. That resource is the limiting factor.
24. **Shelford's Law of Tolerance** (शेलफोर्ड का सहनशीलता नियम) states that each species survives only within a **minimum–maximum range** for each factor. Beyond that range, stress or death follows even if other factors are abundant.
25. **N_2** makes up about **78%** of air, but most life cannot use it directly. Organisms need **nitrogen fixation** first.
26. **Rhizobium** bacteria live in legume root nodules and fix atmospheric N_2. Pulses, gram, and soybean are common legume examples.
27. **Azotobacter** is a free-living soil bacterium that fixes nitrogen.
28. **Cyanobacteria** (blue-green algae) also fix nitrogen in soil and water.

29. **Humus** is organic matter in soil. It comes from decomposed plant and animal material, so it is **not** an abiotic component.
30. **Dissolved Oxygen (DO)** is a key aquatic abiotic indicator. Fish die in polluted rivers when DO falls too low.
31. **Operational environment** is **not** part of this standard classification.
32. **Internal environment** (आंतरिक पर्यावरण) means conditions inside an organism's body, such as blood chemistry and tissue fluids studied in physiology.
33. **External environment** (बाह्य पर्यावरण) means conditions outside the organism. Ecology usually studies the external environment.
34. **World Wetlands Day** is observed on **2 February** (Ramsar (रामसर) Convention anniversary).
35. **CRZ-IA** covers ecologically sensitive areas such as mangroves, coral reefs, and nesting beaches — **no new development** of the usual commercial kind.

Chapter 03: Ecology (पारिस्थितिकी) & Ecosystem (पारिस्थितिकी तंत्र)

🎯 Consolidated Must-Score Facts

1. **Ecology** is the **study** of organism–environment (पर्यावरण) interactions. **Haeckel (1866)** coined the term. **Ramdeo Misra** is called the father of Indian ecology. **Bionomics** (बायोनॉमिक्स) is synonymous with ecology; it does **not** mean "management of life."
2. **Gause's competitive exclusion** says two species with identical niches cannot stably coexist when resources are limited.
3. **Liebig** = limiting factor; **Shelford** = tolerance range; **Lindeman (1942)** ≈ **10%** energy transfer between trophic levels.
4. **NPP = GPP − plant respiration**. Plants capture only about **1%** of incident sunlight on leaves.
5. **Primary succession** (प्राथमिक अनुक्रमण) starts on a lifeless substrate. **Secondary succession** (द्वितीयक अनुक्रमण) starts where **soil already exists** after disturbance.
6. **Eutrophication** (सुपोषण) is nutrient overload that drives algal bloom and a dissolved-oxygen crash.
7. **Upwelling** brings cold, nutrient-rich water to the surface and raises marine productivity.
8. **Eucalyptus** is often treated as an environmental hazard because it draws heavy groundwater and can lower the water table.
9. **Allen's Rule:** colder climates → smaller appendages (less heat loss). **Bergmann's Rule:** colder climates → larger body size (heat conservation). Do not swap the pair.
10. **Gloger's Rule:** darker pigmentation is more common in warm, humid regions.
11. **Limiting similarity** means coexisting species differ enough in niche use that they do not compete for exactly the same resource at the same time.
12. **Ecology** (from Greek *Oikos* = house + *Logos* = study) is the study of relationships between living organisms and their environment.
13. **Ernst Haeckel** coined the term **Ecology** in **1866**.
14. **Bionomics** (Greek *bio* = life; *nomos* = law) is a comprehensive study of an organism in relation to its environment. It is **synonymous with ecology**.
15. **Ramdeo Misra** is regarded as the father of **ecology in India** (ecology teaching at BHU).
16. **Eugene P. Odum** is called the father of modern **ecosystem ecology** (*Fundamentals of Ecology*, 1953).
17. **Arne Næss** coined **deep ecology** in **1973**.
18. **John W. Bennett** used the concept of **Ecological Transition** in *The Ecological Transition: Cultural Anthropology and Human Adaptation*.
19. **Organ** is anatomy, **not** an ecological organisation level.
20. **Autoecology** studies one species in its environment.
21. **Synecology** studies communities and ecosystems together.
22. **A.G. Tansley** introduced the term **ecosystem** in **1935**.
23. **Energy flow** is **unidirectional** (Sun → producers → consumers → heat loss).
24. **Nutrient cycling** is **cyclic** — matter is reused through decomposers.
25. **Natural ecosystems** self-regulate (forest, lake, ocean).
26. **Anthropogenic / artificial ecosystems** depend on human inputs (cropland, rice field, orchard, garden, aquarium, sewage-treatment wetland, spaceship life-support systems).
27. **Mountain ecosystems** show vegetation change with altitude along the Himalaya (हिमालय) belts.
28. **Tundra** (टुंड्रा) and **taiga** are cold-climate terrestrial types. They are not typical of the Indian mainland in Prelims options.
29. **Wetlands**, **estuaries**, and **mangroves** are **transitional**. They are neither purely terrestrial nor purely open aquatic.
30. **Phytoplankton** grow abundantly in the limnetic zone because open water receives enough light for photosynthesis.
31. **Producers (autotrophs)** convert solar energy into food through photosynthesis or chemosynthesis. Green plants, algae, and cyanobacteria are the main producers.
32. **Consumers (heterotrophs)** feed on other organisms. They include herbivores, carnivores, and omnivores arranged in trophic levels.

33. **Decomposers** break down dead organic matter and return minerals to the environment. Bacteria and fungi are the classic decomposers. Viruses are **not** treated as decomposer organisms in standard food-chain questions.
34. **Producers and decomposers** act as intermediaries between biotic and abiotic components.
35. **Detritivores** feed directly on detritus. Earthworms are a common example.

Chapter 04: Food Chain (खाद्य श्रृंखला) & Energy Flow

🎯 Consolidated Must-Score Facts

1. **Trophic levels** are feeding ranks counted from producers: **T1 producers, T2 herbivores, T3–T5 carnivores**.
2. **Humans** are **primary as well as secondary** consumers (plants and meat).
3. **Biomagnification** raises toxin concentration up the chain. For DDT, the **snake** (tertiary consumer) shows higher concentration than grasshopper or toad.
4. **Omnivores** such as bears, crows, and humans link grazing and detritus pathways. They can occupy **more than one trophic level** at once.
5. **Keystone predators** such as tigers regulate entire webs. Removing them causes trophic cascade across several chains.
6. **T1 (Producers)** are autotrophs such as green plants, algae, and phytoplankton. They fix solar energy into chemical energy.
7. **T2 (Primary consumers)** are herbivores that eat producers directly. Examples include deer, zooplankton, and caterpillars.
8. **T3 (Secondary consumers)** are carnivores that eat primary consumers. Examples include frogs and small fish.
9. **T4 (Tertiary consumers)** eat secondary consumers. Examples include snakes and large fish.
10. **T5 (Quaternary consumers)** are top predators with few or no natural enemies. Tigers and eagles appear here when the chain is long enough.
11. **Omnivores** eat both plants and animals. They span **multiple trophic levels** simultaneously.
12. **Decomposers** break down dead matter from **all levels**. They recycle nutrients but are usually drawn **beside** the main chain, not as the top of an energy pyramid.
13. **Detritivores** such as earthworms fragment dead material and bridge the detritus chain toward decomposers.
14. **Biomagnification** means a non-degradable toxin (classic example **DDT**, also mercury) rises in concentration at successive trophic levels and lodges in fatty tissues. The highest concentration appears in top consumers such as a **snake** in a grasshopper–toad–snake chain.
15. **Energy flow** is the movement of solar energy through living systems via food chains and food webs.
16. **Nutrients** such as carbon and nitrogen **cycle** through decomposers and return to producers. **Energy does not cycle.**
17. **Raymond Lindeman (1942)** showed that roughly **10%** of energy or organic matter passes to the next higher trophic level.
18. **Ecological efficiency** equals energy at the higher level divided by energy at the lower level, multiplied by 100. The NCERT standard is about **10%**.
19. **Charles Elton (1927)** developed the early pyramid concept in trophic studies.
20. **Decomposers** are usually omitted from standard pyramid diagrams. They act on all levels and return nutrients to producers.
21. **Biomagnification** of toxins such as DDT increases concentration up the food chain. Top predators suffer most even as energy decreases upward.

Chapter 05: Biodiversity

🎯 Consolidated Must-Score Facts

1. **Walter G. Rosen (1985)** coined the word *biodiversity*. **E.O. Wilson** popularised it. **Whittaker (1960)** advanced **alpha, beta, and gamma** diversity.
2. **Genetic diversity** is variation **within** a species. **Species diversity** is variety **among** species. **Ecosystem diversity** is variety of habitats and ecosystems.
3. **Alpha** = mean species diversity at a local habitat (वास स्थान). **Beta** = change between ecosystems. **Gamma** = overall regional diversity.
4. **Silent Valley (साइलेंट वैली) (Kerala)** is among India's richest rainforest biodiversity sites (NP **1984**; Silent Valley Project = Kerala). **Valley of Flowers** (फूलों की घाटी) is in **Chamoli (चमोली), Uttarakhand (उत्तराखंड)**. Himalayan richness reflects confluence of biogeographic zones.
5. **In-situ** (स्थान पर) = natural habitat (NP, sanctuary, BR, sacred grove). **Ex-situ** (स्थानांतरित) = outside habitat (zoo, botanical garden (वनस्पति उद्यान), seed bank (बीज बैंक)). Cryo-banks use **liquid nitrogen**. Biosphere reserves (जैवमंडल आरक्षित क्षेत्र) best join biodiversity with traditional human life / cultural diversity.
6. **TRAFFIC** (WWF (डब्ल्यूडब्ल्यूएफ)–IUCN alliance, 1976) monitors wildlife trade (पण्याध्यक्ष) — not a UNEP (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) bureau.
7. **Diclofenac** (डाइक्लोफेनाक) veterinary painkiller caused India's vulture crash. **Dugong** = herbivorous marine mammal (Schedule I); Gulf of Mannar / Palk. **Indian wild ass** = Rann of Kachchh. **Lion-tailed macaque** = Western Ghats (Kerala, Karnataka, Tamil Nadu (तमिलनाडु)).
8. **Sea buckthorn** (Ladakh (लद्दाख) / cold Himalaya) checks erosion and is nutritious — not primarily a biodiesel (बायो-डीजल) timber crop.
9. **TEEB** makes nature's economic values visible for decision-makers. **Wetlands International** is an independent NGO (HQ Netherlands), not a Ramsar intergovernmental organ.
10. **CEBPOL** (Centre for Biodiversity Policy and Law) sits with NBA, **Chennai** (India–Norway).
11. **Gulf of Mannar Biosphere Reserve (जैवमंडल आरक्षित क्षेत्र)** is often cited as India's richest marine biodiversity region (21 islands mosaic).
12. **Biodiversity** is the variety and variability of life at **genetic**, **species**, and **ecosystem** (पारिस्थितिकी तंत्र) levels in a given area — the diverse range of plants and animals in an ecosystem.
13. **Walter G. Rosen (1985)** coined the word *biodiversity* (from "biological diversity") for a National Research Council seminar title. **E.O. Wilson** popularised the term thereafter.
14. **International Day for Biological Diversity** is **22 May** (UN moved it from the older **29 December** CBD entry-into-force date). The UN declared **2011–2020** the **Decade on Biodiversity**.
15. **Genetic diversity** is variation in genes and alleles **within** a species.
16. **Species diversity** depends on **species richness** (how many species) and **evenness** (how evenly individuals are spread).
17. **R.H. Whittaker (1960)** advanced the **alpha–beta–gamma** scale for species diversity.
18. **Alpha diversity** (अल्फा विविधता) is the **mean species diversity in habitats at a local scale**.
19. **Beta diversity** (बीटा विविधता) is the comparison / change of species **between** ecosystems.
20. **Gamma diversity** is overall diversity across different ecosystems in a **region**.
21. **Ecosystem diversity** is the variety of habitats and ecosystems in a region such as forests, wetlands, grasslands, and deserts.
22. **17 megadiverse countries** together hold most of the world's species. **India** is one of them.
23. **Russia is not** on the standard megadiverse list of 17.
24. **Silent Valley (Kerala, Palakkad)** is remnant moist evergreen rainforest of the Western Ghats.
25. **Direct values** include food, fuel, fibre, medicine, and timber taken from nature.
26. **Indirect values** include soil formation, prevention of soil erosion, recycling of waste, pollination of crops, climate regulation, and flood control.
27. **Ethical and aesthetic values** cover cultural, spiritual, and recreational benefits.

28. **Option value** is the potential future use of species and genes not yet exploited.
29. **Threatened species** = **VU + EN + CR**.
30. **Endangered** means **EN only**, not all threatened categories.
31. **DD** (Data Deficient) and **NE** (Not Evaluated) mean insufficient data for a full assessment.
32. **IUCN** (आईयूसीएन) is headquartered in **Gland, Switzerland**. It assesses species status but does **not** regulate international trade (**CITES** (साइट्स) does that).
33. **IPBES** (Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services) is headquartered in **Bonn, Germany**. It gives biodiversity assessment reports similar to IPCC (आईपीसीसी) for climate.
34. **State Biodiversity Boards (SBB)** and **Biodiversity Management Committees (BMC)** operate at state and local levels.
35. **Biodiversity Management Committees (BMCs)** are central to Nagoya-style ABS; they may levy **collection fees** on access to biological resources in their jurisdiction.

Chapter 06: Habitat (वास स्थान), Flora (वनस्पति) & Fauna (प्राणीजात)

🎯 Consolidated Must-Score Facts

1. **Habitat loss** is the number-one biodiversity threat. **Fragmentation** splits patches; **corridors** reconnect them.
2. **MISHTI** is the Mangrove Initiative for Shoreline Habitats & Tangible Income from Budget **2023–24** under MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय). It is **not** a 2019 scheme.
3. **NUHHP** is the National Urban (नगरीय) Housing and Habitat Policy of **2007** (MoHUA).
4. **Flora** means plant life of a region. **Fauna** means animal life. Do not swap the two.
5. **BSI (1890, Kolkata)** surveys plants. **ZSI (1916, Kolkata)** surveys animals. Both sit in Kolkata; kingdoms differ.
6. **Water hyacinth** is a common invasive threat in UP village ponds.
7. **Terrestrial habitats** exist on land. Organisms that live on land are called **terrestrial habitants**.
8. **Aquatic habitats** exist in freshwater or marine water. Rivers are lotic and lakes are lentic aquatic habitats.
9. **Transitional habitats** such as wetlands, mangroves, and estuaries are neither purely terrestrial nor purely aquatic.
10. **Abiotic factors** in a habitat include temperature, moisture, light, soil, pH, and salinity.
11. **Biotic factors** include food availability, predators, competitors, and symbiotic partners.
12. **Habitat conservation** protects the natural dwelling place of species so ecological processes can continue.
13. **Habitat loss** completely removes an area's ability to support native species.
14. **Habitat fragmentation** breaks a large habitat into small isolated patches and blocks movement and gene flow.
15. **Habitat degradation** reduces habitat quality without removing it entirely, such as river pollution affecting dolphin habitat.
16. **In-situ conservation** protects species inside their natural habitat. National parks and wildlife sanctuaries are examples.
17. **Wildlife corridors** link fragmented habitats. The **Terai Arc Landscape** connects Dudhwa with Katarniaghat across the Nepal border.
18. **Habitat restoration** rebuilds damaged ecosystems through afforestation, wetland revival, and mangrove planting.
19. **MISHTI** stands for **Mangrove Initiative for Shoreline Habitats and Tangible Income**.
20. **NUHHP** addresses urban habitat quality through housing and green-space planning in cities.
21. **Flora** is the collective **plant** life of a particular region, habitat, or geological period.
22. **Fauna** is the collective **animal** life of a particular region, habitat, or geological period.
23. **Adolf Engler** mapped the world's major **floral regions**.
24. **C.O. Sauer** introduced the concept of **planting hearth**, the region where domesticated plants first originated.
25. **Red Sanders (*Pterocarpus santalinus*)** is a South India dry-deciduous endemic (Palkonda–Seshachalam / southern Eastern Ghats (पूर्वी घाट)) — **not** a tropical rainforest tree.
26. **Taxus** occurs naturally in the Himalaya, is Red Data–listed, and yields **taxol** (anti-cancer; older stems also cite Parkinson's).

Chapter 07: Protected Areas & Conservation

🎯 Consolidated Must-Score Facts

1. **Ramsar (1971)** marks wetlands of international importance. India joined in **1982**. Uttar Pradesh has **12** Ramsar sites.
2. **Rudrasagar** is in **Tripura**. **Sultanpur** is in **Haryana**. Neither is an Uttar Pradesh Ramsar site.
3. **International Tiger Day = 29 July** (St Petersburg 2010). **M-STrIPES** monitors tiger reserves (patrol + ecology (पारिस्थितिकी) GIS tool).
4. **Sathyamangalam TR (TN)** is a classic ecological link between Eastern and Western Ghats (पश्चिमी घाट).
5. **UP teaching row:** Dudhwa NP / TR, Pilibhit TR, Ranipur TR, **12 Ramsar**, Terai corridors, **no marine PA**, **no biosphere reserve** inside present UP.
6. **Three UNESCO instruments:** World Heritage · Biosphere (MAB / WNBR) · Intangible Cultural Heritage — do not force one site into all three without checking the list.
7. **Nokrek** is in **Meghalaya** (Garo (गारो) Hills; Nokrek peak), not Manipur. **Simlipal** is in **Odisha** (ओडिशा). **Agasthyamalai** spans **Kerala and Tamil Nadu (तमिलनाडु)** (Neyyar, Peppara, Shendurney WLS + Kalakad–Mundanthurai TR).
8. **Asan (आसन) Conservation Reserve** in Uttarakhand is a known example and is also a Ramsar site.
9. **Marine Protected Areas (MPAs)** include **Marine National Parks** and **Marine Wildlife Sanctuaries** notified under WPA.
10. **Gahirmatha Marine Sanctuary** in Odisha is famous for olive ridley turtle nesting.
11. **Uttar Pradesh has no marine protected area** because it is landlocked.
12. **World Wetlands Day** (विश्व आर्द्रभूमि दिवस) is **2 February** (रामसर).
13. **Uttar Pradesh has 12 Ramsar sites**.
14. **Rudrasagar Lake** is in **Tripura**, not UP.
15. **Sultanpur National Park** is in **Haryana**, not UP.
16. **UNESCO World Heritage Sites** are places of outstanding universal value under the **World Heritage Convention (1972)**.
17. **Natural** World Heritage Sites are recognised for exceptional natural beauty or biodiversity.
18. **Keoladeo** is both a Ramsar wetland and a UNESCO natural World Heritage Site (प्राकृतिक विश्व धरोहर स्थल).
19. **Uttar Pradesh has no natural UNESCO World Heritage Site**. Nanda Devi is in **Uttarakhand** (उत्तराखंड).

Chapter 08: Wildlife Conservation (वन्यजीव संरक्षण)

🎯 Consolidated Must-Score Facts

1. **Schedule VI** plants cannot be cultivated without a **licence** from the Chief Wildlife Warden (or authorised officer) — not "banned under all circumstances."
2. **NTCA** became statutory through the **2005/2006** WPA amendments and governs **Tiger Reserves only**.
3. **Project Tiger** (प्रोजेक्ट टाइगर) began in **1973**. Corbett was the first tiger reserve. India has **58** tiger reserves. The **2022** census figure often cited is **3,682** tigers.
4. **Project Elephant** (प्रोजेक्ट एलीफेंट) began in **1992** with about **33** elephant reserves and **no WPA statutory basis** — it covers **wild and captive** elephants.
5. **Project Lion (2020)** focuses on Asiatic lions at **Gir (गीर), Gujarat**. Uttar Pradesh (उत्तर प्रदेश) has **no** wild Asiatic lion population (जनसंख्या).
6. **IRV 2020** planned rhino recovery across seven PAs, including **Dudhwa** (दुधवा) in Uttar Pradesh.
7. **Project Snow Leopard (2009)** and **GSLEP (2013)** cover high Himalaya (हिमालय). UP has **no** snow leopard habitat.
8. **Operation Olivia** is an Indian Coast Guard (तटरक्षक बल) nesting patrol on the **Odisha** (ओडिशा) coast.
9. **Whale shark** is India's largest fish; seasonal aggregations off **Gujarat** (Mar–May); protection programme from **2004**.
10. **Kharai camel** (Kutch, Gujarat) can swim in seawater and graze mangroves; can be domesticated — **not** "wild only."
11. **Pugmark** technique estimates (प्राकलन समिति) wild animal populations from footprints.
12. **Spider monkey** is Central/South American — **not** naturally found in India (Star tortoise, Monitor lizard, Pygmy hog are).
13. **Community Reserve** (सामुदायिक आरक्षित क्षेत्र) (WPA): CWLW authority applies; hunting banned; NTFP collection allowed; **traditional agriculture / jhum not** allowed after notification.
14. **Animal Welfare Board of India** (1962) is under the **Prevention of Cruelty to Animals Act, 1960** — **not** EPA 1986 (पर्यावरण संरक्षण 1986). NTCA is statutory; NGRBA was PM-chaired.
15. **New York Declaration on Forests (2014)** is voluntary/non-binding; India was **not** an original signatory.
16. **UNCCD (1994)** (मरुस्थलीकरण रोकथाम संधि) is the sole legally binding desertification (मरुस्थलीकरण) convention; bottom-up local participation; no exclusive South Asia finance monopoly claim.
17. **Wildlife conservation** means protecting wild plants and animals and the habitats they need to survive.
18. **Ex-situ conservation** (बाह्य-स्थाने संरक्षण) uses zoos, captive breeding, and gene banks as backup when wild populations are critically low.
19. **World Wildlife Day** is observed on **3 March**, marking the adoption of CITES (साइट्स) in 1973.
20. **Wildlife Week** in India is observed from **2 to 8 October**.
21. **Operation Olivia** is an **Indian Coast Guard** initiative to protect **Olive Ridley turtles** along the **Odisha coast**.
22. **Wildlife (Protection) Act, 1972** is India's main law for protecting wild animals, birds, and plants.
23. **Schedule I** gives maximum protection. Hunting and trade (पण्याध्यक्ष) are banned for tiger, elephant, rhino, lion, gharial, snow leopard, and Indian vultures.
24. **Schedule V** lists vermin such as the common crow and rat, which may be hunted.
25. **Schedule VI** lists specified plants whose cultivation is **prohibited except with a licence** from the Chief Wildlife Warden or authorised officer (classic pitcher-plant / specified-plant pattern).
26. **Article 48A** directs the State to protect wildlife. **Article 51A(g)** makes it a citizen's duty to protect wildlife.
27. **Animal Welfare Board of India** was established in **1962** under the **Prevention of Cruelty to Animals Act, 1960** — not under EPA 1986.
28. **Project Tiger** was launched in **April 1973** by PM Indira Gandhi (गांधी) at Jim Corbett (जिम कॉर्बेट) National Park.
29. **International Tiger Day** is **29 July** (decided at the **2010 St Petersburg** Tiger Summit).
30. **Nagarjunsagar–Srisailam** holds the largest critical tiger habitat among common options. **Bandipur** (बांदीपुर) pioneered drone monitoring among common stems.
31. **Sathyamangalam TR (Tamil Nadu (तमिलनाडु))** sits at the Eastern–Western Ghats (पश्चिमी घाट) junction.

32. **Madhya Pradesh** has the most tiger reserves among states.
33. **Project Elephant** was launched in **1992** as a Centrally Sponsored Scheme.
34. **Elephant Reserves** are notified by state governments on MoEFCC recommendation.
35. **National Chambal Sanctuary** (UP–MP–Rajasthan) is India's main gharial stronghold.

Chapter 09: Forests & Forest Management

🎯 Consolidated Must-Score Facts

1. **Madhya Pradesh** has the largest forest **area**. **Mizoram** has the highest **percentage** cover among States. **Arunachal Pradesh** often leads **very dense forest** area.
2. **Forest cover ≠ Recorded Forest Area ≠ tree cover**. Recorded forest area is a legal-notification figure near about **23%**.
3. **Champion and Seth (1968)** classify Indian forests into **16 types in 6 groups**. The largest share is **Tropical Dry Deciduous** (about **38%**).
4. **Tropical wet evergreen** (>**200 cm** rain) holds maximum plant diversity / biomass among common options — Western Ghats (पश्चिमी घाट), NE hills, A&N (not Himachal (हिमाचल) as wet evergreen).
5. **Teak** dominates **tropical moist deciduous** forests; **Madhya Pradesh** has the maximum teak area among common options.
6. **Red Sanders (*Pterocarpus santalinus*)** is a dry-deciduous South India endemic (Palkonda–Seshachalam / southern Eastern Ghats (पूर्वी घाट)) — **not** a tropical rainforest tree.
7. **Taxus** (Himalaya (हिमालय); Red Data) yields **taxol** (anti-cancer; also cited vs Parkinson's in older stems).
8. **Amazon rainforest** is often called the **"lungs of the Earth"** (~**20%** of terrestrial photosynthetic oxygen share in coaching figures).
9. **National Forest Policy 1988** targets about **33%** national forest/tree cover and **66%** in hills. **NFP 1952** classes: Protected/Conserved, National, Village Forests, Tree Lands — **not** "National Park" as a forest class.
10. **Chipko (1973, Reni, Uttarakhand (उत्तराखंड))** — Bahuguna, Bhatt, **Gaura Devi**. The **Godavarman (1996)** line broadened "forest."
11. **Social forestry** grows trees on **non-forest** land. **Agroforestry** integrates trees with crops/livestock (**CAFRI Jhansi (झांसी)**).
12. **JFM (1990)** is State–village co-management of **degraded forest**. **FRA 2006** recognises dweller rights (**IFR ~4 ha**).
13. **FCA** controls **diversion**. **FRA** recognises **rights**. **JFM** is executive co-management — three different tools.
14. **Sonbhadra** (सोनभद्र) has UP's highest district forest cover. Districts above **20%** include **Chandauli** and **Shravasti**.
15. **Urbanisation is a cause of deforestation**, not an "impact/effect" of deforestation (classic trap).
16. **Aluminium** is called a **"green metal"** for recyclability/eco profile — not primarily "to save forests by replacing wood."
17. **Living root bridges** are characteristic of **Meghalaya** (मेघालय).
18. **Andaman (अंडमान) & Nicobar (निकोबार)** uniquely combine mangrove + evergreen + deciduous in one option set.
19. **FRI Dehradun (देहरादून) (1906)**; **ICFRE HQ Dehradun**; **IIFM Bhopal**; Survey of India ≠ environment agency.
20. **Global forest types** include tropical rainforest, tropical deciduous, temperate, boreal, Mediterranean scrub, and mangrove.
21. **Reserved Forest** has the strictest protection. Most acts are prohibited unless permitted.
22. **Protected Forest** is state-regulated forest where some grazing, collection, or customary use may continue under rules.
23. **Unclassed Forest** is forest land not yet formally placed in the Reserved or Protected category.
24. **ISFR forest cover** means any patch of **1 hectare or more** with **10% or more** tree canopy, regardless of ownership.
25. **Very Dense Forest** has canopy density **70% or more**. **Moderately Dense Forest** has **40–70%**. **Open Forest** has **10–40%**.
26. **Forest cover** is not the same as **Recorded Forest Area**. Much forest cover exists outside legally notified forest land.
27. **Tree cover** counts scattered trees in orchards and farms that fall below the 1-hectare forest patch rule.
28. **Mangrove and littoral swamp forests** are distinct coastal types with salt-tolerant species such as sundari in the Sundarbans (सुंदरबन).
29. **Champion and Seth (1968)** is India's standard forest type classification with **16 major types** in **6 groups**.
30. **Tropical Wet Evergreen** forest occurs where rainfall exceeds about **200 cm**, mainly on the Western Ghats, Northeast hills (e.g. Arunachal, Mizoram/Khasi (खासी)–Jaintia (जयंतिया) belt), and Andaman & Nicobar — **not** Himachal (pine/temperate belts).
31. **Tropical Semi-Evergreen** forest is a transition type with rainfall around **1500–2500 mm**.
32. **Tropical Moist Deciduous** forest (about **100–200 cm** rain) drops leaves in the dry season; **teak** is the dominant commercial species. **Madhya Pradesh** has the maximum teak area among common state options. Other species: bamboo, sal, shisham, sandalwood, khair, arjun.

33. **Red Sanders (*Pterocarpus santalinus*)** is endemic to dry deciduous hills of South India (Palkonda–Seshachalam / southern Eastern Ghats) — valued red wood; **not** a tropical rainforest species.
34. **Taxus** grows naturally in the Himalaya, is Red Data–listed, and yields **taxol** (anti-cancer; older stems also cite Parkinson's).
35. **Littoral and Swamp** forest includes mangroves on the Sundarbans and Odisha coast (**Sundari** is the famous Sundarbans tree).

Chapter 10: Pollution & Waste Management

🎯 Consolidated Must-Score Facts

1. **Pollution** is harmful addition beyond the environment (पर्यावरण)'s absorption capacity. **Degradation** is the broader fall in environmental quality and productivity.
2. **Biodegradable** pollutants (sewage, faecal matter, urine, domestic waste, rubber in some stems) break down by microbes. **Non-biodegradable** include pesticides, plastics, mercury, asbestos.
3. **Point sources** are chimneys and drains. **Non-point sources** include farm runoff and roads.
4. **Primary pollutants** are emitted directly (SO_2, PM, CO, NOx, lead). **Secondary pollutants** form in air (O_3, PAN, photochemical smog, acid rain (अम्ल वर्षा)).
5. **CPCB** (केंद्रीय प्रदूषण नियंत्रण बोर्ड) sets ambient air and surface-water standards. **SPCB** enforces in States. **CGWB** handles **groundwater** assessment — not CPCB.
6. **CO** binds haemoglobin **~200–300×** more strongly than O_2 (carboxyhaemoglobin). Lead antiknock in petrol harms IQ/brain development.
7. **Fly ash** from coal TPPs: usable in bricks / Portland cement replacement; contains SiO_2, Al_2O_3, CaO **plus toxic metals** — not "oxides only."
8. **Lichens** are SO_2 bio-indicators and die in polluted air. **Radon** is the classic major **indoor** air pollutant (lung cancer risk).
9. **NCAP (2019)** covers non-attainment cities with a **20–30%** PM reduction target. **BS-VI** fuel arrived in **2020**. **WAYU** air-purifying units were installed in **Delhi**.
10. **CCAC** (2012, with UNEP (संयुक्त राष्ट्र पर्यावरण कार्यक्रम)) focuses on **methane, black carbon, HFCs** — not a G20-only club.
11. **Eutrophication** (सुपोषण) is nutrient overload (N/P) → algal bloom → dissolved-oxygen crash — a **water** phenomenon, not air pollution.
12. **Damodar** (दामोदर) is classically called a **biological desert** from mining/industry pollution. Ganga (गंगा) BOD often highest **Kanpur (कानपुर)–Allahabad (इलाहाबाद)** stretch.
13. **Bioremediation** uses microbes to clean pollution; **cannot** readily/completely treat **Cd/Pb** heavy metals. **Oilzapper** (TERI) is for oily sludge/spills. Genetic engineering can create specialised strains (e.g. *Pseudomonas putida*).
14. **3R** priority is **Reduce > Reuse > Recycle**. **Swachh Bharat** began in **2014**. NAQI launched **17 Oct 2014**.
15. **Euro / BS** emission norms limit CO, HC, NOx, PM. Euro-II ultra-low sulphur diesel ≈ **≤0.05% S**. India skipped BS-V to **BS-VI (1 Apr 2020)**.
16. **Asian Brown Cloud** / atmospheric brown cloud is especially noted over **South Asia** (winter aerosols, black carbon, sulphate).
17. **Radioactive pollution** is ionizing radiation beyond safe dose. **Becquerel (Bq)** counts disintegrations per second; **Sievert (Sv)** is biological dose; **half-life** is time for **50%** decay — not complete disappearance.
18. **AERB** (Atomic Energy Regulatory Board) regulates nuclear safety in India. **Chernobyl (26 April 1986)** is the classic nuclear-accident teaching case; **Bhopal (1984)** is MIC industrial gas, not radioactive pollution.
19. **Thermal pollution** raises water temperature (mainly power-plant cooling water). Warm water holds **less DO**; even a **2–3°C** rise can stress aquatic life. **Cooling towers/ponds** mitigate discharge under Water Act consent conditions.
20. **Environmental pollution** is the harmful addition of substances or energy when the rate exceeds the environment's **absorption or assimilative capacity**.
21. **Biodegradable** pollutants (sewage, faecal matter, urine, domestic organic waste) can be broken down by microbes. **Non-biodegradable** pollutants (pesticides, plastics, many metals, asbestos) persist or degrade very slowly.
22. **Point sources** have identifiable outlets such as factory (फैक्टरी) chimneys and sewage pipes.
23. **Non-point sources** are diffuse, such as fertilizer runoff from farms and urban (नगरीय) stormwater.
24. **Primary pollutants** are emitted directly, including **SO_2**, **CO**, **NOx**, and **particulate matter**.
25. **Secondary pollutants** form in the atmosphere, including **ground-level ozone**, **PAN**, photochemical smog, and **acid rain**.
26. **CPCB** sets national standards. **SPCBs** enforce pollution control at the state level.
27. **Environmental degradation** is the decline in environmental quality and productivity from pollution, resource depletion, and ecosystem (पारिस्थितिकी तंत्र) damage.

28. **Urbanization** degrades the environment through sewage overload, heat islands, and waste accumulation.
29. **Mining** removes vegetation and can cause acid drainage and toxic tailings. **Sonbhadra (सोनभद्र) (UP)** faces mining-related degradation.
30. **Intensive agriculture** degrades soil through pesticide buildup, salinization, and organic matter loss.
31. **EIA under EPA 1986** aims to prevent degradation before major projects start.
32. **Air pollution** is the presence of harmful gases, dust, and smoke at levels that injure health or visibility.
33. **Natural sources** include volcanoes and forest fires. **Man-made sources** include fuel combustion, vehicles, and industry.
34. **PM2.5** particles (≤2.5 μm) penetrate deep into the lungs and bloodstream; rain cleans coarser dust better than ultrafines.
35. **PM10** particles (≤10 μm) mainly affect the upper respiratory tract.

Chapter 11: Climate Change (जलवायु परिवर्तन)

🎯 Consolidated Must-Score Facts

1. **Global warming** is the long-term rise in average surface temperature. It is a subset of broader **climate change**, which also includes rainfall, extremes, and sea-level shifts.
2. **Joseph Fourier (1820s)** postulated the greenhouse-effect idea. Sun sends **shortwave** (UV + visible + IR); Earth re-emits **longwave IR**. Ozone absorbs harmful UV; GHGs trap outgoing IR.
3. **Water vapour** is Earth's most abundant natural GHG (NASA: ~half of the natural greenhouse effect) but mainly a **feedback**, not the primary policy driver.
4. **Not GHGs: argon, nitrogen (N_2), oxygen, hydrogen, propane**. **SO_2 / NOx** do **not** contribute to warming **directly** (indirect / other roles).
5. **Direct** GHGs include CO_2, CH_4, N_2O, CFCs, SF_6, NF_3. **Indirect** agents include **NOx, CO, NMVOCs, SO_2**.
6. **Carbon fertilization** = more plant growth from higher atmospheric CO_2 (not the same as ocean acidification (महासागरीय अम्लीकरण) or warming).
7. **Methane ("marsh gas")** sources: paddies, cattle, coal mines, landfills, termites; natural wetlands dominate natural CH_4 (~**75–76%** of natural). **Rice** is a key anthropogenic source of **both CH_4 and N_2O**.
8. **Methane hydrates** under Arctic tundra (टुंड्रा) / seafloor can release CH_4 if warmed; atmospheric CH_4 oxidises to CO_2 in roughly a decade or two.
9. **Adaptation** = coping with impacts. **Mitigation** = cutting emissions and enhancing sinks.
10. **NAPCC (2008)** has **eight** national missions — **not** nuclear power as a listed mission.
11. **LiFE Mission** launched **June 2022**; idea at **COP26 (2021)**, not COP25.
12. **CDM** (Kyoto Art. 12) lets Annex-I parties fund projects in developing countries for **CERs** (1 CER ≈ 1 tCO_2). **Carbon credit** (कार्बन क्रेडिट) concept originates from Kyoto.
13. **Green Climate Fund** was set up at **Cancun COP-16 (2010)**, not Durban. **Earth Hour** (अर्थ आवर) = **WWF** (डब्ल्यूडब्ल्यूएफ), last Saturday of March (lights off ~8:30–9:30 pm).
14. **GHG Protocol** = **WRI + WBCSD** accounting tool. **Blue carbon** = ocean/coastal carbon (mangroves, seagrass, salt (लवणाध्यक्ष) marsh).
15. **Milankovitch** orbital theory (eccentricity, obliquity, precession) — **solar irradiance** is **not** part of that astronomical set.
16. **Ice cores** are the classic **cryogenic** climate archive (Greenland / Antarctica).
17. **Solar flares** are **not** a primary cause of recent climate change in standard NASA/IPCC framing.
18. **Global warming** is the long-term increase in Earth's average near-surface temperature.
19. **Climate change** is broader and includes shifts in rainfall, monsoon timing, cyclones, and extremes.
20. **IPCC AR6** estimates (प्राकलन समिति) warming of about **1.1°C** above the **1850–1900** pre-industrial baseline.
21. **Human activities** — fossil fuels, deforestation, industry, agriculture — are the **dominant cause** of recent warming.
22. **India** has warmed about **0.7°C** since 1901 according to **IMD** data.
23. **Oceans absorb about 90%** of excess heat, causing thermal expansion and marine stress.
24. **Arctic warming** is about **2–3 times faster** than the global average.
25. **Joseph Fourier (1820s)** postulated the greenhouse-effect concept.
26. **Water vapour** is the most abundant natural GHG (about **half** of the natural greenhouse effect in NASA-style framing) but acts mainly as a **feedback**, not a primary emitted driver like fossil CO_2.
27. **Greenhouse gases** absorb and emit infrared radiation in the atmosphere.
28. **CO_2** is the main anthropogenic driver by concentration and total forcing. IPCC-style forcing share is often ~**76%**; older WMO-bulletin style relative shares used in many Indian papers are **CO_2 ~64%, CH_4 ~19%, N_2O ~6%, CFCs + others ~11%**.
29. **CO_2** has **GWP = 1**. Pre-industrial ~**278–280 ppm (~0.03%)**; ~**315 ppm** around 1950s; now **415–420+ ppm**.
30. **Methane (CH_4)** — also called **marsh gas** — has **GWP around 28–36** (100-yr). Sources: paddies, cattle, coal mines, landfills, termites; natural **wetlands** dominate natural CH_4 (~**75–76%**).

31. **Rice** is a major anthropogenic source of **both methane and nitrous oxide** (anaerobic soils + nitrogen fertiliser).
32. **Nitrous oxide (N_2O)** has **GWP around 265** and links strongly to synthetic fertilisers.
33. **Not greenhouse gases: argon, N_2, O_2, hydrogen, propane**. **Nitrogen oxide (NO_x)** is **not** listed among the classic naturally occurring GHGs in UPPCS-style keys.
34. **SO_2 and oxides of nitrogen** do **not** contribute to global warming **directly** (acid rain (अम्ल वर्षा) / air pollution / indirect chemistry roles).
35. **IPCC direct GHGs** include CO_2, CH_4, N_2O, halocarbons, SF_6, NF_3. **Indirect** agents include **NO_x, CO, NMVOCs, SO_2**.

Chapter 12: Ozone Layer

🎯 Consolidated Must-Score Facts

1. **UV-A (315–400 nm)** reaches the surface; **UV-B (280–315 nm)** is mostly blocked by ozone; **UV-C (100–280 nm)** is fully blocked. Ozone blocks ~**93–99%** of the most damaging UV.
2. **Kigali (2016)** phases down **HFCs**. HFCs are not classic ODS, but they have high global-warming potential.
3. **Ozone (O_3)** is a molecule of three oxygen atoms that absorbs harmful ultraviolet radiation.
4. **UV-A** largely reaches the surface. **UV-B** is mostly blocked by ozone and causes sunburn and skin cancer if exposure rises.
5. **UV-C** is fully absorbed by ozone and oxygen under normal conditions.
6. **Column ozone** is measured in **Dobson Units (DU)**. Global average is about **300 DU**.
7. **Stratospheric ozone** is protective. **Tropospheric ozone** is a harmful air pollutant and greenhouse gas.
8. **Ozone depletion** is the sustained reduction of stratospheric ozone below natural levels.
9. **Ozone-Depleting Substances (ODS)** contain chlorine and/or bromine that destroy ozone in the stratosphere.
10. **CFCs** were used in refrigerants, aerosols, foam blowing, and solvents.
11. **One chlorine atom can destroy up to about 100,000 ozone molecules** before removal.
12. **Halons** used in fire suppressants release **bromine**, which is even more destructive per atom.
13. **Polar Stratospheric Clouds (PSCs)** (ध्रुवीय समतापमंडलीय बादल) form at very low temperatures and enable chlorine activation.
14. **Arctic** ozone loss can occur in cold winters but is **less severe and less regular** than Antarctic depletion.
15. **International Ozone Day** is **16 September**, the date the Montreal Protocol was signed.
16. **India ratified in 1992**. The **Ozone Cell under MoEFCC** coordinates compliance.
17. **Montreal controls ozone**. **Kyoto controls climate**. Do not swap them.

Chapter 13: Acid Rain (अम्ल वर्षा)

🎯 Consolidated Must-Score Facts

1. **Acid rain** means wet or dry acidic deposition with pH below **5.6**.
2. **Wet deposition** is acids in rain, snow, or fog. **Dry deposition** is acidic gases and particles settling without rain.
3. **SO_2** oxidises to **H_2SO_4** (sulphuric acid), mainly from coal TPPs, smelters, and refineries.
4. **NOx** oxidises to **HNO_3** (nitric acid), mainly from vehicles and high-temperature combustion.
5. **Lichens** die in high SO_2 and act as pollution bio-indicators.
6. **Mathura (मथुरा) refinery** is a classic SO_2 source in the Taj Trapezium narrative.
7. **Acid rain** is precipitation or atmospheric deposition with **pH below 5.6**, mainly from **sulphuric acid (H_2SO_4)** and **nitric acid (HNO_3)** (नाइट्रिक अम्ल).
8. **Normal unpolluted rain** has pH about **5.6**, not 7, because atmospheric **CO_2** forms weak **carbonic acid (H_2CO_3)**.
9. **Wet deposition** carries acids in rain, snow, fog, or dew.
10. **Dry deposition** means **SO_2**, **NO_2**, and sulphate/nitrate particles settle on soil, leaves, and buildings **without rainfall**.
11. **SO_2** and **NOx** are **primary pollutants**. Strong acids form in the atmosphere, so acid rain is a **secondary pollutant** phenomenon.
12. **Sulphur dioxide (SO_2)** is the main source of **sulphuric acid rain**.
13. **Nitrogen oxides (NOx)** form at **high combustion temperatures** when nitrogen in air reacts with oxygen.
14. **CO_2** dissolves as weak carbonic acid and sets the **pH 5.6 baseline**. It is **not** the main driver of severe acid rain.
15. **Coal thermal power plants** are India's largest **SO_2** point sources.
16. **Vehicles** are a major **NOx** source in cities.
17. **SO_2 is heavier than air** and tends to accumulate near ground-level emission sources.
18. **Aquatic ecosystems** fail when lake/river pH drops **below about 5.0** — fish eggs do not hatch and adult fish die.
19. **Aluminium (Al^{3+})** leached from acidified soil is **toxic to fish gills** and plant roots.
20. **Soil acidification** leaches essential **calcium (Ca^{2+})** and **magnesium (Mg^{2+})** and releases toxic **Al^{3+}**.
21. **Coniferous forests** (pine, spruce, fir) are **most sensitive** to acid rain and acid fog.
22. **Crops** suffer from direct leaf damage and nutrient loss in soil.
23. **Marble and limestone ($CaCO_3$)** react with sulphuric acid to form **gypsum ($CaSO_4$)**, causing yellow crust and pitting.
24. **Metal structures** rust faster under acid deposition.
25. **SO_2 and NOx** cause **respiratory illness** — asthma, bronchitis, and lung irritation from acid aerosols and $PM_{2.5}$.
26. **Lichens** are **bio-indicators of clean air**. They **die in high SO_2** and do **not** thrive in polluted areas.
27. **Flue Gas Desulphurisation (FGD) scrubbers** pass chimney gas through **lime or limestone slurry** to absorb **SO_2** and can remove **90%+** of it.
28. **Coal washing** and **low-sulphur coal** cut sulphur before combustion.
29. **Catalytic converters** in vehicles convert **NO to N_2** and reduce **NOx** that forms nitric acid.
30. **BS-VI norms (2020)** brought **10 ppm sulphur** in petrol and diesel, cutting both SO_2 and NOx precursors.
31. **CNG, LPG, and natural gas** produce **negligible SO_2** and lower NOx than coal.
32. **CPCB (केंद्रीय प्रदूषण नियंत्रण बोर्ड) emission standards** under the Air Act cap SO_2 and NOx from TPPs, cement, steel, and refineries.
33. **Tall chimneys alone are not prevention** — they only disperse pollution farther.
34. **Montreal Protocol controls ozone**, not acid rain — do not confuse treaties.
35. **NCAP (2019)** targets PM reduction in 122 cities with co-benefits for SO_2 and NOx cuts.

Chapter 14: Desertification (मरुस्थलीकरण) & Land Degradation (भू-क्षरण)

🎯 Consolidated Must-Score Facts

1. **Desertification** is land degradation in **arid, semi-arid, and dry sub-humid (ASDM)** areas from climate variation plus human pressure.
2. **Land degradation** is broader: decline in productive capacity on any land through erosion, salinization, waterlogging, mining, or pollution.
3. **Normal geological erosion** is slow and roughly balanced by soil formation. **Accelerated erosion** is human-driven and faster than formation — the focus.
4. **UNCCD** was adopted (अंगीकृत) in **1994**. Desertification and Drought Day is **17 June**.
5. **LDN** (Land Degradation Neutrality) under **SDG 15.3** targets no net loss of healthy land by **2030**.
6. **DPAP** launched in **1973**. **DDP** is **1977–78**. **IWMP** is **2009**. **MGNREGA (2005)** funds bunds and check dams.
7. **Sonbhadra** (सोनभद्र) is a UP mining-related land-degradation hotspot.
8. **Desertification** is **land degradation in arid, semi-arid, and dry sub-humid (ASDM) areas** caused by climate variation and human activities.
9. **Overgrazing** is the **main human cause** in drylands. Livestock remove vegetation, compact soil, and expose topsoil to wind and water erosion.
10. **Deforestation**, **fuelwood collection**, and **unsustainable cropping** remove root binding and reduce local moisture retention.
11. **Improper irrigation** can cause **salinization** and **waterlogging**, turning fertile irrigated land barren — common in parts of Rajasthan canal commands and the Indo-Gangetic fringe.
12. **Drought and climate change (जलवायु परिवर्तन)** kill vegetation and trigger flash erosion on bare soil, creating a feedback loop of further degradation.
13. **Wind erosion** dominates true arid zones like the **Thar Desert**, where dunes migrate and bury fields.
14. **17 June** is observed as **World Desertification and Drought Day**.
15. **SDG 15.3** targets **Land Degradation Neutrality (LDN) (भूमि क्षरण तटस्थता) by 2030** — avoid new degradation and restore degraded land.
16. **Land degradation** is the **broader decline in a land's productive capacity** from natural or human factors.
17. **Desertification** is one **subset** limited to ASDM drylands. **Soil erosion** is one **major process** within degradation.
18. **Human causes dominate** India's degradation statistics: overgrazing, deforestation, unsustainable agriculture, excessive fertilizers, mining, industrial pollution, and urban (नगरीय) sprawl.
19. **Salinization** from excessive canal irrigation affects **Punjab, Haryana, and western UP**.
20. **Mining degradation** is severe in **Sonbhadra (UP), Jharkhand, and Odisha**.
21. **Salinization and waterlogging** can often be reversed through drainage, gypsum, and drip irrigation. **Deep ravines** are hardest to restore.
22. **Afforestation prevents degradation** — it is **not** a cause. Pick it when asked "NOT a cause."
23. **Topsoil (A-horizon)** is the fertile organic-rich layer, usually **15–30 cm deep**. It can take **centuries to form** but may be lost in **one heavy monsoon** on bare slopes.
24. **Normal (geological) erosion** is slow and balanced by soil formation. **Accelerated erosion** from human activity is **10–100× faster** than formation — this is the usual default meaning of "soil erosion."
25. **Water erosion** dominates humid and monsoon India. Raindrop impact dislodges soil and runoff carries it downslope.
26. **Water erosion sequence:Sheet erosion** (thin uniform layer) → **rill erosion** (क्षुद्र सरिता (रिल) अपरदन) (small channels) → **gully erosion** (deep cuts) → **ravines** (extensive badland networks).
27. **Chambal ravines** (चंबल) cover about **3 million hectares** across **MP (Morena, Bhind), UP (Etawah, Agra), and Rajasthan (Dholpur, Karauli)**. They are India's classic **gully/ravine erosion** landscape.
28. **Wind erosion** dominates arid zones through **deflation** (lifting fine particles) and **abrasion** (sandblasting surfaces). It affects the **Thar**, Rann of Kutch, and western Rajasthan-UP margins.

29. **Gravity erosion** (landslides, slumping) matters in **Himalayan and NE hill states**, especially after deforestation and road cutting.
30. **Vegetation cover** is the best natural erosion shield. Deforestation and overgrazing are the top acceleration factors.
31. **Conservation measures:contour bunding/ploughing**, **terrace farming**, **shelter belts/wind breaks**, **afforestation**, **gully plugging**, **check dams**, **strip cropping**, and **cover crops**.
32. **Earthworms** aerate soil, improve structure, and boost fertility — healthy soil resists erosion.

Chapter 15: Environmental Impact Assessment

🎯 Consolidated Must-Score Facts

1. **EIA** studies a project's likely environmental effects **before** approval. The aim is anticipatory prevention, not reactive cleanup.
2. **Environmental Clearance** is mandatory prior clearance for listed projects.
3. **Category A** is appraised centrally by **MoEFCC** (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) through the **EAC**.
4. **Category B** is appraised by **SEIAA** through **SEAC**. **B1** needs full EIA (usually with public hearing); **B2** is simplified.
5. **Scoping** fixes the **Terms of Reference (ToR)** (संदर्भ की शर्तें) for the EIA study.
6. **Rapid EIA** uses one season; **comprehensive EIA** covers all seasons.
7. **NGT (Act 2010)** hears EC and pollution appeals. **PARIVESH** is MoEFCC's online EC portal.
8. **SEA** assesses policies, plans, and programmes upstream of projects. India has project-level EIA law but **no standalone SEA statute**.
9. **Environmental Impact Assessment (EIA)** is a systematic study of a proposed project's likely environmental, social, and health effects **before** major decisions are taken.
10. **Environmental Clearance (EC)** is mandatory for **listed projects** in the Schedule before construction or land preparation.
11. **Category A** projects are large, high-impact projects appraised centrally by **MoEFCC** through the **Expert Appraisal Committee (EAC)**.
12. **Category B** projects are appraised by the **State Environment Impact Assessment Authority (SEIAA)** through the **State Expert Appraisal Committee (SEAC)**.
13. **Screening** applies to **Category B only** and splits projects into **B1** (full EIA required) or **B2** (no full EIA report).
14. **Rapid EIA** uses one season of baseline data. **Comprehensive EIA** covers all seasons and is more reliable for monsoon-dry variation.
15. **CPCB and SPCBs** set pollution standards and monitor compliance. They **do not grant EC**.
16. **PARIVESH** is MoEFCC's online single-window portal for EC applications and tracking.
17. **NGT (National Green Tribunal)**, under the **NGT Act 2010**, hears appeals against EC decisions.
18. **Strategic Environmental Assessment (SEA)** evaluates the environmental consequences of **policies, plans, and programmes (PPP)** at the earliest planning stage.
19. **SEA sits upstream of EIA**. It asks whether a development path is wise **before** individual projects are designed.
20. **EIA** can only accept, modify, or reject **one pre-selected project**. It cannot question whether an entire sector plan — such as many dams in one basin — is environmentally sound.
21. **SEA assesses cumulative impacts** when multiple projects together may overwhelm an ecosystem (पारिस्थितिकी तंत्र) even if each passes individual EIA.
22. **Public participation** in SEA happens when policy choices are still flexible — earlier and cheaper than late-stage project hearings.
23. **Rio Declaration (1992) Principle 17** calls for **EIA** for proposed activities with significant adverse impact. It does **not** mandate SEA by name.
24. **Espoo Convention (1991)** covers **transboundary EIA**. The **SEA Protocol (2003)** signed at **Kiev** extends assessment to plans and programmes.
25. **India has robust project-level EIA** under the 2006 Notification but **no standalone SEA legislation** as of current law.

Chapter 16: Sustainable Development & Environmental Governance

🎯 Consolidated Must-Score Facts

1. **Sustainable development** (Brundtland **1987**, *Our Common Future*) means meeting present needs without compromising future generations' ability to meet theirs.
2. **Natural capital** means renewable and non-renewable environmental resources (air, water, land, minerals, forests). **Roads** are not natural capital.
3. **SDGs** were adopted (अंगीकृत) in **September 2015**, run to **2030**, and include **17 goals** and **169 targets**.
4. **Agenda 21** is Rio **1992**, not 1995.
5. **LiFE Mission** launched in **June 2022**; the idea was floated at **COP26 Glasgow 2021**, not COP25.
6. **Green economy** (UNEP (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) **2011** line) means low-carbon, resource-efficient, socially inclusive growth.
7. **Circular economy** redesigns take-make-dispose into reduce-reuse-recycle loops, often with **EPR** rules. Circular sits inside, but is not identical to, green economy.
8. **Sustainable agriculture** means using land so its **quality remains intact**, with mixed cropping, organic manures, nitrogen-fixing plants, and pest-resistant varieties as the environment-friendly strategy set.
9. **NMSA** (National Mission for Sustainable Agriculture) is a **NAPCC** mission on productivity, water-use efficiency, nutrient management, and livelihood diversification.
10. **Sustainable development** means meeting the needs of the present **without compromising** the ability of future generations to meet their own needs.
11. **Inter-generational equity** protects future generations. **Intra-generational equity** ensures fair sharing among people alive today.
12. **Objectives** include fulfilling **basic needs**, respecting **ecological limits** and carrying capacity, integrating environment into economic planning, and ensuring **public participation**.
13. **Natural capital** includes all renewable and non-renewable environmental resources such as air, water, land, minerals, and forests. **Roads** and other purely human-made assets are **not** natural capital.
14. **Indicators** include **HDI** (UNDP), **ecological footprint** (पारिस्थितिक पदचिह्न), **SDG Index**, **Environmental Performance Index (EPI)**, and **231 SDG indicators**.
15. **NITI Aayog** (नीति आयोग) publishes the **SDG India Index** ranking states on SDG progress.
16. **SDGs replaced MDGs**, which had **8 goals (2000–2015)** and focused mainly on developing countries.
17. **SDGs are universal** — they apply to all countries, including developed ones.
18. **Leave No One Behind** requires progress to reach the poorest and most vulnerable groups.
19. **NITI Aayog** coordinates SDG localisation in India. **MoSPI** maintains the National Indicator Framework.
20. **NITI Aayog's SDG India Index** ranks states and UTs. **Kerala** has repeatedly held the **first** position (including the 2019–20 and 2020–21 editions).
21. **Environmental governance** is the system of rules, institutions, and processes through which environmental decisions are made and enforced.
22. **MoEFCC** (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) is India's nodal ministry for environment, forest, and climate policy.
23. **Rio principles** include **Precautionary principle**, **Polluter Pays**, **Public Trust doctrine**, and **inter-generational equity** — adopted in Indian judgments like *Vellore Citizens' Welfare Forum*.
24. **NGT (2010)** is the specialised tribunal for environmental cases and EC appeals.
25. **Agenda 21 (1992)** is a Rio blueprint for sustainable development planning, including local-level action.
26. **Environmental justice** means fair treatment of all in environmental benefits and burdens — both **distributive** (who gets pollution) and **procedural** (who participates in decisions).
27. **Environmental ethics** asks what moral duties humans owe to nature.
28. **Anthropocentrism** values nature mainly for human use.
29. **Biocentrism** gives intrinsic worth to all living beings.

30. **Ecocentrism** values whole ecosystems.
31. **Aldo Leopold's Land Ethic (1949)** stresses preserving the integrity of the biotic community.
32. **Gandhi's trusteeship** holds that resources are held in trust for society and future generations.
33. **Extended Producer Responsibility (EPR)** (विस्तारित उत्पादक दायित्व) under Plastic, E-Waste, Battery, and Tyre rules makes producers responsible for end-of-life collection.
34. **Circular economy** focuses on **material flows and waste elimination**.
35. **Green economy** is **broader** — it also covers renewable energy (नवीकरणीय ऊर्जा), biodiversity, jobs, and equity. Circular economy is a **component** of green economy.

Chapter 17: Environmental Organizations (India)

🎯 Consolidated Must-Score Facts

1. **MoEFCC** (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) is India's nodal ministry for environment (पर्यावरण), forests, wildlife, and climate change (जलवायु परिवर्तन). It began as MoEF in **1985** and was renamed MoEFCC in **2014**. Headquarters is **New Delhi**.
2. **CPCB** (केंद्रीय प्रदूषण नियंत्रण बोर्ड) was created under the **Water Act, 1974**, sits in **New Delhi**, sets **NAAQS**, and coordinates State Pollution Control Boards. The Air Act **1981** widened its air-pollution role.
3. **UPPCB** is Uttar Pradesh (उत्तर प्रदेश)'s board with headquarters at **Lucknow** (लखनऊ). It enforces norms on Kanpur (कानपुर) tanneries, Noida–Ghaziabad industry, and Ganga (गंगा)–Yamuna (यमुना) discharges.
4. **NGT** was created by the **NGT Act, 2010** (operational **2011**). Headquarters is New Delhi, with regional benches at Bhopal, Pune (पुणे), Kolkata, and Chennai.
5. **WII** (**1982**, Dehradun (देहरादून)) does wildlife research and trains forest officers. It helped frame India's **ten** biogeographic zones and tiger-census methods. It is **not** the plant or animal survey body.
6. **BSI** (**1890**, Kolkata) surveys **plants** and publishes *Flora (वनस्पति) of India* and the plant Red Data Book (रेड डेटा बुक). **ZSI** (**1916**, Kolkata) surveys **animals** and publishes *Fauna (प्राणीजात) of India* and the animal Red Data Book.
7. **FSI** (**1981**, Dehradun) maps forest cover (वन आवरण) and publishes the **India State of Forest Report** every **two years**. ISFR is **not** an annual CPCB product.
8. **NEERI** (**1958**, **Nagpur**) began as **CPHERI** and is a CSIR laboratory for environmental science and engineering under the Ministry of Science and Technology.
9. **CAZRI = Jodhpur**; **IIFM = Bhopal**; operational Antarctica (भारती) stations = **Maitri + Bharati** (मैत्री).
10. **INCOIS = Hyderabad (हैदराबाद)**; ENVIS population (जनसंख्या)–environment centre = **IIPS Mumbai**.
11. **MoEFCC** is India's **nodal ministry** for environment, forests, wildlife, and climate change policy.
12. **Headquarters** is **New Delhi**.
13. **CPCB** was established in **1974** under the **Water (Prevention and Control of Pollution) Act, 1974**.
14. **UPPCB** is Uttar Pradesh's board with HQ at **Lucknow**.
15. **NGT** was established under the **National Green Tribunal Act, 2010** and became operational in **2011**.
16. **Headquarters** is **New Delhi** with regional benches at **Bhopal, Pune, Kolkata, and Chennai**.
17. **WII** was established in **1982** at **Dehradun** under **MoEFCC**.
18. **BSI** was established in **1890** and is headquartered at **Kolkata**.
19. **ZSI** was established in **1916** and is headquartered at **Kolkata**.
20. **FSI** was established in **1981** at **Dehradun** under **MoEFCC**.
21. **NEERI** is India's pioneering CSIR laboratory for **environmental science and engineering**.
22. **Wildlife Institute of India** — **Dehradun**. **Forest Research Institute** — **Dehradun**. **Indian Institute of Forest Management** — **Bhopal** (not Kolkata).
23. **Central Arid Zone Research Institute (CAZRI)** — **Jodhpur** (ICAR; arid-zone research).
24. **Indian Grassland and Fodder Research Institute** — **Jhansi** (झांसी). **IARI** — **New Delhi**.
25. **National Institute of Nutrition** — **Hyderabad**. **National Institute of Ayurveda** — **Jaipur**. **National Institute of Naturopathy** — **Pune**.
26. **Indian Institute of Ecology (पारिस्थितिकी) and Environment** — **New Delhi** (founded **5 June 1980** teaching).
27. **ENVIS Centre on Population and Environment** — **IIPS Mumbai**.
28. **Indian National Centre for Ocean Information Services (INCOIS)** — **Hyderabad** (1999; Ministry of Earth Sciences).
29. **G.B. Pant Institute of Himalayan Environment and Development** — **Almora** (अल्मोड़ा). **Centre for Ecological Sciences** — **Bengaluru**.
30. **Kailash Sacred Landscape Conservation and Development Initiative (KSLCDI)** partners: **India, China, Nepal**.

Chapter 18: Environmental Laws & Policies

🎯 Consolidated Must-Score Facts

1. **Article 48A** (DPSP (राज्य के नीति निदेशक तत्व)) directs the State to protect the environment (पर्यावरण). **Article 51A(g)** is the citizen's Fundamental Duty to protect nature. Do not swap them.
2. **Article 21** (अनुच्छेद 21) has been read to include a healthy environment. **Article 48** is about agriculture and cattle — it is **not** the environment article (**48A** is).
3. **FCA** controls **diversion** of forest land. **FRA** recognises **rights** of forest dwellers. Do not treat them as one Act.
4. **Environmental law** in India combines **constitutional duties**, **statutory Acts**, and **Supreme Court/NGT jurisprudence**.
5. **Article 48A** (DPSP) directs the **State** to protect and improve the environment and safeguard forests and wildlife.
6. **Article 51A(g)** (Fundamental Duty) requires every **citizen** to protect and improve the natural environment.
7. **Article 21** (Right to Life) was expanded by the Supreme Court to include the **right to a wholesome environment** (*M.C. Mehta*, *Subhash Kumar*).
8. **Article 48** deals with **agriculture and animal husbandry** — it is **not** the environment article. Do not confuse with **48A**.
9. **Section 6** empowers the Centre to make **rules** for environmental protection.
10. **Environmental Clearance** under EIA is distinct from SPCB **consent** under Water/Air Acts. Large projects may need **both**.
11. **CGWA** (regulatory authority under EPA) is **not** the same as **CGWB** (assessment board under Jal Shakti (शक्ति)).
12. **MoEFCC** (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) is the nodal ministry for EPA implementation.
13. **Six Schedules** classify species by protection level.
14. **Hunting** is banned for Schedule **I–IV** species with narrow exceptions for research or public safety with permission.
15. **National Park (Section 35)** has the **strictest** protection — no grazing or exploitation.
16. **Wildlife Sanctuary (Section 18)** allows more regulated human use under the Chief Wildlife Warden.
17. **Tiger Reserves** are notified under **Section 38V** (2006 amendment) with **NTCA** oversight.
18. **Prior Central Government approval** is mandatory before using **forest land** for **non-forest purposes** like mining, dams, or industry.
19. **Community Forest Resource (CFR)** (सामुदायिक वन संसाधन अधिकार) rights allow communities to manage and protect forest resources.
20. **State Biodiversity Boards (SBB)** operate at state level.
21. **Biodiversity Management Committees (BMC)** function at local panchayat level.
22. **People's Biodiversity Registers (PBR)** document local biodiversity.

Chapter 19: International Environmental Agreements & Conferences

🎯 Consolidated Must-Score Facts

1. **Stockholm 1972** (स्टॉकहोम सम्मेलन) was the first UN Conference on the Human Environment (पर्यावरण). It led to **UNEP** (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) (Nairobi). World Environment Day (विश्व पर्यावरण दिवस) is **5 June**. The slogan fact is "Only One Earth."
2. **Rio 1992** (UNCED) produced the Rio Declaration, **Agenda 21**, Forest Principles, and opened **UNFCCC** (यूएनएफसीसीसी) and **CBD** for signature. **CITES** is **not** a Rio product — CITES is **1973**.
3. **Agenda 21** is Rio's non-binding sustainable-development action plan for the twenty-first century. It was **not** signed in 1995.
4. **CMS (Bonn)** covers migratory species. **Basel** covers hazardous **waste**. **Rotterdam** covers hazardous **chemicals** (PIC). **Stockholm POPs** covers persistent organic pollutants. **Minamata** covers **mercury**.
5. **CBD** has three pillars: conserve biodiversity, sustainable use, and fair benefit-sharing. **Nagoya Protocol (2010)** details access and benefit-sharing.
6. **Ramsar** (रामसर) (**1971**) is the wetlands treaty under the Wise Use idea. World Wetlands Day (विश्व आर्द्रभूमि दिवस) is **2 February** (रामसर).
7. **CITES** (**1973**, Washington) regulates trade through Appendices **I–III**. IUCN (आईयूसीएन)'s Red List assesses extinction risk; CITES regulates trade — different jobs.
8. **GEF** was established in **1991** on the eve of Rio as a finance mechanism for major environmental conventions.
9. **UNCCD (1994)** (मरुस्थलीकरण रोकथाम संधि) is the sole legally binding convention linking environment, development, and sustainable land management; it stresses bottom-up local participation against desertification (मरुस्थलीकरण).
10. **World Environment Day** is celebrated on **5 June** every year, marking the Stockholm opening day.
11. **Agenda 21** is a non-binding action plan for **sustainable development in the 21st century** — adopted at **Rio 1992**, not 1995.
12. **Local Agenda 21** encourages cities and local bodies to prepare their own SD plans.
13. **Earth Summit +5** was the UN General Assembly special session (पंगु सत्र) held in **New York in June 1997** to review Rio progress.
14. **Johannesburg Summit (2002)** was Rio+10 on sustainable development and reaffirmed Agenda 21.
15. **Rio+20 (2012)** — the United Nations Conference on Sustainable Development — reviewed SD progress and produced the outcome document ***The Future We Want***.
16. **PAGE (Partnership for Action on Green Economy (हरित अर्थव्यवस्था))** was created as the UN response to Rio+20's call for green-economy transitions.
17. **SDGs were adopted in 2015** at the UN General Assembly.
18. **Earth Hour** (अर्थ आवर) is a **WWF** (डब्ल्यूडब्ल्यूएफ) climate-awareness event (usually last Saturday of March). **GHG Protocol** = **WRI + WBCSD**.
19. **International Ozone Day** is **16 September** (Montreal signing date).
20. **TRAFFIC** (WWF–IUCN alliance) monitors wildlife trade for sustainability — it is **not** a UNEP bureau.
21. **BirdLife International** identifies Important Bird and Biodiversity Areas; biodiversity **hotspots** were framed by **Norman Myers**, not BirdLife.
22. **CITES (1973, Washington)** regulates **international trade** in endangered wild fauna (प्राणीजात) and flora (वनस्पति).
23. **Appendix I** — trade banned except in exceptional cases.
24. **Appendix II** — regulated trade with permits.
25. **Appendix III** — species listed by a member country requesting cooperation.
26. **World Wetlands Day** is **2 February**.

Chapter 20: Climate & Environmental Institutions

🎯 Consolidated Must-Score Facts

1. **IPCC** sits in **Geneva**, issues Assessment Reports, and synthesises climate science. It does **not** set NDCs and is **not** the UNFCCC treaty body.
2. **UNEP** (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) headquarters is **Nairobi**. Flagship products include the **Emissions Gap Report** (उत्सर्जन अंतराल रिपोर्ट) and GEO. UNEP was created after Stockholm **1972**.
3. **UNDP** headquarters is **New York**. It publishes the **Human Development Report** (from **1990**) and the **HDI**. HDR is UNDP, not UNEP.
4. **IUCN** (आईयूसीएन) headquarters is **Gland**. It publishes the **Red List** of extinction risk. IUCN assesses risk; it does **not** regulate wildlife trade (पण्याध्यक्ष) — that is **CITES** (साइट्स).
5. **WWF** is a conservation NGO also linked to Gland, known for the panda logo and the **Living Planet Report / Living Planet Index**.
6. **FAO** headquarters is **Rome**. Environment-facing outputs include the **State of the World's Forests (SOFO)** and agriculture–forest links. FAO partners UNEP on some chemical-convention work such as Rotterdam.
7. **Headquarters** is in **Geneva, Switzerland**.
8. **Working Group I** covers physical science. **WG II** covers impacts and adaptation. **WG III** covers mitigation.
9. **UNFCCC** (यूएनएफसीसीसी) is the **treaty body** where countries negotiate **NDCs** at **COP** meetings. IPCC informs but does not decide policy.
10. **UNEP** was created by the **Stockholm Conference (1972)** (स्टॉकहोम सम्मेलन) as the UN's principal environmental authority.
11. **Headquarters** is in **Nairobi, Kenya** — the only major UN environment HQ in the Global South.
12. **GEO (Global Environment Outlook)** is UNEP's comprehensive state-of-environment assessment.
13. **UNDP** is the UN's global **development** network working on poverty, governance, and SDG implementation.
14. **Headquarters** is in **New York**.
15. **HDI** combines **life expectancy**, **education**, and **GNI per capita (PPP)**.
16. **IUCN** is a global conservation union of states and NGOs, founded in **1948**.
17. **Headquarters** is in **Gland, Switzerland**.
18. **Threatened species** (संकटग्रस्त प्रजातियां) means **Vulnerable + Endangered + Critically Endangered (VU + EN + CR)**.
19. **WWF (World Wide Fund for Nature)** is a global conservation **NGO**, not a UN agency.
20. **Headquarters** is also in **Gland, Switzerland** — same city as IUCN but **different organisation and role**.
21. **FAO (Food and Agriculture Organization)** is a UN specialised agency for **food and agriculture**.
22. **Headquarters** is in **Rome, Italy**.

Chapter 21: Biodiversity Conservation Methods

🎯 Consolidated Must-Score Facts

1. **In-situ** (स्थान पर) conservation protects biodiversity **in the natural habitat (वास स्थान)**. It is the preferred primary strategy because the whole ecosystem (पारिस्थितिकी तंत्र) and natural selection stay intact.
2. **Ex-situ** (स्थानांतरित) conservation protects biodiversity **outside** the natural habitat — zoos, botanic gardens, seed banks, gene banks, cryobanks. It is a backup, not a replacement for in-situ.
3. **Orthodox seeds** (सामान्य (रूढ़िवादी) बीज) (wheat, rice) tolerate drying and cold storage around **−18°C**. **Recalcitrant seeds** (कठिन (असहनीय) बीज) (mango, coconut) cannot take simple dry seed-bank storage and need living collections or cryo routes.
4. **Cryopreservation** stores material at about **−196°C** in liquid nitrogen.
5. **CZA** is the statutory zoo regulator under **MoEFCC** (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय).
6. **In-situ conservation** protects species **where they naturally live**, keeping habitat, food webs, pollinators, and competitors intact.
7. **National Parks** offer strict habitat protection under **Wildlife (Protection) Act, 1972**, with human activity restricted. **Dudhwa NP (UP)** is a Terai (तराई) in-situ example.
8. **Wildlife Sanctuaries** protect habitat with slightly more flexible human use than NPs but remain **in-situ**. **Katerniaghat WLS (UP)** protects gharial and Gangetic dolphin in natural rivers.
9. **Biosphere Reserves** follow the UNESCO **MAB** model with **core**, **buffer**, and **transition** zones, combining strict protection with sustainable community use.
10. **Conservation Reserves** and **Community Reserves** (WPA amendments) protect habitat patches with community participation.
11. **Sacred groves** (deorai, sarna (सरना), kavu) are traditional **in-situ** community forests protected by local belief systems.
12. **Ramsar wetlands** and **marine protected areas** are specialised **in-situ** tools for aquatic and coastal ecosystems.
13. **Wildlife corridors** link fragmented habitats at landscape scale. The **Terai Arc Landscape** connects Dudhwa–Corbett habitats.
14. **Limitation:** In-situ alone cannot save species whose habitat is completely destroyed or whose wild population (जनसंख्या) is critically small without ex-situ backup.
15. **Ex-situ conservation** removes or copies genetic material **outside the natural ecosystem** — zoos, seed banks, gene banks, captive breeding, tissue culture.
16. **When essential:** critically small wild populations, **habitat already lost**, **reintroduction stock** needed (gharial, vulture), or **Extinct in the Wild (EW)** species surviving only in collections.
17. **Major ex-situ types:** zoological parks, botanical gardens, seed banks, gene banks, captive breeding centres, tissue culture, cryopreservation.
18. **Genetic bottleneck risk:** small captive groups lose diversity through inbreeding — zoos rotate breeding pairs and maintain studbooks.
19. **Reintroduction link:** ex-situ breeding succeeds only if **in-situ habitat** is restored. Cheetah reintroduction required **Kuno NP** habitat preparation alongside captive stock.
20. **IUCN (आईयूसीएन) Red List "Extinct in the Wild (EW)"** marks species surviving only in zoos, botanic gardens, or seed banks with no viable wild population.
21. **CBD + BDA 2002:** India regulates access to genetic resources; ex-situ collections at NBPGR/NBAGR support national biodiversity obligations.
22. **ICAR-NBPGR** at **Pusa Campus, New Delhi** (est. **1978**) is India's national **plant** gene bank. It holds one of the world's largest plant collections.
23. **NBAGR, Karnal (Haryana)** (est. **1984**) conserves **animal** genetic resources — semen, embryos, and DNA of indigenous livestock breeds (Sahiwal, Gir (गीर), Red Sindhi).
24. **Cryopreservation** uses **liquid nitrogen at -196°C**, stopping biological activity for decades-long storage of embryos, sperm, pollen, and tissue.
25. **Active collection** holds frequently accessed seeds for breeders. **Base collection** is long-term duplicate storage with minimal disturbance.
26. **Trap: NBPGR vs BSI** — NBPGR **stores** genetic resources (Delhi). **BSI** does plant **survey/taxonomy** (Kolkata).

27. **Trap: NBPGR vs NBAGR** — **NBPGR = plants (Delhi)**; **NBAGR = animals (Karnal)**.
28. **Orthodox seeds** (wheat, rice, maize, millets, pulses) tolerate desiccation and cold — suitable for **decades of storage**.
29. **Recalcitrant seeds** (mango, jackfruit, coconut, sal) **cannot tolerate drying** and need **botanical garden living collections** or **cryopreservation** instead of conventional seed banking.
30. **NBPGR** operates the **National Active Collection** (breeder access) and **National Gene Bank** (long-term base store) with **4 lakh+ accessions** including landraces and crop wild relatives.
31. **Svalbard Global Seed Vault (Norway, 2008)** stores **duplicate copies** of national collections as Arctic "doomsday backup" — not ownership transfer.
32. **Regeneration** is mandatory when germination rates fall — gene banking is dynamic, not store-and-forget.
33. **Community seed banks** (on-farm) preserve landraces in active cultivation — different from **formal ex-situ cold storage** at NBPGR. Notes usually mean NBPGR-style cold storage.
34. **Botanical gardens** maintain **living plant collections** outside wild habitat for conservation, research, education, and display.
35. **Other major gardens:** Lalbagh (Bengaluru, 1760), **NBRI garden (Lucknow)**, Empress Garden (Pune (पुणे)).

Chapter 22: Species & Ecology (पारिस्थितिकी)

🎯 Consolidated Must-Score Facts

1. **Threatened** on IUCN means **VU + EN + CR** only — not "Endangered" alone.
2. **Ramsar (रामसर) (1971)** is the wetlands treaty under Wise Use. World Wetlands Day (विश्व आर्द्रभूमि दिवस) is **2 February** (रामसर).
3. **Mangroves** are salt-tolerant coastal forests with **pneumatophores** (वात-मूल) and often viviparous seeds. **Sundarbans** (सुंदरबन) is the classic largest mangrove belt; Rhizophora and Avicennia are common genera.
4. **MISHTI** is the Union Budget **2023–24** mangrove restoration push. Do not date it as 2019.
5. **Coral reefs** are built from calcium-carbonate skeletons and live with symbiotic **zooxanthellae** (ज़ूजैंथेली). **Bleaching** happens when heat stress expels the algae.
6. **Grasslands** are grass-dominated systems maintained by fire and grazing. UP's Terai (तराई) is a foothill grassland–forest mosaic; **Dudhwa** (दुधवा) holds barasingha (swamp deer). Kaziranga (काजीरंगा) is the floodplain grassland rhino fact.
7. **ZSI (Zoological Survey of India)** publishes the **Red Data Book of Indian Animals**.
8. **Red Data Book ≠ IUCN Red List:** Red Data Book is **national documentation**; IUCN Red List is a **global standardised assessment** (EX through NE).
9. **Red Data Book ≠ CITES:** Red Data Book documents status; **CITES** regulates **international trade** in listed species.
10. **Extinct (EX):** No reasonable doubt the last individual has died.
11. **Extinct in the Wild (EW):** Survives only in **captivity, cultivation, or seed banks** — no viable wild population.
12. **Critically Endangered (CR) (अति संकटग्रस्त):** Extremely high extinction risk — e.g., **gharial**, **Great Indian Bustard**.
13. **Endangered (EN) (लुप्तप्राय):** Very high risk — e.g., **Gangetic dolphin**, **tiger** (improved from CR in some assessments).
14. **Vulnerable (VU):** High risk but lower urgency than EN/CR — e.g., **sloth bear**.
15. **Near Threatened (NT):** Close to qualifying as threatened but not yet.
16. **Least Concern (LC):** Widespread, low extinction risk.
17. **Data Deficient (DD):** Insufficient data — not the same as safe.
18. **Not Evaluated (NE):** Not yet assessed — majority of world's species.
19. **Threatened = VU + EN + CR** — the most tested formula. **Endangered alone is only one-third of threatened.**
20. **Risk hierarchy** among living categories: **CR > EN > VU > NT > LC**.
21. **IUCN assesses extinction risk.CITES** regulates **trade** — separate bodies, separate functions.
22. **Wetlands** are ecosystems where **water saturates the soil** — marshes, swamps, bogs, floodplains, lake margins — permanent or seasonal.
23. **India** has **75+ Ramsar sites** (2024 milestone). **Amrit Dharohar Scheme (2023)** supports wetland conservation and livelihoods.
24. **UP Ramsar sites:Sarsai Nawar** (Etawah (इटावा)), **Samaspur Bird Sanctuary** (Raebareli), **Nawabganj** (Unnao), **Upper Ganga River**, **Bakhira**, **Haiderpur**, **Parvati Agra Bird Sanctuary**.
25. **NOT in UP:Rudrasagar Lake** = **Tripura**. **Sultanpur National Park** = **Gurugram, Haryana**. Both are Ramsar sites outside UP.
26. **Threats:** drainage for agriculture, pollution (eutrophication (सुपोषण)), invasive species (आक्रामक विदेशी प्रजाति) (**water hyacinth** chokes UP ponds), hydrological alteration.
27. **Mangroves** are **salt-tolerant** trees and shrubs growing in **intertidal mudflats** of tropical and subtropical coasts.
28. **Pneumatophores** (vertical aerial breathing roots) and **prop roots** (Rhizophora) enable gas exchange in waterlogged anaerobic soil.
29. **Viviparous germination:** seeds germinate **while attached to the parent tree** and drop as ready seedlings (propagules).
30. **Ecological roles:** coastal protection against cyclones/tsunamis, fish nursery grounds, sediment trapping, shoreline stability, **blue carbon** (नीला कार्बन) sequestration.
31. **Coastal Odisha** is India's most cyclone-prone Bay of Bengal coast; large mangrove loss in the **Mahanadi (महानदी) delta (डेल्टा)** increases cyclone risk (classic A/R pair).

32. **Sundarbans** (West Bengal + Bangladesh) is the **largest mangrove forest** — **Sundari** tree is the flagship associate; royal Bengal tiger and estuarine crocodile.
33. **Bhitarkanika** (Kendrapara, Odisha) sits on the **Brahmani–Baitarni** delta — **not** Vansadhara–Subarnarekha, and **not** West Bengal. Ramsar site (**2002**).
34. **Coringa** mangrove (Andhra Pradesh) lies in the **Godavari** (गोदावरी) delta — do not confuse with Coondapur/Honnavar (Karnataka coast names).
35. **Andaman & Nicobar** uniquely combine **mangrove + evergreen + deciduous** forests in one geography.

Chapter 23: Renewable Energy (नवीकरणीय ऊर्जा)

🎯 Consolidated Must-Score Facts

1. **Renewable** energy replenishes on a human timescale — sun, wind, water, biomass, geothermal. **Nuclear** is low-carbon but usually **not** counted as renewable MCQs.
2. **Solar PV** turns sunlight directly into electricity. **CSP** (सीएसपी) uses mirrors to make heat, then steam and turbines.
3. **Bhadla** in Rajasthan is India's mega solar-park fact (not Gujarat as "largest"). **PM-KUSUM** supports farmer solar pumps. **OSOWOG** is One Sun One World One Grid.
4. **Domestic Content Requirement (DCR)** in solar news means mandating domestically manufactured PV cells/modules.
5. **Cochin International Airport** = world's first fully solar-powered airport. **Diu** = first Indian UT / smart city running **100% solar** (daytime / UT stems). **Surat** = first district with **100% solar-powered PHCs**.
6. **Biomass** is energy from organic matter. **Biogas** (बायोगैस) from anaerobic digestion is mainly **methane (CH_4) and carbon dioxide (CO_2) (कार्बन डाइऑक्साइड)** — not a butane/carbon-monoxide pair.
7. **Geothermal** uses Earth's internal heat. Indian provinces include **Himalaya (हिमालय), Cambay, West Coast, SONATA (Son–Narmada (नर्मदा)–Tapi (तापी)), Godavari (गोदावरी), Mahanadi (महानदी)** — **not Ganga (गंगा) Delta (डेल्टा)**. Sites: **Puga, Manikaran, Tattapani**.
8. **Tidal** energy comes from Moon-driven sea-level change. India's best potential is often locked to the **Gulf of Khambhat**, then Gulf of Kutch.
9. **Small hydro** under MNRE is **≤25 MW**. Above that is treated as large hydro. Himalayan streams hold large SHP potential.
10. **Green hydrogen** is H_2 from water electrolysis powered by **renewable** electricity. Grey hydrogen comes from natural gas; blue adds CCS (सुरक्षा कैबिनेट समिति) to gas routes.
11. **Solar energy** is radiant light and heat from the Sun — harnessed via **photovoltaic (PV)** panels or **concentrated solar power (CSP)** (संकेंद्रित सौर ताप).
12. **PV cells** (silicon semiconductors) convert photons to DC electricity; an inverter converts to AC for the grid.
13. **CSP** uses mirrors to heat fluid, produce steam, and drive a turbine — can store heat for after-sunset generation.
14. **National Solar Mission (2010)** is a pillar of **NAPCC (2008)** — also called **JNNSM** from January 2010 — target scaled to **100 GW solar**, achieved around **2022–2024**.
15. **500 GW non-fossil fuel capacity by 2030** (COP26, 2021) includes solar, wind, hydro, nuclear, and biomass — solar remains the largest growth component.
16. **International Solar Alliance (ISA)** (अंतरराष्ट्रीय सौर गठबंधन) launched **30 November 2015** at **COP21 Paris** by **India and France** — treaty-based IGO, HQ **Gurugram, Haryana**.
17. **Domestic Content Requirement (DCR)** in solar RfS means using domestically manufactured PV cells and modules.
18. **OSOWOG (One Sun One World One Grid)** and **World Solar Bank** concepts are linked to ISA's cross-border solar grid vision.
19. **PM-KUSUM (2019):** solar pumps for farmers and grid-connected solar on barren/fallow land — MNRE nodal.
20. **Cochin International Airport** was the **first airport in the world** fully powered by solar energy.
21. **Diu** became the first Indian UT / smart-city example to run on **100% solar** (daytime / UT stems). **Surat** became the first district with **100% solar-powered Primary Health Centres**.
22. **Bhadla (Rajasthan)** is India's mega solar park fact — not "Gujarat has the largest solar park" in newer stems. India's large floating solar example: **Ramagundam (Telangana)**.
23. **Wind energy** converts the **kinetic energy of moving air** into electricity using **wind turbines**.
24. **Tamil Nadu** is historically India's leading wind state (**Muppandal wind farm**). **Gujarat, Karnataka, Maharashtra, Rajasthan** follow.
25. **Onshore** turbines dominate India. **Offshore wind** potential on **Gujarat and Tamil Nadu coasts** — stronger winds but higher cost.
26. **Intermittency:** wind is variable — needs **grid integration**, **hybrid wind+solar**, or **storage** (batteries, pumped hydro).
27. **Capacity factor:** wind farms typically produce **~25–35%** of nameplate capacity annually — location-dependent.
28. **UP has limited wind potential** compared to western and coastal states — focus is national leaders.

29. **Biomass energy** is stored **chemical energy in organic matter** — plants, crop residue, animal waste, wood — released by burning, fermentation, or **anaerobic digestion**.
30. **Anaerobic digestion:** bacteria break organic matter **without oxygen** in a digester → **biogas** + **slurry** (organic fertiliser).
31. **Biogas composition:Methane (CH_4) ~50–70%** + **Carbon dioxide (CO_2) ~25–45%** + traces of H_2S and moisture.
32. **Butane** is an LPG component — **not** a major biogas constituent. **Carbon monoxide** is incomplete combustion product — not main biogas component.
33. **Bagasse** (sugarcane crush residue) is burned in **UP sugar mills** (Meerut (मेरठ), Muzaffarnagar belt) for **cogeneration** — electricity + process heat.
34. **GOBAR-Dhan / SATAT:** government push for community biogas and **bio-CNG** — oil marketing companies procure compressed biogas.
35. **E20 ethanol blending target by 2025** — ethanol from sugarcane, maize, and agri-waste reduces petrol imports.

Chapter 24: Disaster & Environment (पर्यावरण)

🎯 Consolidated Must-Score Facts

1. **NDMA** (आपदा प्राधिकरण) is under the Disaster Management Act **2005**. The **Prime Minister** chairs NDMA. **NDRF** is the response force. Sendai Framework for disaster risk reduction is **2015**.
2. **DPAP** (Drought Prone Areas Programme) is **1973**. Do not date DPAP as **1952**. A rough programme spine is CDP (सीडीपी) 1952 (सामुदायिक विकास) → DPAP 1973 → FWP 1977 → TRYSEM 1979.
3. **Piral Lao–Paise Pao** in Uttarakhand (उत्तराखंड) pays for pine-needle collection to cut fire risk and raise income. It is an Uttarakhand flagship, not a default UP scheme name.
4. **Riverine floods** swell slowly over days–weeks on plains. **Flash floods** follow cloudbursts in hills within hours. **Urban flooding** is waterlogging from impervious surfaces and clogged drains.
5. **UP flood rivers:** Ganga, Yamuna, Ghaghara (Saryu), Rapti, Gandak, Gomti** — monsoon inundation of the Gangetic floodplain.
6. **Human amplification:** floodplain encroachment, embankments that raise silted beds, and **deforestation** in catchments that speeds runoff.
7. **Terai flash risk:** Himalayan/Nepal-side catchments can inundate **Pilibhit, Lakhimpur Kheri, Bahraich** rapidly after extreme rain.
8. **Natural flood benefit:** controlled flooding deposits **alluvial (जलोढ़) silt** that renews soil fertility — the problem is uncontrolled flooding of encroached zones.
9. **Institutions:NDMA (Disaster Management Act 2005)** — **Prime Minister** is Ex-officio Chair; **NDRF** for response; **IMD/CWC (केंद्रीय जल आयोग)** for forecasting; **Sendai Framework 2015** for disaster risk reduction.
10. **Mitigation:** floodplain zoning, wetland restoration (natural sponges), catchment afforestation, early warning.
11. **Drought** develops slowly — weeks to months — unlike sudden floods.
12. **Meteorological drought:** rainfall below long-term normal.
13. **Agricultural drought:** soil moisture insufficient for crops (even if some rain occurs).
14. **Hydrological drought:** rivers, lakes, and groundwater below normal after prolonged dry spell.
15. **Bundelkhand (UP):** seven districts — **Jhansi (झांसी), Lalitpur, Jalaun, Banda, Hamirpur, Mahoba, Chitrakoot (चित्रकूट)** — chronic drought–desertification (मरुस्थलीकरण) hotspot.
16. **Drought Prone Areas Programme (DPAP)** launched in **1973** — soil conservation, water harvesting, afforestation in dryland districts.
17. **Feedback loop:** drought kills vegetation → soil exposed → erosion → worse water retention → next drought worse.
18. **Human triggers:** deforestation (roots bind soil), road cutting/blasting, unplanned hillside construction, mining — landslides are **not purely natural**.
19. **Earthquake link:** Himalayan seismic shaking can trigger widespread mass failures.
20. **Downstream risk:** landslide debris can dam (दाम -) rivers; dam-break floods then hit plains including UP Terai.
21. **Mitigation:** bio-engineering, retaining walls, afforestation, land-use zoning on steep slopes, GSI hazard zonation maps.
22. **Eye:** calm, clear, low-pressure centre. **Eyewall:** ring of **strongest winds and rain** around the eye. Trap: strongest winds are **not** in the eye.
23. **Storm surge:** cyclone winds push seawater onshore — often deadlier than wind on flat coasts (Odisha, WB, Bangladesh).
24. **Bay of Bengal** produces **more frequent and intense** cyclones than the **Arabian Sea** (warmer, shallower, moister waters).
25. **Mangroves** (e.g. Sundarbans (सुंदरबन)) absorb surge energy — coastal buffer. Deforestation of mangroves increases damage.
26. **IMD plains criteria:** maximum temperature **≥40°C** with departure **+4.5°C to +6.4°C** = heat wave; departure **≥+6.4°C** or absolute (निरपेक्ष) temp **≥47°C** = **severe heat wave**.
27. **Urban heat island:** cities stay hotter than surrounding rural areas due to concrete, asphalt, waste heat, and lost greenery — amplifies heat-wave mortality.
28. **Climate link:** warming makes heat waves more frequent, longer, and hotter (IPCC (आईपीसीसी) AR6).
29. **Compound risk:** heat + drought raises crop failure and forest-fire fuel dryness; heat + power surge risks blackouts.

30. **Health:** heatstroke, dehydration — elderly, outdoor workers, children most vulnerable.
31. **Adaptation:** cool roofs, urban greening, IMD heat warnings, Heat Action Plans (Ahmedabad (अहमदाबाद) model).
32. **Surface fire** burns litter/grass; **crown fire** burns canopy (most destructive); **ground fire** smoulders in organic soil.
33. **Chir pine** forests (Uttarakhand, Himachal (हिमाचल)) accumulate **dry pine needles** — major summer fuel load.
34. **Environmental damage:** biodiversity loss, soil organic matter destruction, erosion, **PM2.5 smoke**, CO_2 release; post-fire slopes raise landslide/flood risk.
35. **FSI** uses satellite (MODIS/VIIRS) for real-time forest fire monitoring and state alerts.

Chapter 25: Current Environmental Issues

🎯 Consolidated Must-Score Facts

1. **Microplastics** are plastic fragments **smaller than 5 mm**. Primary sources include microbeads; secondary sources come from larger plastic breaking down. They are not "fast biodegradable."
2. **Carbon trading** is the market that buys and sells allowances or credits (cap-and-trade). The **EU ETS** is the largest compliance market. India's **CCTS (2023)** under the Energy Conservation Act line is steered with **BEE**.
3. **PAT** creates energy-efficiency certificates (**ESCerts**). PAT is **not** identical to carbon credits under CCTS.
4. **Green Credit Programme (2023)** rewards eco-actions. It is **not** the same instrument as a carbon credit.
5. **Net zero** means anthropogenic emissions balanced by removals. India's target is **2070** (COP26 Glasgow **2021**). China is often locked at **2060**; EU/UK at **2050**. Net zero is not gross zero.
6. **Green finance** is capital for climate and environment (पर्यावरण) projects — green bonds, climate funds, GCF. India issued **Sovereign Green Bonds** in FY **2022–23**; SEBI's Green Bond Framework is **2023**.
7. **Blue economy** is sustainable ocean and coastal use — fisheries, ports, tourism, offshore renewables — aligned with **SDG 14**. **Blue carbon** (नीला कार्बन) is carbon stored in mangroves and seagrass.
8. **MISHTI (2023 Budget)** restores mangroves and sits beside Sagarmala and Deep Ocean Mission in the blue–coastal policy neighbourhood.
9. **LiFE** (Lifestyle for Environment (जीवनशैली पर्यावरण)) is a MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) behaviour-change mission. The idea was pitched at **COP26 Glasgow (2021)** and launched in **June 2022**. It is **not** a COP-25 Madrid product.
10. **Microplastics** are plastic particles **smaller than 5 mm** — primary (manufactured small) or secondary (fragmented from larger plastic).
11. **Primary:** microbeads in cosmetics, plastic **nurdles**, industrial abrasives.
12. **Secondary:** bags, bottles, fishing nets, and **tyre wear** breaking down — most environmental mass is secondary.
13. **Major sources:** single-use packaging, **synthetic textile fibres** (laundry effluent), tyre abrasion, fishing gear, agricultural plastic mulch.
14. **Pathways:** land → rivers → sea. **Ganga and Yamuna (यमुना)** studies show microplastic contamination downstream of urban (नगरीय) centres (Kanpur (कानपुर), Varanasi (वाराणसी), Lucknow (लखनऊ)).
15. **Health concern:** detected in blood, lungs, placenta — potential carrier for toxins; research ongoing.
16. **Persistence trap:** polyethylene/polypropylene persist **decades to centuries** — photodegradation makes **smaller** particles, not harmless minerals.
17. **Global policy:** UNEA **2022** resolution launched **Global Plastics Treaty** negotiations (INC process).
18. **India:** Plastic Waste Management Rules (EPR); **ban on identified single-use plastic items (July 2022)**; National Dashboard for EPR.
19. **Carbon credit** is a tradable certificate representing **one tonne of CO_2 equivalent (tCO_2e)** reduced, avoided, or removed.
20. **Project-based generation:** verified emission-reduction project (solar displacing coal, biogas (बायोगैस), afforestation) → accredited body issues credits → sold to buyer.
21. **Kyoto CDM (Clean Development Mechanism):** developed countries fund projects in developing countries → **Certified Emission Reductions (CERs)**. India hosted many wind/solar CDM projects.
22. **Voluntary Carbon Market:** companies buy credits for net-zero pledges — **Verra (VCS), Gold Standard** standards.
23. **Nature-based credits:** afforestation, REDD+, **mangrove restoration** — permanence risk if forest burns or is cleared.
24. **Avoidance vs removal:** avoidance prevents emissions that would occur; removal pulls CO_2 from atmosphere (afforestation, DAC).
25. **Additionality:** project must prove reduction would not happen without credit revenue.
26. **Green Credit Programme (2023)** rewards tree plantation, water conservation, waste management — **NOT the same as carbon credit** (broader eco-action incentive, not strictly 1 tCO_2e unit).
27. **Carbon trading** is the **market mechanism** to buy and sell emission **allowances or credits** under a regulatory cap.

28. **Cap-and-trade:** regulator sets total emissions cap → firms get allowances → efficient firms sell surplus to high emitters → cap met at lowest cost.
29. **Carbon credit vs carbon trading:** credit = **unit** (1 tCO_2e); trading = **market** exchanging those units.
30. **EU Emissions Trading System (EU ETS)** is the world's **largest compliance carbon market**.
31. **Kyoto flexible mechanisms:** (1) Emissions Trading among Annex I countries, (2) **CDM**, (3) Joint Implementation.
32. **India Carbon Credit Trading Scheme (CCTS), 2023** under **Energy Conservation Act 2001** — **Bureau of Energy Efficiency (BEE)** (ऊर्जा दक्षता ब्यूरो) as designated authority.
33. **PAT (Perform, Achieve and Trade (पण्याध्यक्ष)), 2012:** trades **energy efficiency certificates (ESCerts)** — parallel market-based compliance, **not identical to carbon credits**.
34. **Compliance vs voluntary:** compliance markets are legally mandated (EU ETS, future CCTS); voluntary markets are optional corporate purchases.
35. **Net zero:** anthropogenic GHG emissions are **balanced by anthropogenic removals** over a specified period — not absolute (निरपेक्ष) gross zero.

Chapter 26: Global Environmental Geography

🎯 Consolidated Must-Score Facts

1. **Köppen** letters: **A** tropical, **B** arid, **C** temperate, **D** cold, **E** polar. India is largely monsoon (**Am** type language); Thar (थार) is hot desert (**BWh**); Gobi is cold desert (**BWk**).
2. **Sahara** is the largest hot desert. **Atacama** is among the driest. **Gobi** spans **Mongolia and China**.
3. **Permafrost** (स्थायी हिम मृदा) appears in **taiga and tundra**. It is **not** a savanna feature.
4. **Global environmental geography** studies *where* environmental systems occur and *why* — linking physical geography (climate, relief, oceans) to biomes, biodiversity, and degradation.
5. **Climate** (temperature + precipitation) is the primary control on biome distribution.
6. **Latitude:** equatorial regions receive direct sunlight year-round → high productivity and biodiversity; polar regions receive oblique rays → low productivity.
7. **Continentality:** interior continents (central Australia, central Asia) have extreme temperatures and aridity.
8. **Altitude:** Himalaya (हिमालय), Andes, Alps show vertical zonation — tropical forest at base, alpine/nival at summit.
9. **Ocean currents:** warm currents (Gulf Stream) mild climate; cold currents (Humboldt) support fisheries and coastal deserts (Atacama).
10. **Subtropical highs (horse latitudes (हॉर्स अक्षांश) ~20–35° N/S)** suppress rainfall — major desert belt (Sahara, Arabian, Thar, Australian interior).
11. **Latitudinal biodiversity gradient:** species richness highest near equator, decreases toward poles.
12. **Tropical rainforest:** hot, wet year-round; layered canopy; highest biodiversity — Amazon (**Selva**), Congo, SE Asia, Western Ghats (पश्चिमी घाट).
13. **Tropical deciduous:** wet–dry monsoon — sal/teak forests of peninsular India.
14. **Desert:** typically **<25 cm** rain; sparse xerophytes — Sahara, Arabian, Gobi, Thar, Atacama.
15. **Savanna:** tropical grassland with **scattered trees** and distinct wet/dry seasons — East Africa ("Land of Big Games"); Gir (गीर) (Gujarat) is India patch.
16. **Temperate grassland:** few trees; fertile soils — Prairies (N America), Steppes (Eurasia), **Pampas (Argentina)**, **Puszta (Hungary)**.
17. **Taiga (boreal):** cold winters; spruce/fir/pine; **permafrost patches** — Siberia, Canada, Scandinavia.
18. **Tundra:** short summer; moss/lichen; **continuous permafrost** — Arctic coasts; no trees.
19. **Mediterranean (chaparral):** wet winters, dry summers; fire-adapted shrubs — California, Mediterranean Basin, Cape Town.
20. **Mangrove / wetland:transitional** terrestrial–aquatic — not purely terrestrial.
21. **Köppen Group A (Tropical):** all months ≥18°C — Af rainforest, **Am monsoon**, Aw savanna. India mainly **Am**.
22. **Group B (Arid):** evaporation > precipitation — BW desert, BS steppe. Thar **BWh** (hot); Gobi **BWk** (cold).
23. **Group C (Temperate):** mild winters — Mediterranean Csa, humid subtropical Cwa.
24. **Group D (Cold/Continental):** cold winters — taiga belt Siberia/Canada.
25. **Group E (Polar):** always cold — tundra ET, ice cap EF (Antarctica).
26. **ITCZ** seasonal shift drives monsoon and equatorial rain; failure causes Sahel drought (सूखा).
27. **Sahara (Africa):** largest **hot desert**.
28. **Atacama (Chile):** among the **driest** deserts on Earth — cold Humboldt Current + rain shadow (छाया).
29. **Gobi:** cold desert extending mainly into **Mongolia and China**.
30. **Kalahari:** southern Africa — **Bushman (San)** habitat.
31. **Thar:** India–Pakistan hot desert — subtropical high + rain shadow.
32. **Australian interior:** desert/semi-desert under subtropical high and continentality.
33. **Amazon (Brazil):** equatorial rainforest called **Selva** — among Earth's richest biodiversity zones.
34. **Terra Roxa:** fertile dark volcanic (basalt) soil of Brazil — coffee/soy region (often tested with Selva).

35. **World Floral Regions** classification is associated with **Adolf Engler**.

Chapter 27: Water Resources & Water Conservation

🎯 Consolidated Must-Score Facts

1. **Freshwater is not a biotic resource.** Biotic resources come from living/organic material (forests, animals, fish); **coal and petroleum** are classed biotic because they form from decayed organic matter.
2. **World Water Day (विश्व जल दिवस) = 22 March** (UN observance from **1993**; UN-Water since **2003**). Theme **2024** = *Leveraging Water for Peace* (also taught as prosperity and peace).
3. **Rajendra Singh** ("Jal Purush" / Waterman of India; **Tarun Bharat Sangh**, Alwar) won **Ramon Magsaysay 2001** and **Stockholm Water Prize 2015**.
4. **IWRM** follows Dublin Principles (**1992**) and treats the **river basin** as the planning unit. National Water Policy **2012** priority language puts drinking first, then ecology (पारिस्थितिकी), agriculture, and industry.
5. **National Water Mission** (NAPCC, from **30 June 2008**) aims at conservation, less wastage, equitable distribution via IWRM. Five goals include public water database + climate impact assessment, citizen/state conservation action, focus on over-exploited areas, **+20% water-use efficiency**, and basin-level IWRM. Urban (नगरीय) wastewater recycling and ocean-water tech for coastal cities fit NWM; **Himalayan–peninsular river linking and free borewell reimbursement do not**.
6. **Rainwater harvesting** captures rain for use or recharge — rooftops, pits, check dams, johads. Tamil Nadu (नाडु)'s mandatory RWH (**2003**) is a state flagship fact.
7. **Yamuna (यमुना) Action Plan** launched **1993** (YAP-I to ~2003; YAP-II Dec 2004; **YAP-III 7 May 2016** under Namami Gange).
8. **Ganges River Dolphin** (*Platanista gangetica*) = **National Aquatic Animal (2009)**; dams, nets, agrochemicals threaten it — crocodile increase is **not** a standard decline cause in UPSC (संघ लोक सेवा आयोग) framing.
9. **NLCP (June 2001)** restored urban/semi-urban lakes; merged with NWCP into **NPCA (Feb 2013)**. NLCP lakes include Bhoj, Sukhna, Pichola, Bhimtal — **Chilika is not NLCP**.
10. **CGWB** prepares the Annual Groundwater Quality Report. **CGWA** (groundwater regulation) is under **EPA 1986** (पर्यावरण संरक्षण 1986). India has the **largest groundwater-irrigated area** in the world. Arsenic drinking-water teaching often points to **Kolkata**.
11. **Atal Bhujal Yojana (2019)** is a groundwater management scheme tag.
12. **Jal Jeevan Mission** launched in **August 2019** for rural household taps (Har Ghar Jal). Do not date JJM as **2014**.
13. **Lisbon Declaration** (UN Ocean Conference) is about **oceans**, not freshwater glaciers.
14. **Surface water:** rivers, lakes, reservoirs — highly seasonal under monsoon.
15. **Groundwater:** India's largest **extractable** freshwater source (~45% irrigation, ~80% rural domestic).
16. **Ganga–Brahmaputra (ब्रह्मपुत्र)** system carries about **40%** of India's river water.
17. **Solar radiation** plays the main role in driving the water cycle by evaporating water from oceans, lakes, and rivers.
18. **Humidity** measures water-vapour content of air and is measured with a **hygrometer** or psychrometer.
19. **Biotic resources** come from living or organic material (forests, animals, fish) and from materials obtained from them. **Coal and petroleum** are counted as biotic because they form from decayed organic matter.
20. **Freshwater / pure water is not a biotic resource** — it is an abiotic natural resource, and usable drinking freshwater is scarce, so conservation techniques matter.
21. **World Water Day (22 March)** highlights freshwater and sustainable management (UN designation; first observed **1993**; **UN-Water** coordination from **2003**). Theme **2024**: *Leveraging Water for Peace*.
22. **Rajendra Singh** of Alwar (Rajasthan), called **Jal Purush / Waterman of India**, runs **Tarun Bharat Sangh** for community water harvesting; awards include **Ramon Magsaysay (2001)** and **Stockholm Water Prize (2015)**.
23. **IWRM** coordinates water, land, and related resources within a **river basin** to balance economy, society, and ecosystems.
24. **Dublin Principles (1992):** freshwater is finite; participatory management; women central; water has **economic value**.
25. **National Water Policy 2012:** river basin as planning unit; allocation priority **drinking water > ecology > agriculture > industry**.
26. **Ecological flows (e-flows):** minimum release below dams for river ecology.
27. **Conjunctive use:** coordinated surface water + groundwater management.

28. **Ministry of Jal Shakti (May 2019)** merged Water Resources with Drinking Water & Sanitation — institutional home for IWRM.
29. **National Water Mission (NWM)** is one of eight **NAPCC** missions (plan launched **30 June 2008**). Goal: conserve water, cut wastage, and ensure more equitable distribution across and within states through integrated water resource development and management.
30. **Rainwater harvesting** collects rooftop or surface runoff for **direct use** or **aquifer recharge** before it drains away.
31. **Rooftop:** roof → filter → tank (or recharge pit).
32. **Surface:** check dams, percolation tanks, farm ponds, contour trenches.
33. **Traditional:** Rajasthan **johad** (community pond) and **taanka** (cistern).
34. **Benefits:** reduces urban flood peaks, recharges groundwater, supplements municipal supply.
35. **Tamil Nadu (2003)** first made RWH **mandatory** in new buildings — Chennai model widely cited.

Chapter 28: Renewable & Non-Renewable Energy (अनवीकरणीय ऊर्जा)

🎯 Consolidated Must-Score Facts

1. **Renewable** sources replenish on a human timescale (sun, wind, water, biomass, geothermal). **Non-renewable** sources are finite stocks (coal, oil, gas, uranium). Notes usually treat **nuclear as non-renewable** even though it is low-carbon.
2. **Energy crisis** in bank language means the danger of exhaustion of **fossil fuels** (coal, oil, gas) — not merely "short hydro" or malnutrition.
3. **Uranium is not a fossil fuel** — nuclear fuel, non-renewable. Coal / petroleum / natural gas are fossils.
4. **Biogas** major constituents are **methane and carbon dioxide**. Butane or carbon monoxide as the major pair is wrong.
5. **Bioethanol** (maize, sugarcane, etc.) blends with petrol. **Biodiesel** (Jatropha, Pongamia, UCO) blends with diesel. Biofuels are eco-friendly and can ease crisis but are often **not cost-effective** vs fossils.
6. **OTEC** uses the temperature difference between warm surface water and cold deep water — not the same as tidal barrage power.
7. **Energy Conservation Act 2001** created **BEE**. **PAT (2012)** trades **ESCerts**. Star labels and UJALA LEDs are efficiency/conservation tools. PAT is **not** CCTS carbon credits.
8. **Hydrogen** is the least-polluting common fuel option (burns to water); called a **future fuel**. National Hydrogen Energy Board set up **3 October 2003**.
9. **Bioasphalt** is non-petroleum / renewable-based paving binder — eco-friendlier road surfacing.
10. **SATAT** promotes **compressed biogas (CBG)** for transport; oil-marketing companies procure CBG. **GOBAR-Dhan** is the community/dung biogas neighbour scheme — do not invent butane as a major biogas gas.
11. **Renewable energy** comes from sources replenished continuously on a human timescale — sun, wind, flowing water, tides, biomass, geothermal heat.
12. **Energy crisis** commonly means the danger of exhaustion of **fossil fuels** such as coal and petroleum.
13. **Nuclear** is usually classified as **non-renewable** (finite uranium + waste).
14. **Non-renewable** sources form over geological time — use depletes the stock.
15. **Coal:** India's primary electricity fuel; large domestic reserves in Jharkhand, Odisha, Chhattisgarh, West Bengal — eastern belt.
16. **Petroleum:** India imports ~**85%** of crude — main energy-security vulnerability.
17. **Natural gas:** cleaner than coal; used in power, fertiliser, CNG; partly imported (LNG).
18. **Uranium/thorium:** nuclear fuel — low operational carbon, finite ore, radioactive waste — **not a fossil fuel**.
19. **Non-conventional** sources are newer/alternative sources — solar, wind, biomass, biogas, tidal, geothermal, ocean energy, small hydro.
20. **PV** converts sunlight directly to electricity; **CSP** uses mirrors to heat fluid for steam turbines.
21. **National Solar Mission (2010)** under NAPCC — solar capacity scaled toward ~100 GW by early 2020s.
22. **ISA (2015):** India–France treaty IGO, HQ **Gurugram**; OSOWOG linked to ISA.
23. **PM-KUSUM (2019):** solar pumps and farmer solarisation.
24. **Bagasse** cogeneration is major in western UP sugar mills.
25. **Coal and nuclear are NOT biomass** sources.
26. **Biofuels** are fuels produced from biomass — bioethanol, biodiesel, biogas/CBG, advanced (2G) fuels.
27. **National Policy on Biofuels, 2018** sets blending roadmap and feedstock categories.
28. **1st generation:** sugarcane molasses ethanol; edible-oil biodiesel — food vs fuel concern.
29. **2G:** agri-waste, residues — reduces food competition.
30. **Maize** is a classic **ethanol** crop; **Jatropha / Pongamia / sunflower** are classic **biodiesel** crops — do not swap.
31. **Butane** and **carbon monoxide** are not major biogas components.
32. **Bioethanol** is alcohol produced mainly by fermenting sugarcane molasses (also maize, 2G waste).
33. **Biodiesel** is fatty acid methyl ester (**FAME**) made by **transesterification** of vegetable oils + methanol + catalyst → biodiesel + glycerol.

34. **Pongamia** grows in arid / saline (लवणीय) / alkaline soils; seeds are lipid-rich with nearly half **oleic acid** — classic dual-statement biodiesel stem with Jatropha.
35. **Ramganga (रामगंगा) (~198 MW)** is **large hydro**, not SHP.

Chapter 29: Environmental Research & Institutions

🎯 Consolidated Must-Score Facts

1. **BSI (1890, Kolkata)** surveys **plants** (*Flora* (वनस्पति), plant Red Data Book (रेड डेटा बुक)). **ZSI (1916, Kolkata)** surveys **animals** (*Fauna* (प्राणीजात), animal Red Data Book).
2. **FSI (1981, Dehradun)** maps forest cover (वन आवरण) and publishes **ISFR every two years**. **FRI (1906, Dehradun)** does forestry **research** under the ICFRE world — it does **not** publish ISFR.
3. **WII (1982, Dehradun)** does wildlife research, IFS/officer training, and tiger-census protocols.
4. **NBA** under the Biological Diversity Act sits in **Chennai**, not Delhi. It handles foreign access and ABS at the national level.
5. **NBPGR (1978, New Delhi, ICAR)** is the plant gene bank. It is **not** Kolkata BSI and **not** Karnal **NBAGR** (animals).
6. **BSI** is the national organisation for survey, taxonomy, and documentation of **Indian plant diversity**.
7. **Does not** survey animals (ZSI), map forest cover (FSI), or store crop gene banks (NBPGR).
8. **ZSI** documents **Indian animal (faunal) diversity** — taxonomy and inventories.
9. **ZSI ≠ WII:** ZSI = species inventory/taxonomy; WII = applied wildlife ecology (पारिस्थितिकी) and training.
10. **FSI** monitors India's **forest and tree cover (वृक्ष आवरण)** using satellite imagery and ground verification.
11. **ISFR 2023:** India forest cover **21.76%**; UP **6.24%**.
12. **FRI** is India's premier **forestry research** institute — silviculture, pathology, entomology, wood science, NTFPs.
13. **FRI ≠ FSI:** both in Dehradun — FRI = **how to grow/manage forests**; FSI = **how much forest exists on the map**.
14. **WII** is MoEFCC's autonomous institute for **wildlife research, training, and conservation science**.
15. **Does not** do national plant/animal taxonomy (BSI/ZSI) or publish ISFR (FSI).
16. **NBA** is the national statutory body under the **Biological Diversity Act, 2002**.
17. **Does not** prepare People's Biodiversity Registers — that is **BMC**.
18. **Uttar Pradesh State Biodiversity Board (UPSBB)** sits at **Lucknow** (लखनऊ).
19. **SBB ≠ SPCB:** SBB = biodiversity (BDA); SPCB = pollution (Water/Air Acts).
20. **BMC** is the **local** committee under BDA 2002 — at panchayat, municipality, or cantonment level.
21. **PBR is not prepared by NBA or BSI**.
22. **NBPGR** conserves **plant genetic resources** — seeds, tissue, DNA — for food and agriculture.
23. **Ex-situ** (स्थानांतरित) conservation — complementary to in-situ (स्थान पर) protected areas.
24. **NBPGR ≠ BSI:** BSI surveys/taxonomy (Kolkata); NBPGR stores germplasm (Delhi).
25. **NBPGR ≠ NBAGR:** animals/livestock gene bank is **NBAGR, Karnal**.

Chapter 30: Indian Environmental Movements

🎯 Consolidated Must-Score Facts

1. **Bishnoi** (बिश्नोई) resistance at **Khejarli, Rajasthan (1730)** is led in memory by **Amrita Devi** (अमृता देवी) and the **363** martyrs who died protecting khejri trees. It is centuries older than Chipko.
2. **Chipko (1973)** began at **Reni, Uttarakhand** (then Uttar Pradesh (उत्तर प्रदेश)). Leaders include **Sunderlal Bahuguna, Chandi Prasad Bhatt (चंडी प्रसाद भट्ट), and Gaura Devi**. Method: embracing trees against logging.
3. **Silent Valley (1970s)** (शांत घाटी) in **Kerala (Palakkad)** opposed a hydro project in rainforest. **KSSP** and scientists were central. It is **not** a Tamil Nadu (नाडु) movement.
4. **Jungle Bachao Andolan (1982)** (जंगल बचाओ) in **Singhbhum, Jharkhand** was tribal (आदिवासी) resistance for forest rights.
5. **Appiko (1983)** in **Karnataka (Sirsi)** was led by **Panduranga Hegde**. It is the southern tree-embrace movement. Appiko is **Karnataka only** — not Uttarakhand, UP, or Kerala.
6. **Narmada Bachao Andolan (from 1985)** spans **Madhya Pradesh, Gujarat, and Maharashtra**, with **Medha Patkar** (मेधा पाटकर) and **Baba Amte** as leader facts. It is anti-large-dam displacement politics.
7. **Save Western Ghats (from about 2011)** is multi-state ecology (पारिस्थितिकी) politics around the **Gadgil** and **Kasturirangan** reports, mining, and ESZ questions.
8. **Maiti Movement** (मैती आंदोलन) (Uttarakhand; pioneered by **Kalyan Singh Rawat**, mid-1990s) asks newly married couples to plant a sapling at the bride's parental home — separate from Chipko.
9. Assertion (A): Chipko began at Reni in then Uttar Pradesh (now Uttarakhand).
10. Reason (R): Appiko and Silent Valley are UP movements.
11. Assertion (A): "NBA" as Narmada Bachao Andolan is not the National Biodiversity Authority.
12. Reason (R): Same letters can mean different bodies.
13. Assertion (A): Tree-hugging method travelled from Chipko north to Appiko south.
14. Reason (R): Same method family, different states and years.
15. Assertion (A): Gadgil is often read as stronger ecological protection; Kasturirangan as later recalibration.
16. Reason (R): Both sit inside Save Western Ghats politics.
17. Assertion (A): Match movement → year → state → leader before answering "which state" stems.
18. Reason (R): State alone often decides the key.
19. Assertion (A): Khejarli martyrdom is Rajasthan Bishnoi memory.
20. Reason (R): It is Uttarakhand Chipko's founding event.

Chapter 31: Environmental Literature & Awareness

🎯 Consolidated Must-Score Facts

1. ***Silent Spring* (1962)** by **Rachel Carson** exposed pesticide/DDT harms. It is not Hardin's commons essay.
2. **"Tragedy of the Commons" (1968)** by **Garrett Hardin** is about overuse of shared resources.
3. ***Limits to Growth* (1972)** is the **Club of Rome** systems study. ***Our Common Future* (1987)** is the **Brundtland Report** that popularised sustainable development. Brundtland ≠ Limits to Growth.
4. ***Small Is Beautiful* (1973)** by **E.F. Schumacher** critiques mass-production economics — "economics as if people mattered." It is not Carson's pesticide book.
5. **Eco-mark (1991)** is the BIS earthen-pot label for certified environment-friendly consumer goods. It is **not** AGMARK and not a FAO mark.
6. **Indira Gandhi Paryavaran Puraskar (1987, MoEF)** honours measurable contribution to environment protection. **Rajiv Gandhi Environment Award** recognises clean technology and industrial pollution reduction.
7. **Tyler Prize** (University of Southern California) is often called the "Nobel of Environment" for environmental science, health, and energy.
8. **Environmental literature** includes books and landmark reports that shaped ecological thought and policy.
9. **Rachel Carson, *Silent Spring* (1962):** exposed DDT/pesticide harm to birds, food chains, and health — triggered the modern environmental movement.
10. **Garrett Hardin, "Tragedy of the Commons" (1968):** argued individuals overuse shared resources (pasture, fisheries, air) for private gain.
11. **Club of Rome, *The Limits to Growth* (1972):** modelled exponential growth hitting planetary resource limits — **not** the Brundtland Report.
12. **Brundtland Commission, *Our Common Future* (1987):** defined and **popularised Sustainable Development** — meeting present needs without compromising future generations.
13. **W.M. Adams, *Green Development* (first edition 1990):** environment and sustainability in the developing world.
14. **Masanobu Fukuoka** was a Japanese farmer-philosopher who pioneered **natural farming** and re-vegetation of desertified lands.
15. **Aldo Leopold, *A Sand County Almanac* (1949):** "Land Ethic" — moral duty to the biotic community.
16. **Paul Ehrlich, *The Population (जनसंख्या) Bomb* (1968):** population growth vs resource limits debate.
17. **Indian voices:** Gandhi (गांधी) *Hind Swaraj (स्वराज)* (हिंद स्वराज) (1909) — restraint vs reckless industrialism; Tagore/Shantiniketan — nature-linked education; Bahuguna — "ecology (पारिस्थितिकी) is permanent economy"; CSE (Anil Agarwal, Sunita Narain) — *Down To Earth*.
18. **Environmental Education (EE)** builds knowledge, attitudes, skills, and participation for environmental protection.
19. **Stockholm 1972** (स्टॉकहोम सम्मेलन) put EE on the global agenda; **Tbilisi Declaration (1977)** became the first intergovernmental EE framework.
20. **Rio 1992 Agenda 21 Chapter 36** called for reorienting education toward sustainable development.
21. **NEP 2020** integrates environmental awareness and ESD across stages through experiential, multidisciplinary learning.
22. **National Green Corps (NGC) (नेशनल ग्रीन कॉर्प्स) / ECO Clubs (MoEFCC):** school/college clubs for plantation, waste management, biodiversity surveys, energy audits.
23. **Paryavaran Mitra (मित्र -):** CEE Ahmedabad (अहमदाबाद) + MoEFCC — student/teacher green ambassadors.
24. **Green Skill Development Programme (GSDP), 2017:** MoEFCC + Skill Ministry — vocational green skills.
25. **Environmental awareness** is public understanding and concern that drives behaviour change and civic action.
26. **Article 51A(g)** (42nd (42वां) Amendment, 1976): Fundamental Duty to protect and improve forests, lakes, rivers, and wildlife — pairs with **Art 48A** (state duty).
27. **National Environment Awareness Campaign (NEAC):** MoEFCC-funded NGO/school campaigns since the 1980s.
28. **Van Mahotsav:** tree-plantation week in **July**, popularised by **K.M. Munshi (1950)**.

29. **LiFE Mission:** idea at **COP26 (2021)**; launched **June 2022** — 75 pro-planet lifestyle actions (awareness/nudge, not school curriculum).
30. **Eco-mark / ECOMARC** is India's eco-labelling scheme launched in **1991** by the Government of India and administered by the **Bureau of Indian Standards (BIS)**.
31. **AGMARK** is a quality certification mark for agricultural produce under the **Agricultural Produce (Grading and Marking) Act, 1937**, issued through the Directorate of Marketing and Inspection (Ministry of Agriculture) — **not FAO**.
32. **BIS / ISI** marks cover many industrial and consumer products (including mandatory certification for automotive tyres and tubes in classic stems).
33. **Greenwashing** means conveying a **false or misleading impression** that a company (कंपनी)'s products or practices are eco-friendly.
34. **Greendex** (ग्रीनडेक्स) (National Geographic / GlobeScan) measures **environmentally sustainable consumer behaviour** across countries — not carbon-credit sales volume.
35. **Plachimada** (Palakkad, **Kerala**) became a landmark local struggle against groundwater contamination linked to a Coca-Cola bottling plant.

Chapter 32: Environmental Days

🎯 Consolidated Must-Score Facts

1. **World Wetlands Day** falls on **2 February** (रामसर) and marks the **Ramsar (रामसर) Convention (1971)** (रामसर कन्वेंशन). It is not 2 March or 3 March.
2. **World Wildlife Day** falls on **3 March** and marks the adoption of **CITES (साइट्स) (1973)** (साइट्स). It is **not** 22 March.
3. **International Day of Forests** falls on **21 March**. It was proclaimed by the UN General Assembly in **2012**.
4. **World Water Day** falls on **22 March**. The classic trap is pairing Wildlife Day with 22 March.
5. **Earth Day** falls on **22 April**. It began in the United States in **1970** as a civic teach-in movement.
6. **International Day for Biological Diversity** falls on **22 May** and commemorates the CBD text adopted (अंगीकृत) at Nairobi in **1992**.
7. **World Environment Day** falls on **5 June**. It marks the opening of the **Stockholm Conference (1972)** (स्टॉकहोम सम्मेलन) and is run by **UNEP** (संयुक्त राष्ट्र पर्यावरण कार्यक्रम).
8. **World Ozone Day** falls on **16 September** and marks the **Montreal Protocol (मॉन्ट्रियल प्रोटोकॉल) (1987)** (मॉन्ट्रियल प्रोटोकॉल).
9. **World Toilet Day = 19 November** (UN, from 2013). **World No Tobacco Day = 31 May** (not 5 May). **World Habitat (वास स्थान) Day** = first Monday of October (not a fixed 5 October).
10. **Swachh Bharat Mission** launched **2 October 2014** (Gandhi (गांधी) Jayanti). Sierra Leone observes national cleanliness on the last Saturday of every month.
11. **International Tiger Day** falls on **29 July** and marks the **St Petersburg Tiger Summit (2010)**. It is **not** World Wildlife Day (3 March).
12. **World Environment Day (WED)** is the UN's main annual day for environmental awareness (पर्यावरण जागरूकता) and action.
13. **2022** also marked **50 years** since Stockholm 1972, which designated **5 June** as World Environment Day.
14. **World Wetlands Day** marks adoption of the **Ramsar Convention on Wetlands** at Ramsar, Iran (**2 February 1971**).
15. **Earth Day** promotes environmental protection, climate action, and sustainability every **22 April**.
16. **International Day for Biological Diversity (IDB)** promotes biodiversity conservation awareness every **22 May**.
17. **International Day of Forests** raises awareness of forests and sustainable forest management every **21 March**.
18. **World Water Day** focuses on freshwater resources, access, and sustainable water management every **22 March**.
19. **World Ozone Day** (International Day for the Preservation of the Ozone Layer) is observed every **16 September**.
20. **World Wildlife Day** celebrates wild fauna and flora and fights wildlife crime every **3 March**.
21. **Highest-yield trap:** listing Wildlife Day as **22 March** (that is Water Day) — wrong pair in **
22. **International Tiger Day** is observed every **29 July**.
23. **World Toilet Day = 19 November** (UNGA 2013; sanitation for all / end open defecation messaging).
24. **World No Tobacco Day = 31 May** (WHO) — not 5 May. Themes rotate (e.g. tobacco–environment / grow food not tobacco).
25. **World Habitat Day** = **first Monday of October** (UN; not a fixed calendar date like 5 October).
26. **World Cities Day = 31 October**. **World Population Day = 11 July**. **World Health Day = 7 April**.
27. **Earth Overshoot Day** marks when humanity's annual demand exceeds Earth's regenerative capacity — an environmental-degradation awareness marker (Global Footprint Network partnership framing).
28. **Swachh Bharat Mission** launched **2 October 2014**. Sierra Leone's national cleanliness day is the last Saturday of each month.

Chapter 33: National Parks & Protected Areas (Advanced)

🎯 Consolidated Must-Score Facts

1. **Valmiki** (वाल्मीकि) National Park is in **Bihar**. **Sultanpur** National Park is in **Haryana**. **Corbett** is in **Uttarakhand** (उत्तराखंड). None is a UP national park.
2. **Rudrasagar** Lake is in **Tripura**. **Sultanpur** NP is in **Haryana**. **Samaspur** and **Sarsai Nawar** are in Uttar Pradesh.
3. **Gir (गीर) (Gujarat)** holds the only wild **Asiatic lions**. **Kaziranga (काजीरंगा) (Assam)** is the classic one-horned rhino stronghold.
4. **Pilibhit (2014)** and **Ranipur (2023)** are tiger reserves. They are **not** additional full state national parks of UP.
5. **Rajaji (राजाजी) NP (UK)** = Dehradun (देहरादून)–Haridwar (हरिद्वार)–Pauri (पौड़ी) (Rajaji + Motichur + Chilla merged **1983**); Tiger Reserve **2015**. Named after C. Rajagopalachari.
6. **Pin Valley = Himachal (हिमाचल) (Lahaul–Spiti)**, not J&K. **Van Vihar = Bhopal**. **Salim Ali NP = Srinagar (ex–City Forest)**. **Nagarhole = Rajiv Gandhi NP (Karnataka)**.
7. **Indravati = Bastar (Chhattisgarh)** NP 1981 / TR 1983. **Bandhavgarh (Umaria, MP)** famous for white tiger association. **Betla/Palamu = Jharkhand**.
8. **Keibul Lamjao (Manipur)** on **Loktak** is the only floating NP (**phumdis**; Sangai (सांगाइ) deer). **Valley of Flowers** (फूलों की घाटी) lies wholly in temperate alpine zone.
9. **Govind NP (Uttarkashi (उत्तरकाशी))** includes the source region of the **Tons** (टौंस) (Yamuna (यमुना) tributary). **Great Himalayan NP = Kullu (HP)**.
10. **Wild Ass Sanctuary (Rann of Kachchh, Gujarat)** is India's largest wildlife sanctuary. **Kuno–Palpur (MP)** is the Asiatic lion second-home site.
11. **Sathyamangalam TR** links Eastern and Western Ghats. **Nagarjunsagar–Srisailam** holds the largest critical tiger habitat (वास स्थान) among common options.
12. **Uttar Pradesh has one National Park — Dudhwa** — in **Lakhimpur Kheri** on the Indo-Nepal Terai border.
13. **Pilibhit TR (2014)** and **Ranipur TR (2023)** are tiger reserves — not additional full state NPs.
14. **Terai trio with Dudhwa:Katarniaghat** (Bahraich) and **Kishanpur** (Lakhimpur Kheri) — tiger, gharial, dolphin corridor.
15. **Hastinapur WLS** (~2,073 sq km) is UP's **largest**, spanning Meerut (मेरठ)–Muzaffarnagar–Ghaziabad–Bijnor–Amroha.
16. **National Chambal WLS** (Agra–Etawah (इटावा) + MP–Rajasthan) — gharial, dolphin, Indian skimmer.
17. **Chandraprabha WLS** is in **Chandauli** (1957; among earliest in UP).
18. **Katarniaghat** was established in **1975**.
19. **Largest NP:Hemis**, Ladakh (~3,350 sq km) — snow leopard country.
20. **Most NPs:Madhya Pradesh** (~11).
21. **Gulf of Mannar Marine NP, Tamil Nadu** — coral islands, dugong; sits inside Gulf of Mannar Biosphere Reserve (जैवमंडल आरक्षित क्षेत्र).
22. **Mahatma Gandhi Marine NP (Wandoor)** and **Rani Jhansi Marine NP** are in the **Andaman** Islands.
23. **World's first National Park** is **Yellowstone NP, USA (1872)** — signed under President **Ulysses S. Grant**.
24. **Keoladeo** (केवलादेव) often carries **NP + Ramsar + UNESCO WH** together.
25. **Manas** was removed from the WH danger list in **2011** after recovery.
26. **Uttar Pradesh has no natural UNESCO World Heritage Site** — nearest are Valley of Flowers / Nanda Devi in Uttarakhand.
27. **Coastal NPs** protect mangrove, estuary, delta (डेल्टा), intertidal, and reef ecosystems; all marine NPs are coastal, but not every coastal NP is a "Marine NP" label.
28. **Gahirmatha** = marine **sanctuary** (turtle rookery), not NP.

Chapter 34: Biosphere Reserves (जैवमंडल आरक्षित क्षेत्र)

🎯 Consolidated Must-Score Facts

1. **Nokrek** is in **Meghalaya** (मेघालय). Pairing Nokrek with **Manipur** is always wrong.
2. **Manas (Assam)** remains a national Biosphere Reserve but was **removed from the UNESCO World Network in 2011**.
3. **Uttar Pradesh has zero Biosphere Reserves.** Nanda Devi (नंदा देवी) BR is in **Uttarakhand** (उत्तराखंड), not UP.
4. **Panna (Madhya Pradesh, 2011)** is India's newest notified Biosphere Reserve.
5. **Three zones: Core** (no exploitative use / genetic diversity (आनुवंशिक विविधता)) → **Buffer** (research, education, regulated eco-tourism and grazing) → **Transition** (settlements, crop cultivation, livelihoods).
6. **Triple mandate:** conservation + sustainable development + logistic support (research/monitoring/education).
7. **Agasthyamalai** includes Neyyar, Peppara, Shendurney WLS and **Kalakad–Mundanthurai Tiger Reserve**.
8. **Gulf of Mannar** is often the largest **UNESCO-certified** Indian BR by area; **Great Rann of Kutch** is the largest **national** BR.
9. **Bundala** Biosphere Reserve is in **Sri Lanka**, not India.
10. **Largest BR:Great Rann of Kutch**, Gujarat. **Smallest:Nokrek**, Meghalaya.
11. **Newest notified:Panna**, Madhya Pradesh (2011).
12. **Manas (Assam)** remains a **national BR** but was **removed from UNESCO WNBR in 2011**.
13. **Madhya Pradesh** has three BRs: Pachmarhi, Achanakmar–Amarkantak (with Chhattisgarh), Panna.
14. **Uttar Pradesh has no notified Biosphere Reserve.**

Chapter 35: Climate Change (जलवायु परिवर्तन) (Advanced)

🎯 Consolidated Must-Score Facts

1. **Climate change** is a long-term shift in temperature, rainfall, and extremes. **Global warming** is the temperature-rise part only.
2. **Green Credit (2023)** (ग्रीन क्रेडिट) and **PAT ESCert** are **not** the same thing as a Kyoto-style carbon credit.
3. **Mitigation** cuts emissions and grows sinks. **Adaptation** adjusts society and systems to climate impacts.
4. **Kyoto** (क्योटो) is climate mitigation (जलवायु शमन). **Montreal** is ozone protection. Do not swap the two treaties.
5. **LiFE** was floated as an idea at **COP-26 (2021)** and launched in **June 2022**. COP-25 Madrid is the false date trap.
6. **Milankovitch** = eccentricity, obliquity, precession — **not** solar irradiance. **Ice cores** = cryogenic climate archive.
7. **GCF** = Cancun **2010**. **CDM/CERs** = Kyoto. **GHG Protocol** = WRI + WBCSD. **Earth Hour** (अर्थ आवर) = WWF (डब्ल्यूडब्ल्यूएफ).
8. **Methane hydrates**, paddy+fertiliser (CH_4+N_2O), and wetland CH_4 are high-yield source traps.
9. **Carbon fertilization** ≠ ocean acidification (महासागरीय अम्लीकरण).
10. **Forest Carbon Partnership Facility** and **BioCarbon Fund ISFL** are **World Bank** programmes — FCPF does **not** fund universities/individuals for research grants.
11. **Climate Action Tracker** = Climate Analytics + New Climate Institute (not IPCC/UNEP wing).
12. **Deep Carbon Observatory** maps deep-Earth carbon (physics, reservoirs, deep energy, deep life).
13. **Common Carbon Metric** (UNEP-linked) measures building-operation carbon footprints.
14. **Climate change** is a long-term change in the climate system — temperature, precipitation, extremes, sea level — lasting decades or longer.
15. **Radiative forcing** (W/m^2) measures the net energy imbalance driving warming.
16. **WMO** publishes the **State of the Global Climate** report.
17. **NOAA** and **NASA GISS** maintain key global temperature datasets.
18. **Global Carbon Project** issues the annual **Global Carbon Budget**.
19. **Emissions Gap Report** is published by **UNEP** — it measures the gap between NDCs and 1.5–2°C pathways.
20. **Energy and electricity** (coal, oil, gas) dominate CO_2.
21. **Industry** adds process CO_2 (cement, steel) plus F-gases.
22. **AFOLU** (agriculture, forestry, land use): **CH_4** from livestock/rice; **N_2O** from fertilisers; CO_2 from deforestation.
23. **Rice** is a major anthropogenic source of **both methane and nitrous oxide**.
24. **Transport** and **buildings** add fossil CO_2; **waste** adds landfill methane.
25. **Argon is not a greenhouse gas.**
26. **Joseph Fourier (1820s)** postulated the greenhouse-effect concept.
27. **Astronomical (Milankovitch) climate theory** uses eccentricity, obliquity, and precession — **solar irradiance is not** listed in that set. **Ice cores** are the classic cryogenic climate indicator.
28. **Global warming** means long-term rise in global mean surface temperature.
29. **Fast cycle:** atmosphere ↔ plants ↔ soils ↔ surface ocean (years to decades).
30. **Slow cycle:** rock weathering, sediment burial, fossil carbon (millions of years).
31. **Photosynthesis consumes** CO_2 and releases oxygen. It does **not** add carbon dioxide to the carbon cycle on Earth.
32. **Carbon trading** is buying and selling emission allowances or carbon credits.
33. **Cap-and-trade** sets an emissions cap and lets the market discover the price.
34. **PAT** trades **ESCerts** for energy efficiency — related market tool, **not** the same as carbon credits.
35. **Mitigation** reduces GHG emissions and/or enhances sinks to limit future warming.

Chapter 36: Atmosphere

🎯 Consolidated Must-Score Facts

1. **Nitrogen gas (N_2) is not a greenhouse gas.Nitrous oxide (N_2O)** is a greenhouse gas. Do not confuse the two.
2. **Argon is not a greenhouse gas.** Oxygen and helium are also not greenhouse gases.
3. **Dry air near the surface: Nitrogen ~78.08%**, **Oxygen ~20.95%**, **Argon ~0.93%**, **CO_2 ~0.04%**.
4. **Permanent gases** (N_2, O_2, Ar) stay roughly constant in the homosphere.
5. **Variable gases** include water vapour (0–4%), CO_2, CH_4, O_3, and aerosols.
6. **N_2 is not a greenhouse gas**; **N_2O** is. Do not confuse N_2 with nitrogen oxides or nitrous oxide.
7. **Argon is not a GHG** (also covered in Topic 34 greenhouse-gas questions).
8. **Water vapour** is the strongest natural tropospheric greenhouse gas by volume, but mainly a climate **feedback**.
9. Assertion (A): Dry air is about N_2 78%, O_2 21%, argon ~0.93%, CO_2 ~0.04%.
10. Reason (R): Argon is a major greenhouse gas.
11. Assertion (A): Stratospheric temperature rises with height largely because ozone absorbs UV.
12. Reason (R): The ozone layer's maximum concentration is in the stratosphere.
13. Assertion (A): Commercial jets often cruise near the tropopause.
14. Reason (R): The tropopause sits atop the troposphere neighbourhood.
15. Assertion (A): Do not place the ozone layer in the troposphere on a match list.
16. Reason (R): Maximum ozone concentration is in the stratosphere.
17. Assertion (A): Temperature falls with height in the troposphere.
18. Reason (R): Temperature rises with height in the stratosphere.
19. Assertion (A): Oxygen and helium are not greenhouse gases.
20. Reason (R): Only CO_2 among all atmospheric gases can ever affect radiation.

Chapter 37: Ozone Layer

🎯 Consolidated Must-Score Facts

1. **Ozone (O_3)** is triatomic oxygen. **Good ozone** (अच्छा ओजोन) sits in the **stratosphere** (समतापमंडल) (about **15–30/35 km**). Ground-level ozone is a harmful pollutant.
2. **HFCs** are high-GWP climate gases under **Kigali**. They are **not** classic ozone-depleting substances.
3. **Montreal = ozone. Kyoto (क्योटो) = climate.** Carbon trading ≠ Montreal.
4. **UV-A (315–400 nm)** largely reaches the surface; **UV-B (280–315 nm)** is mostly blocked by ozone; **UV-C (100–280 nm)** is blocked and does not reach the surface under normal conditions.
5. **G.W. Kent Moore** (University of Toronto) led the **2005** report of an ozone "halo" / low-ozone feature over the **Tibetan Plateau**.
6. **Ozone (O_3)** is triatomic oxygen that absorbs harmful solar ultraviolet radiation.
7. **UV-A (315–400 nm)** largely reaches the surface; **UV-B (280–315 nm)** is mostly blocked by ozone; **UV-C (100–280 nm)** is blocked by O_2/O_3 and does not reach the surface under normal conditions.
8. **Ozone depletion** is a sustained fall in stratospheric ozone below natural levels.
9. **Rowland–Molina (1974)** linked CFCs to ozone loss; Nobel Prize in Chemistry **1995**.
10. **ODS** are chemicals controlled under the Montreal Protocol because they destroy stratospheric ozone.
11. **CFCs:** refrigerants, aerosol propellants, foam blowing, electronics cleaning — long-lived chlorine sources.
12. **Halons (e.g. Halon-1211, Halon-1301):** fire suppressants — bromine; often higher ozone-depleting impact per molecule.
13. **ODP (Ozone Depleting Potential)** ranks destructiveness relative (सापेक्ष) to **CFC-11 = 1**.
14. **HFCs** were ozone-friendly replacements but have high **GWP**; controlled later by the **Kigali Amendment** (climate, not classic ODS).
15. **Nitrous oxide (N_2O)** also contributes to ozone chemistry / long-term ozone loss discussions, but CFCs/halons dominate classic "ozone hole" stems.
16. **16 September** is observed as **International Day for the Preservation of the Ozone Layer (World Ozone Day)** — UN observance from **1994**.

Chapter 38: Greenhouse Gases

🎯 Consolidated Must-Score Facts

1. **CO_2** has **GWP = 1**. It dominates long-term forcing from fossil fuels, cement, and deforestation.
2. **Methane (CH_4)** has GWP about **28–36**, a shorter lifetime near **12 years**, and strong links to paddies, cattle, landfills, termites, and wetlands (**marsh gas**).
3. **Nitrous oxide (N_2O)** is "laughing gas," has very high GWP (about **265**), and is strongly linked to **fertilizer** use. **Rice** emits both CH_4 and N_2O.
4. **Water vapour** is the most abundant natural greenhouse gas (~half of natural effect), but it mainly acts as a **feedback**, not the primary anthropogenic driver.
5. **Tropospheric ozone** (क्षोभमंडलीय ओजोन) is a greenhouse gas and pollutant. **Stratospheric ozone** (समतापमंडलीय ओजोन) is the UV shield.
6. **CFCs** are both greenhouse gases and ozone-depleting substances and fall under Montreal controls.
7. **HFCs** are greenhouse gases only (not classic ODS) and are phased down under **Kigali 2016**.
8. **GWP** compares heat-trapping power to CO_2 over a stated horizon, usually **100 years**. Methane's short-horizon GWP is much higher than CO_2 — CH_4 is more potent per molecule.
9. **CO_2e** converts all greenhouse gases into carbon-dioxide equivalent using GWP.
10. **Direct** GHGs: CO_2, CH_4, N_2O, CFCs, SF_6, NF_3. **Indirect:** NO_x, CO, NMVOCs, SO_2.
11. **Carbon fertilization** = higher plant growth from elevated CO_2.
12. **Methane hydrates** in Arctic/seafloor can release CH_4 if warmed; CH_4 oxidises to CO_2 in about a decade or two.
13. **China** is the largest CO_2 emitter; India ~3rd absolute (निरपेक्ष); **Bhutan** often carbon-negative.
14. **CO_2** is the main anthropogenic GHG by total forcing (~**76%** IPCC-style). Many Indian papers also use WMO-style relative shares: **~64% CO_2, ~19% CH_4, ~6% N_2O, ~11% CFCs + others**.
15. **GWP = 1** — the reference gas for converting other gases to **CO_2e**.
16. **Carbon fertilization** increases plant growth from higher CO_2 — distinct from warming and ocean acidification (महासागरीय अम्लीकरण).
17. **Methane** is the second major anthropogenic GHG by forcing (~**16%** IPCC-style; ~**19%** in WMO-style relative-share stems).
18. **GWP ≈ 28–36** (100-year); atmospheric lifetime ~**12 years**. Short-horizon GWP is much higher than CO_2.
19. **Livestock** enteric fermentation (cattle/buffalo) is a major Indian source.
20. **Methane hydrates** under Arctic tundra (टुंड्रा) and seafloor can release CH_4 if warmed; CH_4 oxidises to CO_2 in roughly a decade or two.
21. **CNG** is mainly **methane** (~90%).
22. **Biogas** is mainly **methane + carbon dioxide**.
23. **N_2O** is a long-lived GHG with **GWP ≈ 265** (100-year) and lifetime ~**100+ years**.
24. **Water vapour** is the strongest contributor to the **natural** greenhouse effect by volume.
25. **Tropospheric (ground-level) ozone** is both an air pollutant and a greenhouse gas.
26. **Stratospheric ozone** is the UV shield — its ma role is protection, not climate forcing.
27. **CFCs** are human-made compounds once used in refrigeration, aerosols, and foam blowing.
28. **HFCs** replaced many CFCs/HCFCs in air-conditioning and refrigeration.

Chapter 39: Pollution (Advanced)

🎯 Consolidated Must-Score Facts

1. **Primary** air pollutants are emitted directly (SO_2, PM, CO, NOx, lead). **Secondary** pollutants form in air (O_3, PAN, photochemical smog, acid mist).
2. **Lichens** are pollution indicators and do **not** grow well in badly polluted air.
3. **NCAP (2019)** targets non-attainment cities with about a **20–30%** particulate cut goal.
4. **London smog** is SO_2 plus smoke. **Photochemical smog** needs NOx, VOCs, and sunlight and yields **ozone + PAN**; **PAN** is a strong eye irritant.
5. **Fly ash** is a thermal-power primary pollutant usable in bricks/cement; oxides SiO_2/Al_2O_3/CaO **plus toxic metals**. **BS-VI** began **1 Apr 2020**.
6. **SAFAR** (सफर) is an IITM air-quality and weather forecast system. **WAYU** units were installed in **Delhi**.
7. **CO** forms carboxyhaemoglobin (**~200–300×** affinity vs O_2). **Radon** is the top indoor air-pollutant teaching fact.
8. **BOD** measures organic pollution through microbial oxygen demand. High BOD means worse water quality.
9. **COD** is usually greater than or equal to BOD. Clean water often shows DO above about **6 mg/L** and BOD below about **3 mg/L**.
10. **Eutrophication** (सुपोषण) is nutrient (N/P) overload → algal bloom → dissolved-oxygen crash — water, not air.
11. **Damodar** (दामोदर) = biological desert. Ganga (गंगा) BOD peak classically **Kanpur (कानपुर)–Allahabad (इलाहाबाद)**. Leather industry = high chemical pollution.
12. **Moradabad** was flagged among the world's noisiest cities in UNEP (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) Frontier reporting.
13. **CCAC** focuses on **methane, black carbon, HFCs** — not a G20-only initiative.
14. **Asian Brown Cloud** is especially associated with **South Asia**. Bag filter ≠ cyclone divider.
15. **Bioremediation** cleans via microbes; **Cd/Pb** not readily treated; **Oilzapper** (TERI) for oily waste.
16. **Bhopal MIC (Dec 1984)**; **Chernobyl (1986)** radioactive disaster teaching pair.
17. **SWM Rules 2016** give elaborate landfill/site criteria. **R2 Code** = responsible electronics recycling.
18. **H-CNG** reduces CO/HC/CO_2 vs CNG but does **not eliminate** CO; up to ~**1/5** H_2 by volume; not cheaper than CNG.
19. **Point sources** are chimneys, stacks, and pipes; **non-point sources** include roads, farms, and open burning.
20. **Air quality** is judged against CPCB **National Ambient Air Quality Standards (NAAQS)**.
21. **Lichens** are sensitive bio-indicators of air pollution, especially SO_2 stress.
22. **SAFAR** (System of Air Quality and Weather Forecasting and Research) is run with **IITM** support for forecast products.
23. **NCAP** (राष्ट्रीय स्वच्छ वायु कार्यक्रम) launched in **2019** to improve air quality in non-attainment cities.
24. **Smog** is a secondary air pollution mixture; classical **London smog** is linked to **SO_2 + smoke** in cold, humid air.
25. **Photochemical smog** forms when **NOx and VOCs/hydrocarbons** react in strong sunlight.
26. **PAN (peroxyacetyl nitrate)** is a powerful **eye irritant** (lachrymator) and damages plant chloroplasts.
27. **Fly ash** pollution is caused by **thermal power plants**.
28. **Cyclone divider** is **not** a standard instrumental device name for air-pollution control.
29. **Catalytic converters** use **platinum, palladium, and rhodium** to convert CO → CO_2, NOx → N_2, and HC → CO_2 + H_2O.
30. **H-CNG** (hydrogen-enriched CNG) can cut CO/HC/CO_2 versus CNG; hydrogen up to about **one-fifth** by volume can be blended; it does **not eliminate** CO and is **not** cheaper than CNG.
31. **CCAC** (Climate and Clean Air Coalition (गठबंधन), **2012**, with UNEP) focuses on **methane, black carbon, and HFCs** — statement that it is a unique G20-only initiative is false.
32. **Asian Brown Cloud** / atmospheric brown cloud is especially noted over **South Asia** (often Jan–Mar), rich in aerosols including black carbon and sulphate.
33. **BOD** measures oxygen used by microbes decomposing organic matter — a standard **aquatic pollution assay**.
34. **Eutrophication:** excess **N and P** → algal bloom → DO crash → fish kill. Eutrophication is **not** an air-pollution process.

35. **Arsenic** contamination is classically associated with the **Bangladesh** / lower Gangetic basin story; first Indian notice often dated to **West Bengal, 1978**.

Chapter 40: Acid Rain (अम्ल वर्षा)

🎯 Consolidated Must-Score Facts

1. **Acid rain** means acidic deposition with **pH below 5.6**.
2. **Wet deposition** arrives with rain, snow, or fog. **Dry deposition** settles as acidic gases or particles without rain.
3. **CO_2 → H_2CO_3** sets the weak baseline. Carbon dioxide alone is **not** the main culprit of severe acid rain.
4. **Lichens** die in high SO_2 and act as pollution bio-indicators — they indicate pollution but do **not** thrive in polluted air.
5. **Mathura (मथुरा) refinery** is a classic SO_2 source in the Taj Trapezium narrative.
6. **Acid rain** means precipitation or atmospheric deposition with **pH below 5.6**, mainly from **sulphuric acid (H_2SO_4)** and **nitric acid (HNO_3)** (नाइट्रिक अम्ल).
7. **Normal unpolluted rain** is already slightly acidic (~**pH 5.6**) because **CO_2** forms weak **carbonic acid (H_2CO_3)** — it is **not** pH 7.
8. **SO_2** from burning sulphur-rich **coal/oil** in thermal plants, refineries, smelters, brick kilns, and generators forms **H_2SO_4**.
9. **NOx** from vehicles, power plants, and high-temperature combustion forms **HNO_3**.
10. **CO_2** only sets the weak baseline (~5.6); it is **not** the main cause of damaging acid rain.
11. **SO_2 is heavier than air** and can linger near ground-level sources.
12. **Conifers** (pine, spruce) are especially sensitive to acid fog and soil change.
13. **Lichens** die in high-SO_2 air and act as pollution bio-indicators (Topic 38 crossover).
14. **Flue Gas Desulphurisation (FGD)** scrubbers use lime/limestone to cut SO_2 from power-plant stacks.
15. **Taj Trapezium Zone (~10,400 km^2)** — Supreme Court-driven controls (1996 onward) forced gas switch/relocation of polluting units around Agra.

Chapter 41: Desertification (मरुस्थलीकरण)

🎯 Consolidated Must-Score Facts

1. **Desertification** is land degradation in **arid, semi-arid, and dry sub-humid (ASDM)** drylands. It is not merely "the desert expanding."
2. **Land degradation** is the broader productivity decline and can occur in any climate zone.
3. **DPAP** began in **1973**. Drought and desertification reinforce each other in a feedback loop.
4. **Overgrazing** is the top dryland cause in many coaching facts. Afforestation helps prevent further degradation.
5. **Land Degradation Neutrality (LDN)** (भूमि क्षरण तटस्थता) targets **no net loss** of healthy land by **2030** (SDG **15.3**).
6. **Desertification** is land degradation in **arid, semi-arid, and dry sub-humid (ASDM)** areas from climate variation and human activities.
7. **Overgrazing** is the leading human cause in drylands; deforestation and unsustainable cropping follow.
8. **Wind erosion** dominates true arid zones such as the **Thar** (थार).
9. **UNCCD** adopted **17 June 1994** (Paris); force **1996**.
10. **17 June** = World Desertification and Drought Day.
11. **SDG 15.3** targets **Land Degradation Neutrality (LDN) by 2030**.
12. **Land degradation** is any lasting decline in the land's productive capacity.
13. **Desertification** is the ASDM subset; erosion, salinity, waterlogging, mining, and pollution are processes under the broader term.
14. **Afforestation reduces** degradation — it is not a cause.
15. **Drought** is a prolonged water deficit relative (सापेक्ष) to normal supply — not the same as desertification, though the two reinforce each other.
16. **Meteorological drought:** rainfall well below normal (IMD often uses **< 75% of Long Period Average** as a severe marker).
17. **Hydrological drought (जल-वैज्ञानिक सूखा):** rivers, reservoirs, and groundwater fall.
18. **Agricultural drought (कृषि सूखा):** soil moisture fails crops even if some rain fell.
19. **Drought Prone Areas Programme (DPAP)** launched **1973–74** for dryland watershed-type works.
20. **Normal (geological) erosion** is slow and roughly balances soil formation.
21. **Accelerated erosion** is human-driven and faster than formation — the focus.
22. **Water erosion ladder:sheet → rill → gully → ravine**.
23. **Chambal ravines** (चंबल) (MP–UP–Rajasthan, including Etawah (इटावा)–Agra belt) are the classic Indian ravine badlands.
24. **Wind erosion:deflation** (lifting fine particles) and **abrasion** (sandblasting) — strong in the Thar.

Chapter 42: Environmental Monitoring (पर्यावरणीय निगरानी)

🎯 Consolidated Must-Score Facts

1. **Lichens** are classic bio-indicators for sulphur dioxide stress.
2. **Monitoring** means systematic repeated measurement against standards. It is not the same as an audit.
3. **CPCB** (केंद्रीय प्रदूषण नियंत्रण बोर्ड) runs **NAMP** (manual air), **CAAQMS** (real-time air), and **NWMP** (surface water).
4. **CGWB** monitors groundwater quality and levels and publishes the annual groundwater quality report.
5. **SAFAR** (सफर) is the IITM air-quality and weather forecast system associated with Harsh Vardhan's launch messaging in **July 2018**.
6. **NCAP (राष्ट्रीय स्वच्छ वायु कार्यक्रम) (2019)** expands city monitoring and action. **SAMEER** is an AQI public app.
7. **Form V** Environmental Statement under EPA Rule **14** goes to the SPCB yearly.
8. **ISO 14001** is the Environmental Management System standard.
9. **BOD** is microbial oxygen demand. **COD** is chemical oxygen demand. COD is usually ≥ BOD.
10. **Environmental indicators** turn raw data into decision signals for policy and public warning.
11. **PSR model:Pressure** (emissions, resource use) → **State** (ambient quality) → **Response** (treatment capacity, action plans).
12. **Ringelmann scale (रिंगेलमैन पैमाना) (0–5)** measures **smoke plume density/opacity**, not water, fog, or noise.
13. **Environmental monitoring** is systematic, repeated measurement of air, water, soil, noise, and related parameters against standards.
14. **CPCB** runs national **air** and **surface-water** programmes and sets ambient/effluent standards with SPCBs.
15. **NAMP** = manual air stations for long-term trends; **CAAQMS** = continuous real-time ambient air (feeds AQI boards).
16. **NWMP** = National Water Quality Monitoring Programme for rivers/lakes.
17. **CGWB** monitors **groundwater** levels and quality; publishes the **Annual Groundwater Quality Report** — not CPCB.
18. **SAFAR** (IITM) combines air-quality monitoring with weather forecasting; inaugurated by **Dr. Harsh Vardhan** (July 2018).
19. **NCAP (2019)** expands monitoring and action in non-attainment cities (including multiple UP cities).
20. **SAMEER** app disseminates CPCB AQI; **PARIVESH** tracks EC compliance reporting.
21. **UPPCB** issues Consent to Establish/Operate and inspects industries in Uttar Pradesh (उत्तर प्रदेश).
22. **Environmental audit** is a systematic, documented check of whether an organisation meets environmental laws, consent conditions, and internal EMS goals.
23. **ISO 14001** is the voluntary international standard for an **Environmental Management System (EMS)**.

Chapter 43: International Environmental Organizations

🎯 Consolidated Must-Score Facts

1. **UNEP** was created after **Stockholm 1972** (स्टॉकहोम सम्मेलन). Its HQ is **Nairobi**. Flagship products include the **Emissions Gap Report** (उत्सर्जन अंतराल रिपोर्ट) and **GEO**.
2. **UNDP** sits in **New York** and publishes the **Human Development Report / HDI** (from **1990**).
3. **FAO** sits in **Rome** and publishes **SOFO**. It co-hosts Rotterdam work with UNEP.
4. **UNESCO** sits in **Paris** and runs World Heritage (विश्व धरोहर) and the **MAB** Biosphere Reserve (जैवमंडल आरक्षित क्षेत्र) programme.
5. **IUCN** (आईयूसीएन) sits in **Gland** and publishes the **Red List** of extinction risk.
6. **WWF** also sits in **Gland** and publishes the **Living Planet Report / LPI**.
7. **IPCC** sits in **Geneva**. It was created by **WMO + UNEP** in **1988** and shared the **2007** Nobel Peace Prize.
8. **WMO** headquarters is **Geneva**. It grew from the International Meteorological Organization (**1873**) and became WMO on **17 March 1951**. It co-created the **IPCC** with UNEP in **1988**.
9. **Greenpeace International** is an environmental NGO headquartered in **Amsterdam**, Netherlands.
10. **OPCW** implements the **Chemical Weapons Convention** (force **29 April 1997**); HQ is **The Hague**. It won the Nobel Peace Prize in **2013**. It is **not** an EU, NATO, or WHO body.
11. **UN-Habitat** (UN Human Settlements Programme; HQ **Nairobi**, from **1978**) promotes sustainable towns and shelter. Partners are **not** governments-only.
12. **UNEP (United Nations Environment Programme)** is the UN's principal environment coordination body, created after the **Stockholm Conference (1972)** (स्टॉकहोम सम्मेलन).
13. **HQ: Nairobi, Kenya** — not Geneva or New York.
14. **GEF (Global Environment Facility, 1991)** finances projects under major environment conventions; UNEP is an implementing agency. Donors are **not** "developed countries only."
15. **UNDP (United Nations Development Programme)** is the UN's development network agency.
16. **FAO (Food and Agriculture Organization)** is the UN specialised agency for food and agriculture.
17. **UNESCO** leads education, science, and culture — environment enters via heritage and biosphere programmes.
18. **Limits to Growth** is **Club of Rome**, not UNESCO.
19. **IUCN (International Union for Conservation of Nature)** is a global conservation union of governments and NGOs, founded **1948**.
20. **HQ: Gland, Switzerland**.
21. **Threatened** categories commonly = **VU + EN + CR**.
22. **WWF (World Wide Fund for Nature)** is a global conservation **NGO**, not a UN agency.
23. **HQ: Gland, Switzerland** — same city as IUCN, different organisation.
24. **IPCC (Intergovernmental Panel on Climate Change (जलवायु परिवर्तन))** was established in **1988** by **WMO** and **UNEP**.
25. **UNFCCC/COP** is the policy–treaty track; IPCC informs it.
26. **WMO headquarters is Geneva, Switzerland.** It co-created the **IPCC** with UNEP in **1988**.
27. **Greenpeace International** is an environmental NGO headquartered in **Amsterdam, Netherlands**.
28. **OPCW (Organisation for the Prohibition (प्रतिषेध) of Chemical Weapons)** implements the **Chemical Weapons Convention** (force **29 April 1997**); HQ **The Hague**, Netherlands. It verifies destruction/non-re-emergence of chemical weapons and assists states against chemical threats — **not** an EU/NATO/WHO body. Nobel Peace Prize **2013**.
29. **UN-Habitat** (UN Human Settlements Programme; HQ **Nairobi**, from **1978**) is mandated to promote socially and environmentally sustainable towns/cities and adequate shelter; partners include governments, local authorities, and non-state actors; contributes to poverty reduction and safe water/sanitation access. Partners are **not** governments-only.

Chapter 44: International Environmental Agreements

🎯 Consolidated Must-Score Facts

1. **Kyoto** (क्योटो) set binding targets mainly for **Annex-I** parties. **Paris** uses universal **NDCs** for all parties.
2. **Montreal** protects the ozone layer. **Kyoto / Paris / UNFCCC** are climate instruments. Never swap Montreal for climate mitigation (जलवायु शमन).
3. **CBD (1992)** is the biodiversity convention. **Cartagena** is the biosafety protocol under CBD.
4. **Ramsar (1971)** covers wetlands. **CITES (1973)** (साइट्स) covers wildlife **trade** (पण्याध्यक्ष). **CMS / Bonn (1979)** covers **migratory** species.
5. **Basel (1989)** covers hazardous **waste**. **Rotterdam (1998)** covers hazardous **chemicals** through the PIC procedure (प्रक्रिया).
6. **Stockholm Convention (2001)** (स्टॉकहोम संधि) covers **POPs** such as DDT and PCBs. It is **not** the Stockholm Conference of **1972**.
7. **Minamata (2013)** covers **mercury**.
8. **CBD** opened at **Rio 1992** with three objectives: **conservation**, **sustainable use**, and **fair benefit-sharing** of genetic resources.
9. **Cartagena Protocol** = biosafety / LMOs.
10. **Nagoya Protocol (2010)** = Access and Benefit Sharing (ABS).
11. **World Wetlands Day (विश्व आर्द्रभूमि दिवस) = 2 February** (Ramsar adoption date).
12. **Montreux Record** = Ramsar sites with adverse / likely adverse ecological change from human interference.
13. **CITES** regulates **international trade** in endangered wild fauna (प्राणीजात) and flora (वनस्पति).
14. **World Wildlife Day (विश्व वन्यजीव दिवस) = 3 March** (CITES adoption date).
15. **Not** the same as the **Stockholm Conference 1972** that created UNEP.

Chapter 45: Current Environmental Issues

🎯 Consolidated Must-Score Facts

1. **Net zero** balances remaining emissions with removals. India's year fact is **2070**. It is not gross zero and not automatically **2050**.
2. **Green hydrogen** uses electrolysis powered by renewables. India's mission (January **2023**) targets about **5 MMT** by **2030**.
3. **Grey hydrogen** reforms natural gas. **Blue hydrogen** adds CCS (सुरक्षा कैबिनेट समिति) to grey pathways. Do not call grey "green."
4. **Microplastics** are plastic pieces smaller than **5 mm**. They are primary (made small) or secondary (fragmented) and are not fast biodegradable.
5. **Climate refugees** are mostly **internally** displaced. They are **not** a separate legal category under the **1951** Refugee Convention.
6. **Nature-based Solutions** protect, manage, or restore ecosystems (IUCN (आईयूसीएन) framing). They are not geoengineering (भू-इंजीनियरिंग) (cirrus thinning / sulphate aerosols are geoengineering ideas for cooling, not NbS).
7. **LiFE** was an idea at **COP-26 (2021)** and launched in **June 2022** under MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) with a long action list. **COP-25 Madrid** is the false trap.
8. **Scope 1** covers **direct** emissions from owned sources (boilers, fleet fuel).
9. **Scope 2** covers **purchased electricity, heat, or steam**.
10. **Scope 3** covers other indirect emissions (supply chain, product use, travel, waste) — often the largest share and hardest to measure.
11. **Biocapacity** is nature's regenerative capacity; if footprint exceeds biocapacity, there is an **ecological deficit**.
12. **Earth Overshoot Day** marks when annual demand exceeds that year's regeneration.
13. **Net zero** means anthropogenic emissions are **balanced by anthropogenic removals** over a period — residual emissions may remain if matched by sinks or removals.
14. **Gross zero** means eliminating virtually all emissions — harder than net zero.
15. **UNEP Emissions Gap Report** measures the gap between pledges and 1.5°C/2°C pathways.
16. **Green finance** (हरित वित्त) (sovereign green bonds, SEBI green-bond rules) funds this pathway — detailed market tools sit with Topic 24.
17. **Blue economy** is sustainable use of ocean and coastal resources for growth and jobs while protecting marine ecosystems.
18. **SDG 14** = **Life Below Water**.
19. **Blue carbon** is carbon stored in mangroves, seagrass, and salt (लवणाध्यक्ष) marshes.
20. **MISHTI** (Mangrove Initiative for Shoreline Habitats and Tangible Income) was announced in **Union Budget 2023–24** to restore mangroves for shoreline protection and income.
21. **Sagarmala** is port-led coastal development; **Deep Ocean Mission** covers deep-sea exploration (Samudrayaan).
22. **Plastic Waste Management Rules 2016** (amended 2022) and related EPR rules make producers responsible for end-of-life collection.
23. **Green hydrogen** is H_2 from **water electrolysis** powered by **renewable electricity**.
24. **Grey hydrogen** from natural-gas reforming dominates today's supply and is **not** climate-friendly.
25. **Blue hydrogen** is grey production plus **CCS** — a debated transition label.
26. **National Green Hydrogen Mission (January 2023)**, coordinated under **MNRE**, targets about **5 MMT/year by 2030** for refining, ammonia/fertiliser, steel, and heavy transport.
27. **Microplastics** are plastic particles **smaller than 5 mm**.
28. **Primary** types are manufactured small (microbeads, nurdles).
29. **Secondary** types form when larger plastics fragment (bags, nets, tyre wear) — most environmental mass is secondary.
30. **Climate refugees / climate migrants** are people forced to move by climate impacts — cyclones, floods, drought (सूखा), sea-level rise, desertification (मरुस्थलीकरण).
31. **Nature-based Solutions** protect, sustainably manage, or restore ecosystems to address societal challenges (climate, disasters, food, water) while aiding biodiversity and well-being.
32. **MISHTI mangrove restoration** is a flagship Indian NbS example (Budget **2023–24**).

33. **LiFE (Lifestyle for Environment)** nudges individual and community behaviour toward mindful consumption.
34. **Mission LiFE** was formally launched in **June 2022**; **MoEFCC** (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) is nodal.

Chapter 46: Environment (पर्यावरण) PYQ Trend Analysis

🎯 Consolidated Must-Score Facts

1. **2023** was the peak tagged year (**25** questions). **2025** stayed strong at **19**.
2. **2020** and **2024** dips in tagged count still hide cross-subject environment leakage in Economy and Geography.
3. **DPAP = 1973**. India **net zero = 2070**. **GEF = 1991**. **Montreal = ozone**. **Kyoto (क्योटो)/Paris = climate**.
4. **Early period (2018–20):** average **12 questions** per paper.
5. **Recent period (2023–25):** average **17.7 questions** per paper, **+47% increase**.
6. ** planning rule: **budget** 15–20 environment marks** in every attempt. Treating env as a 8-question afterthought is outdated.
7. **2020 and 2024 dips** are anomalies. Cross-subject env-adjacent questions (Economy reports, Geography protected areas) keep true environment ROI higher than tagged count alone.
8. **2-statement correct?** — Scheme years, treaty purposes, pollution facts
9. **A/R** — SDGs, overgrazing→desertification, ecological concepts
10. **Match List-I / II** — SDG ↔ goal; org ↔ report; erosion ↔ landform
11. **NOT in UP / NOT in India** — Ramsar, sanctuaries, biosphere reserves
12. **NOT correctly matched** — Treaty ↔ wrong purpose; wrong HQ; wrong year
13. **Direct MCQ** — Movement identification; report publisher; single fact